FT-ITC ANALYSIS · USER MANUAL

FT-ITC Analysis
User manual

Processing, fitting, and interpreting isothermal titration calorimetry experiments.

VERSION 1.5.0 · FOR MACOS, WINDOWS, AND LINUX
Manual ready.

Print / Save as PDF ↓

FT-ITC ANALYSIS · USER MANUAL

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CHAPTER 01

FT-ITC Analysis user manual

Start here for a tour of FT-ITC Analysis and the conventions used throughout this manual.

Last verified: 2026-09-23

FT-ITC Analysis is a desktop application for processing, fitting, comparing, and presenting isothermal titration calorimetry (ITC) experiments.

This manual is for ITC practitioners who are new to FT-ITC Analysis. It explains how to use the application and provides guidance on scientific interpretation; no programming knowledge is required.

These instructions apply on macOS, Windows, and Linux. The manual treats the supported desktop applications as one product. A **Platform note** appears only when an operating-system or interface difference changes how you complete a task.

> **VERIFIED:**
> This edition reflects FT-ITC Analysis 1.5.0 and was verified on the date shown for each page.

What the application does

FT-ITC Analysis supports the complete workflow from compatible instrument data to a saved analysis project and publication figures:

1. Import a raw thermogram, integrated heats, or an existing project.
2. Review and edit experiment details, concentrations, comments, and attributes.
3. Process thermograms using baseline models, spline editing, and manual or automatic adjustment of integration regions.
4. Fit one or more experiments with a supported model.
5. Review fitted parameters, uncertainties, residuals, and result validity.
6. Save portable projects in the `.ftxtc` format.
7. Export figures, numerical data, and result tables.

Raw input files are read, not rewritten. Desktop analysis is local: experiment data are not uploaded during ordinary analysis. An optional launch-time online check retrieves version and citation information. Optional automated interpretation sends the selected report evidence and supplied context to the hosted interpretation service at app.ft-itc.org, which uses OpenAI; usage metadata are retained without automatic expiry. The separate browser-based Project Viewer at the same host uploads a project for viewing and parsing; opening a project there does not submit it for automated interpretation. See Privacy and online checks and Analysis Report.

Product tour

The data list contains loaded experiments and completed Analysis Results. Selecting an experiment exposes four shared task views:

- **Overview** summarizes the experiment and provides access to its details.
- **Process Data** builds and refines baselines, adjusts injection regions, and converts a raw thermogram into integrated heats with estimated uncertainties.
- **Analyze Data** fits a single experiment or multiple experiments.
- **Final Figure** presents the thermogram, heats, fitted curve, residuals, and annotations.

Selecting an Analysis Result opens its result workspace, with a parameter summary, member fits, parameter correlation, uncertainty display, and any compatible advanced analyses. The menus provide project operations, experiment management, export commands, preferences, additional tools, citation information, and support links. The FT-ITC Analysis website, latest release, and software DOI provide project, installation, and citation context.

Manual conventions

The shortest route to a first result is Quick start, which uses your own compatible data. The relevant task chapter provides setting details and diagnostic guidance when a workflow needs closer inspection.

Manual directory

- Quick start
- Installation, files, and projects
- Workspace
- Processing
- Single-experiment fitting
- Multiple-experiment fitting
- Results and advanced analyses
- Figures and export
- Tools
- Settings and defaults
- Glossary and equations

The manual uses these callouts:

> **NOTE:**
> A detail that affects how the application behaves.

> **CAUTION:**
> A condition that can produce a failed task, invalid result, or misleading output.

> **INTERPRETATION:**
> Scientific context for judging output; it is not an automated conclusion made by the application.

> **RECOMMENDATION:**
> A defensible working practice rather than a software requirement.

> **PLATFORM NOTE:**
> A genuine operating-system or interface difference that changes how a task is completed.

> **CALCULATION:**
> A compact relationship used by FT-ITC Analysis or needed to interpret its output.

Interface labels appear in **bold**. A path such as **File > Save As...** means choose the menu and then the command. The names of views and controls are used instead of position-dependent phrases, so the instructions remain usable when the window size or platform changes.

Help and support

Help and Guide opens an in-app manual, and **Technical Details** opens the in-app processing and scientific notes. **Citation** presents the current paper and versioned software citations and supports copying or exporting BibTeX. **Contact Support...** helps you contact support by email and prepares a diagnostic report containing the application version, operating system, recent activity, and full application log. **Copy Support Report** places that report on the clipboard. Official links include the FT-ITC Analysis website, latest release, project viewer, and software DOI 10.5281/zenodo.14832177.

CHAPTER 02

Quick start

Open your own ITC dataset, process it, fit a one-set-of-sites model, review the result, save the project, and export a figure.

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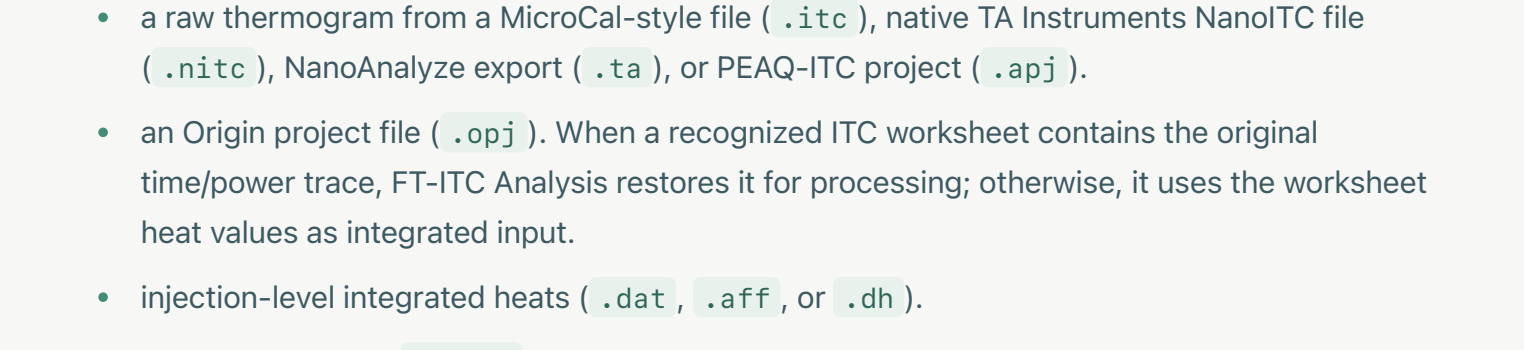
This procedure takes a compatible file through an ordinary one-set-of-sites analysis. Use a dataset for which that model is scientifically plausible. The goal is to learn the application workflow, not to prescribe the correct model for every interaction.


1
Open
raw data, heats, or project
2
Check details
concentrations, temperature, metadata
3
Process
baseline and integration
4
Fit
model, constraints, uncertainty
5
Review
curve, residuals, validity
6
Preserve
save project and export output

The overall analysis workflow. Integrated-heat imports enter after thermogram processing.

Before you begin

Have one of these inputs available:

- a raw thermogram from a MicroCal-style file (`.itc`), native TA Instruments NanoITC file (`.nitc`), NanoAnalyze export (`.ta`), or PEAQ-ITC project (`.apj`).
- an Origin project file (`.opj`). When a recognized ITC worksheet contains the original time/power trace, FT-ITC Analysis restores it for processing; otherwise, it uses the worksheet heat values as integrated input.
- injection-level integrated heats (`.dat`, `.aff`, or `.dh`).
- an FT-ITC project (`.ftxtc`).

Supported formats and project behavior are described in Installation, files, and projects.

> **NOTE:**
> Integrated-heat files and Origin projects without an original time/power trace skip baseline correction and peak integration; begin with experiment details and fitting.

1. Open your data

Launch FT-ITC Analysis and choose **Open File...** on the welcome screen, choose **File > Open...**, or drag compatible files into the application window. Select your file and confirm the open operation.

The experiment appears in the data list. Select it and open **Overview** to orient yourself in the imported experiment. Drag any non-button area of a data or result row to change its position. This order is used throughout the application and is retained when the project is saved as `.ftxtc`.

2. Edit experiment details

Open **Details...** for the selected experiment. Concentration entries and the experiment date/time can be changed here when needed. Comments and attributes relevant to later analysis can also be added or edited. A changed date is marked as user-modified in the experiment overview.

Apply corrections only when you have an independent experimental basis. Concentration entries influence the calculated concentration ratio and fitted parameters.

3. Process a raw thermogram

If the import contains a thermogram, open **Process Data**.

1. If the trace shows a smooth global drift, try **Polynomial** baseline. For more complicated baseline shapes, choose **Spline**; for local baseline behavior, choose **Segmented**. Keep the default integration settings initially.
2. Inspect whether the baseline represents the signal between injections rather than the peaks.
3. Inspect the start and end of every integration region. A region should include the injection response without extending unnecessarily into baseline noise. Zoom to a peak and adjust the integration end point. Use **Space** to copy the integration length to the next injection; the start is also copied when **Copy start time to next** is enabled.

These baseline alternatives are explained in Processing.

> **INTERPRETATION:**
> A visually smooth baseline is generally desired. As a rough rule of thumb, aim for about 20% of the data points to be baseline data; this is guidance, not a hard threshold. Apparent jumps in the baseline may indicate insufficient equilibration time between injections. Major spikes may require advanced baseline editing; the **Spline** baseline type provides more flexibility.

4. Fit one set of sites

Open **Analyze Data**, choose **Single experiment**, and select **One-Set-Of-Sites**.

1. Review the initial parameter values. Use physically plausible orders of magnitude for affinity, enthalpy, and stoichiometry.
2. Choose an optimizer. **Levenberg-Marquardt** is efficient near a suitable solution; **Nelder-Mead** can be useful when the starting surface is less cooperative.
3. Optionally enable **Weight by injection error** when the integration uncertainties are meaningful for the dataset.
4. Choose **None**, **Bootstrap residuals**, **Leave-one-out**, or **Profile likelihood** for error estimation.
5. Choose **Run Fit**.

Create analysis result determines whether a usable fit also creates a separate Analysis Result. Enable it to continue through the result workspace in the next step; otherwise, the fitted solution remains attached to the Experiment Data.

If the fit fails or reaches a limit, the possible causes are described under Fit availability and non-convergence.

5. Review the result

Inspect the fitted curve together with the residuals. When **Create analysis result** is enabled, select the resulting **Analysis Result** and check:

- the included experiment and injections;
- fitted parameter values and units;
- convergence and fit loss;
- whether weighting and uncertainty estimation match your intention;
- parameter uncertainty or confidence intervals, when calculated;
- result validity.

When **Create analysis result** is disabled, the fitted solution remains available in **Analyze Data** rather than as a stored Analysis Result.

> **INTERPRETATION:**
> Parameter precision does not establish model adequacy. Look for structured residuals, sensitivity to initial values, values at limits, excessive parameter correlation, or disagreement with known stoichiometry and experimental conditions.

6. Save a portable project

Choose **File > Save As...** and save the project as `.ftxtc`. The project contains imported data, experiment details, processing state, fits, results, and available advanced-analysis output. Keep the original instrument files as source records even though the project is portable.

If autosave is enabled, it supplements normal saving; it is not a replacement for a named project file.

7. Export a figure

Return to the experiment and open **Final Figure**. Choose the elements you need, such as the thermogram, integrated heats, fitted curve, residuals, confidence band, or parameter information. Review axis ranges and labels, then use the figure export or print command provided by the view.

Use **Analysis Result Exporter...** when you need numerical result tables rather than a graphic. For injection-level heats, choose **File > Export Integrated Peaks...**. The same output is available through **File > Export Data...** by choosing **Integrated Peaks**.

The figure and table workflows continue in Figures and export.

CHAPTER 03

Installation, files, and projects

Install FT-ITC Analysis, open supported data, save portable projects, and use autosave or recovery safely.

Last verified: 2026-09-23

Install the application

Use the latest FT-ITC Analysis release for current packages and release notes. Verify that the file came from this official release page before accepting an operating-system security prompt. Package options may change between releases. For installation help, contact `support@ft-itc.org`.

macOS

Open the DMG, drag **FT-ITC Analysis** to **Applications**, and eject the disk image. If macOS blocks a verified download, review it in **System Settings > Privacy & Security** and use **Open Anyway** only after confirming its source.

Windows

Run the supplied Windows x64 `.exe` installer and follow the setup prompts. The installer registers `.ftxtc` project associations. If Windows displays an **Unknown publisher** or Microsoft Defender SmartScreen warning, continue only after confirming that the installer came from the official FT-ITC release page.

Linux

Install the supplied `.deb` package matching the system architecture (AMD64 or ARM64, when available) on a compatible Debian-based distribution. Package trust and dependency behavior depend on the channel through which the package was obtained. Review the current release notes for known platform limitations.

> **PLATFORM NOTE:**
> Package installation and file-association prompts are controlled by the operating system. The analysis workflow is otherwise shared; the manual notes the few interface labels that differ between desktop editions.

Supported input formats

EXTENSION	SOURCE	IMPORTED CONTENT
`.itc`	MicroCal-style raw data	Raw thermogram and injections
`.nitc`	TA Instruments NanoITC native data	Raw thermogram and injections
`.ta`	TA Instruments/NanoAnalyze export	Raw thermogram and injections
`.apj`	PEAQ-ITC project	Raw thermogram and injections from the first experiment
`.opj`	Origin project file	Raw thermogram or integrated heats, according to the recognized worksheet
`.dat`	Integrated-heats table	Integrated heats and injections
`.aff`	Integrated-heats table	Integrated heats and injections
`.dh`	Fixed-layout integrated-heats file	Integrated heats, injections, and experiment metadata
`.ftxtc`	Current FT-ITC project	Stored project state
`.ftitc`	Legacy FT-ITC project	Readable for migration; not written by current versions

NanoAnalyze `.ta` files are raw thermogram exports. FT-ITC Analysis restores their time/power data and injection information as unprocessed Experiment Data.

If a MicroCal `.itc` file contains multiple concatenated runs, FT-ITC Analysis asks how to calculate concentration changes between segments. Choose **Use MicroCal Concat** for the standard MicroCal progression or **Use Back-Mixing Compensation** to enter a dead volume, mixing fraction, and whether overflow was removed. The prompt begins with the instrument-compatible MicroCal choice; closing it or choosing **Use MicroCal Concat** uses that choice. See Experiment Merger for the tandem concentration and back-mixing methods.

PEAQ-ITC `.apj` projects contain a raw thermogram as well as injection and analysis information. FT-ITC Analysis imports the raw thermogram and injection information from the first experiment in the project. Integrated heats, processing choices, and fitted results produced by PEAQ are not imported; process and fit the restored raw data in FT-ITC Analysis.

Native NanoITC `.nitc` imports restore the raw thermogram, injection schedule, concentrations, cell volume, temperature and stirring information, and available source details. They open as unprocessed Experiment Data and follow the normal thermogram-processing workflow.

Origin `.opj` files are general project containers. FT-ITC Analysis searches them for the first recognized ITC worksheet and can restore either its original time/power trace or its integrated heats. When a raw trace is available, it is imported as an unprocessed thermogram and is authoritative even if the worksheet also contains integrated heats. If no usable trace is present, the worksheet heat values are used as integrated input and **Process Data** is skipped. ResultsLog text is retained in the experiment comments as source history. Origin baseline processing, fitted models, and Fit/DY columns are not imported or converted into native FT-ITC fits. Newer `.opju` files are not supported.

Delimited `.dat` and `.aff` inputs must provide positive `INJV` injection volumes and at least one usable heat column. FT-ITC Analysis prefers a complete `DH` column as absolute injection heat. If `DH` is absent or incomplete, it accepts a complete `NDH` column as normalized heat per mole; `NDH` may be absent for the automatically excluded first injection. The separate `.dh` format uses a fixed metadata-and-injection layout rather than the delimited column layout used by `.dat` and `.aff` files.

The first injection is imported but excluded from fitting by default because its heat is commonly unreliable. You can include it manually if appropriate. Its injection volume still contributes to the concentration progression used for later injections.

These files do not encode an unambiguous heat unit. For `DH`, select the absolute-energy unit used by that column. For an `NDH`-based import, select the energy unit in the per-mole values (for example, select **calorie** for cal/mol). The reader converts normalized heat to absolute injection heat using the syringe concentration and injection volume. If the syringe concentration cannot be inferred from `Xt` / `Mt`, it must be supplied before an NDH-based import can continue; canceling that prompt skips only the current file. Reuse the selected unit for the remaining files only when every file in that import operation uses the same heat unit and heat-column convention.

For `.dat` and `.aff`, **Recalculate concentrations and ratios** uses the injection volumes and selected dilution model to calculate the concentration progression instead of retaining the table's progression. The checkbox starts at the corresponding preference, but your choice applies only to the current import. **Use selected action for remaining files** reuses the unit and reprocessing choice for later `.dat` and `.aff` files in the same import. A `.dh` file contains metadata and injection volumes and heats, without a concentration progression to retain, so its concentrations and ratios are always calculated; its prompt has no reprocessing checkbox.

When `Mt` / `Xt` concentration values are available, the reader uses their progression and the selected dilution model to infer cell volume and syringe concentration. This inference does not depend on the heat values. Each injection row stores the concentrations before that injection; an optional final row without an injection stores the concentrations after the last injection. `Mt` and `Xt` are interpreted as mM in normal application imports. If the concentration sequence is absent, malformed, or internally inconsistent, `DH` heat and injection-volume rows remain importable; an `NDH`-based import first requires a syringe concentration, and validation asks for any other unresolved metadata instead of silently guessing it. These formats contain no thermogram, so importing them cannot reconstruct a baseline or processing-derived injection uncertainties.

Open files

Choose **File > Open...**, use the welcome-screen action, or drag files into the application. Multiple supported files can be opened together. Files opened into a populated document are added to its existing Data / Results list; this includes `.ftxtc` projects. Clear the current document first if you want to open a project by itself.

> **CAUTION:**
> After an `.ftxtc` project is added to an existing document, that opened project becomes the document's current save destination. Use **Save As...** before saving if you do not intend to replace it with the combined document.

> **CAUTION:**
> Appending can create similarly named experiments or results. Confirm the data list and details before fitting or exporting.

Save projects

Choose **File > Save** to update the current named project, or **File > Save As...** to choose a new name or location. Use the current `.ftxtc` format for ongoing work.

An `.ftxtc` project preserves the data and metadata needed to continue analysis, including thermograms where imported, concentrations and uncertainties, attributes and comments, injection inclusion, processing state, fit solutions, Analysis Results, and completed derived analyses. The package is portable and does not depend on the original raw-file path for ordinary reopening.

Save Selected... writes selected project content when you need a smaller handoff. Confirm the selection before saving and reopen the result if the subset is critical. Saving the selected experiments saves only the experiment with any solution. Saving the selected Analysis Results saves the result along with the involved experiments.

> **RECOMMENDATION:**
> Save a processed version of the project before fitting if you want a reusable starting point. After fitting, save the project again—under a new name if you want to preserve the processed-only version—to retain the fitted solutions and Analysis Results.

Autosave and recovery

Autosave behavior is configured in **Preferences...**. When recovery data is available after an interrupted session, the application offers a recovery path. Open the recovered document, inspect the data list and recent changes, then save it as a named `.ftxtc` project.

Recovery mode is designed to salvage valid project components when possible. If an experiment is unavailable, its saved fit is omitted; Analysis Results that depend on that fit are also omitted rather than being restored with a scientifically different set of members. Unaffected experiments, fits, and results can still be recovered. When recovery was required, the application displays a warning and records the individual recovery issues in the application log. A recovered project can be detached from its former save location and marked as changed. Use **Save As...** rather than assuming the damaged or interrupted file was repaired in place.

> **CAUTION:**
> Recovery cannot guarantee that every optional result or saved analysis survived. Confirm experiment counts, processing, fits, and result validity before continuing.

Remove and clear content

Removing an experiment or Analysis Result changes only the open document; it does not delete the original raw file. **Remove All Data/Results** clears the current document after confirmation. **Clear Processing/Results** removes all Analysis Results from the open document.

Saving after removal makes the removal part of the saved project. Use **Save As...** first if you want to preserve the original project version.

Privacy and online checks

Processing, fitting, saving, recovery, export, and printing run locally. Optional automated interpretation in **Analysis Report** sends the selected report evidence, question, and context to an online service when you request generation. See Analysis Report for details.

If **Check for updates and online resources on launch** is enabled, the application checks for release and citation updates; it does not upload experiment data. Disable this setting to prevent launch-time checks; it does not control automated interpretation requests. A failed or disabled check does not prevent local processing, fitting, or saving.

Update safely

Before installing a new application version, save important projects and retain the current installer when reproducibility policy requires it. After updating, open a copy of a representative project, confirm processing and results, and save only when you intend the project to be written by the new version.

Online data flow and retention

Automated interpretation: Choosing **Generate** sends the selected report evidence and your question or context to FT-ITC's online interpretation service at app.ft-itc.org. The service uses OpenAI to generate the interpretation. Sent evidence can include experiment and result names, comments, conditions, concentrations, saved fits, uncertainty summaries, diagnostics, and injection data. Supporting experiments are included when selected. Compressed thermograms are off when the dialog opens and are sent only if you opt in. **Save package** creates a local copy for review without sending it. Opening Preferences or the generation dialog can check service availability and account access without sending scientific evidence. A supplied access code is sent to verify access.

Project Viewer: Opening a project in the browser-based Project Viewer uploads the complete `.ftxtc` file, including its embedded data and comments, to app.ft-itc.org for display. This does not submit the project for automated interpretation or send it to OpenAI. The viewer has a 50 MB upload limit. Closing the viewer does not retract an upload or guarantee that temporary processing copies or infrastructure records have been deleted; the full service has no verified end-to-end deletion schedule.

Retention: FT-ITC's interpretation service retains usage metadata that can be linked to requests or accounts, without automatic expiry. These usage records do not contain the complete scientific report evidence or generated interpretation. There is no in-app control to delete service records. Canceling generation or deleting a local report does not retract information already sent. Data sent to OpenAI is subject to its applicable data controls and retention policies; see OpenAI data controls. No zero-retention or fixed-deletion guarantee is made for the complete service flow.

CHAPTER 04

Workspace

The application window, Experiment Data capabilities, and Analysis Result capabilities.

Last verified: 2026-09-23

Application window

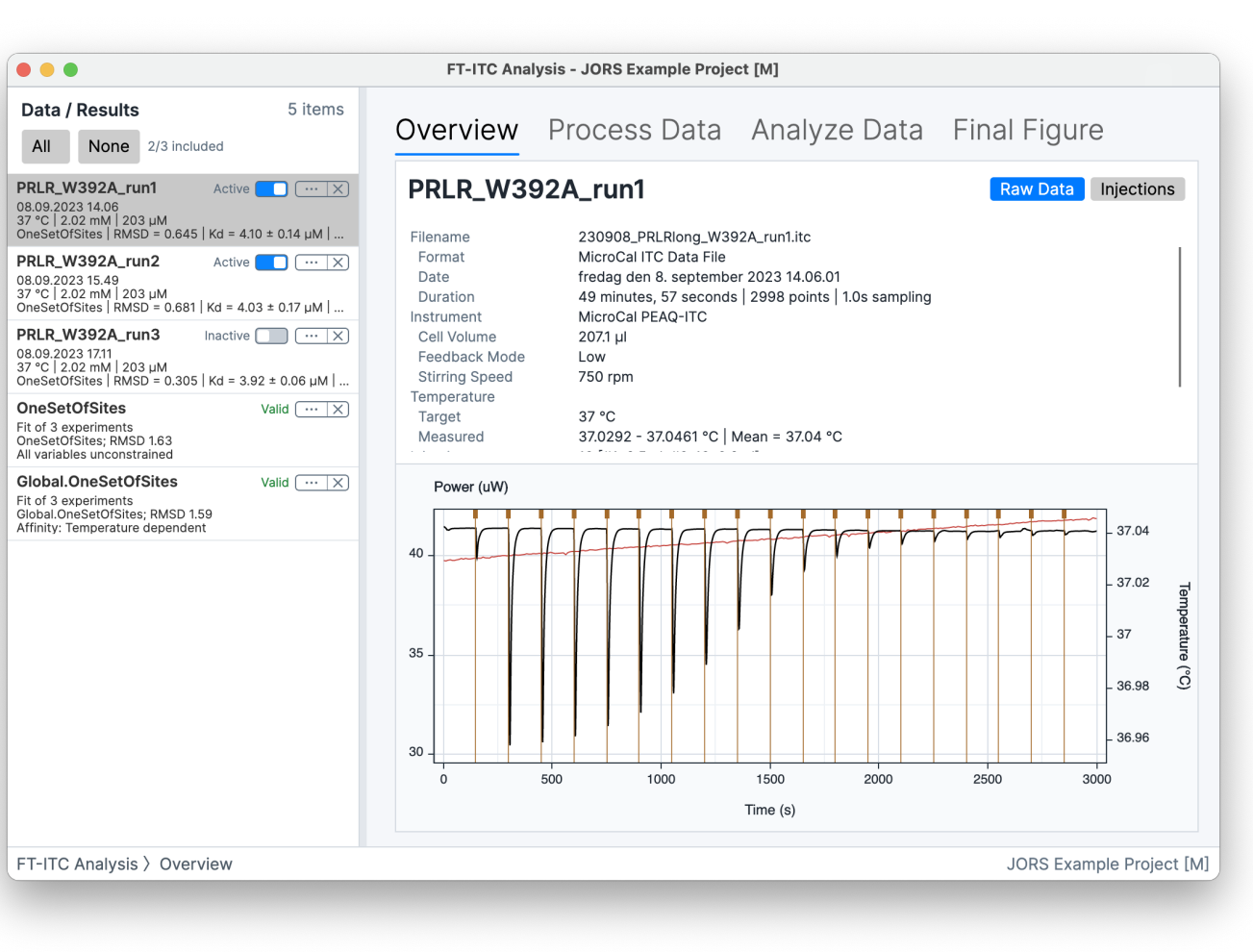
The **Data / Results** list is the project navigator. It contains two types of project item: Experiment Data and completed Analysis Results. Selecting an item shows it in the workspace; selection does not change whether an experiment is Active.

Only Experiment Data has an **Active** toggle. Active experiments participate in operations that use a group of datasets, such as multiple-experiment fitting, processing propagation, and coordinated export. The toggle becomes available after the experiment has been processed. **Enable All**, **Disable All**, and **Invert Active** change several experiments. The **Sort** commands order the list by name, date, temperature, type, ionic strength, or protonation enthalpy. Multiple-experiment fitting is described in Multiple-experiment fitting, and the processing prerequisite is covered in Processing.

The rest of the window follows the selected item and task:

- The workflow navigator switches between the available stages of an experiment.
- The workspace contains the graphs, tables, and controls for the current task.
- Workflow controls provide options for the view or analysis currently open.
- Item actions apply to the selected Experiment Data or Analysis Result.
- The application menu contains project commands, tools, preferences, and help.

Experiment Data



FT-ITC Analysis - JORS Example Project [M]

Data / Results — 5 items — 2/3 included

PRLR_W392A_run1 — Active — 08.09.2023 14.06 — 37 °C | 2.02 mM | 203 µM — OneSetOfSites | RMSD = 0.645 | Kd = 4.10 ± 0.14 µM | ...
PRLR_W392A_run2 — Active — 08.09.2023 15.49 — 37 °C | 2.02 mM | 203 µM — OneSetOfSites | RMSD = 0.681 | Kd = 4.03 ± 0.17 µM | ...
PRLR_W392A_run3 — Inactive — 08.09.2023 17.11 — 37 °C | 2.02 mM | 203 µM — OneSetOfSites | RMSD = 0.305 | Kd = 3.92 ± 0.06 µM | ...
OneSetOfSites — Valid — Fit of 3 experiments — OneSetOfSites; RMSD 1.63 — All variables unconstrained
Global.OneSetOfSites — Valid — Fit of 3 experiments — Global.OneSetOfSites; RMSD 1.59 — Affinity: Temperature dependent

PRLR_W392A_run1

Filename	230908_PRLRlong_W392A_run1.itc
Format	MicroCal ITC Data File
Date	fredag den 8. september 2023 14.06.01
Duration	49 minutes, 57 seconds | 2998 points | 1.0s sampling
Instrument	MicroCal PEAQ-ITC
Cell Volume	207.1 µl
Feedback Mode	Low
Stirring Speed	750 rpm
Temperature	
Target	37 °C
Measured	37.0292 - 37.0461 °C | Mean = 37.04 °C

Selecting Experiment Data opens its experiment workflow in the shared workspace.

Experiment Data represents an imported or application-created dataset. Its list entry provides identifying information and a summary of its processing or fitted state.

Experiment workflow

Four task views are available for Experiment Data:

- **Overview** presents the imported data and available experiment information.
- **Process Data** turns a raw thermogram into integrated heats through baseline modeling and injection-region control.
- **Analyze Data** configures and runs single- or multiple-experiment fitting.
- **Final Figure** prepares the thermogram, integrated heats, fit, residuals, and annotations for presentation.

Details and attributes

The **Details...** view contains editable concentrations, comments, and experiment attributes. The macOS sheet has separate **Details** and **Attributes** tabs: Details contains the experiment name, date/time, conditions, concentrations, and comments; Attributes contains optional metadata. Attributes describe conditions and analysis inputs such as buffer, salt, ionic strength, competitor, or prebound species.

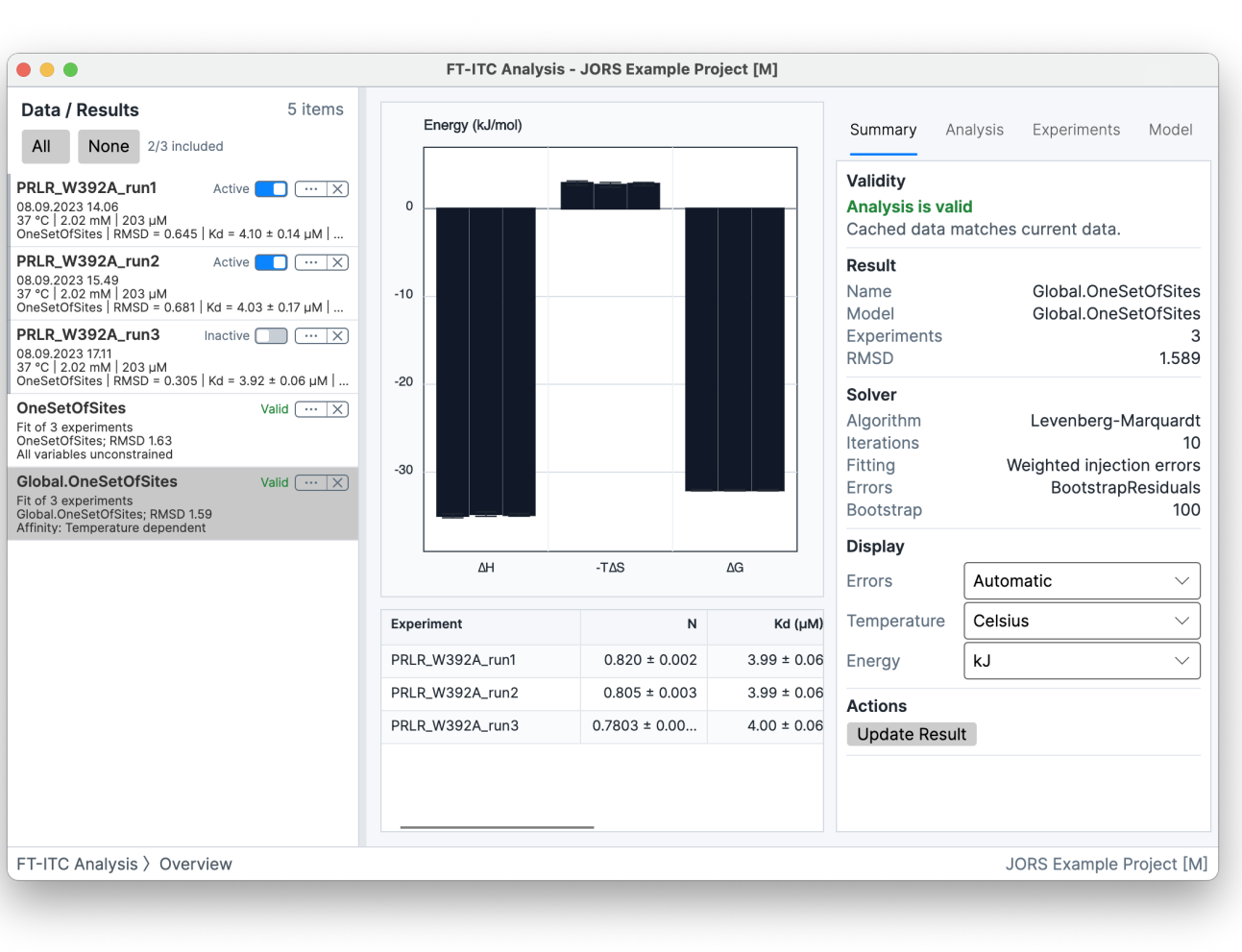
> **PLATFORM NOTE (MACOS):**
> Edit the date and time using separate native controls, including seconds. Use **Add Attribute** on the Attributes tab to add a row. Each tab scrolls independently while **Cancel** and **Apply** remain visible at the bottom. Cancel discards edits; Apply saves them.

Copy Attributes to All copies all attributes from the selected experiment to every other experiment. **Attribute Operations...** copies either one attribute or all attributes to **All other experiments**, **Active experiments**, a **Specific experiment**, or **Experiment names containing...**. The name option targets every other experiment whose name contains the entered text, without regard to capitalization. **Clear Attributes** removes all attributes from the selected experiment after confirmation.

Experiment Data actions

- **Details...** opens the editable experiment information and attributes.
- **Duplicate Data** creates another project item from the selected Experiment Data.
- **Export Selected Data...** exports the selected dataset.
- **Clear Solution** removes the fitted solution currently attached to the Experiment Data.
- **Remove Data** removes the item from the open project without deleting its source file.

Analysis Result



FT-ITC Analysis - JORS Example Project [M]

Data / Results — 5 items — 2/3 included

Validity
Analysis is valid
Cached data matches current data.

Result	
Name	Global.OneSetOfSites
Model	Global.OneSetOfSites
Experiments	3
RMSD	1.589

Solver	
Algorithm	Levenberg-Marquardt
Iterations	10
Fitting	Weighted injection errors
Errors	BootstrapResiduals
Bootstrap	100

Display	
Errors	Automatic
Temperature	Celsius
Energy	kJ

Experiment	N	Kd (µM)
PRLR_W392A_run1	0.820 ± 0.002	3.99 ± 0.06
PRLR_W392A_run2	0.805 ± 0.003	3.99 ± 0.06
PRLR_W392A_run3	0.7803 ± 0.00...	4.00 ± 0.06

An Analysis Result opens its dedicated result workspace when selected.

An Analysis Result stores a fit for one or more experiments together with its model, constraints, fitting settings, uncertainty output, and member solutions. Its list entry summarizes the stored fit.

Result workspace

The result workspace provides these views:

- **Summary** compares fitted and derived parameters across the result and provides its member table.
- **Fit** shows the stored fitted curve and residuals for a selected member experiment.
- **Correlation** shows parameter correlations calculated from residual-bootstrap refits when sufficient bootstrap information is available.
- **Temperature**, **Salt**, and **Protonation** appear when the result and its experiment information support those analyses.

The inspector organizes result information under **Summary**, **Analysis**, **Experiments**, and **Model**. See Results and advanced analyses for interpretation and prerequisites.

Analysis Result actions

Changes to the fit-relevant state of Experiment Data stored in an Analysis Result can invalidate the result. The validity indicator shows whether the stored result still matches the current data.

- **Details...** edits the result name and comments.
- **Copy Result Table** copies the current result table; **Analysis Result Exporter...** provides controlled table export.
- **Update Result** reruns the stored model and fitting settings using the current member experiments and replaces the result only after a successful fit. For a residual-bootstrap result, the update dialog keeps the stored bootstrap count by default and offers larger supported iteration presets. Changing that count affects only this rerun; the stored parameter limits and other fitting settings remain unchanged.
- **Set Active Experiments** makes the result's member experiments Active.
- **Load Solutions to Experiments** attaches the stored member solutions to their corresponding Experiment Data.
- **Export Associated Final Figures...** exports figures for the member experiments.
- **Remove Result** removes the Analysis Result from the open project.

Separate tools

The **Tools** menu opens separate windows for tasks outside the main experiment and result workflows. These include **Experiment Designer...**, **Buffer Subtraction...**, and **Experiment Merger...**. Their features are described in Tools.

CHAPTER 05

Processing

Baseline models, integration regions, injection uncertainty, and processing propagation and locking.

Last verified: 2026-09-23

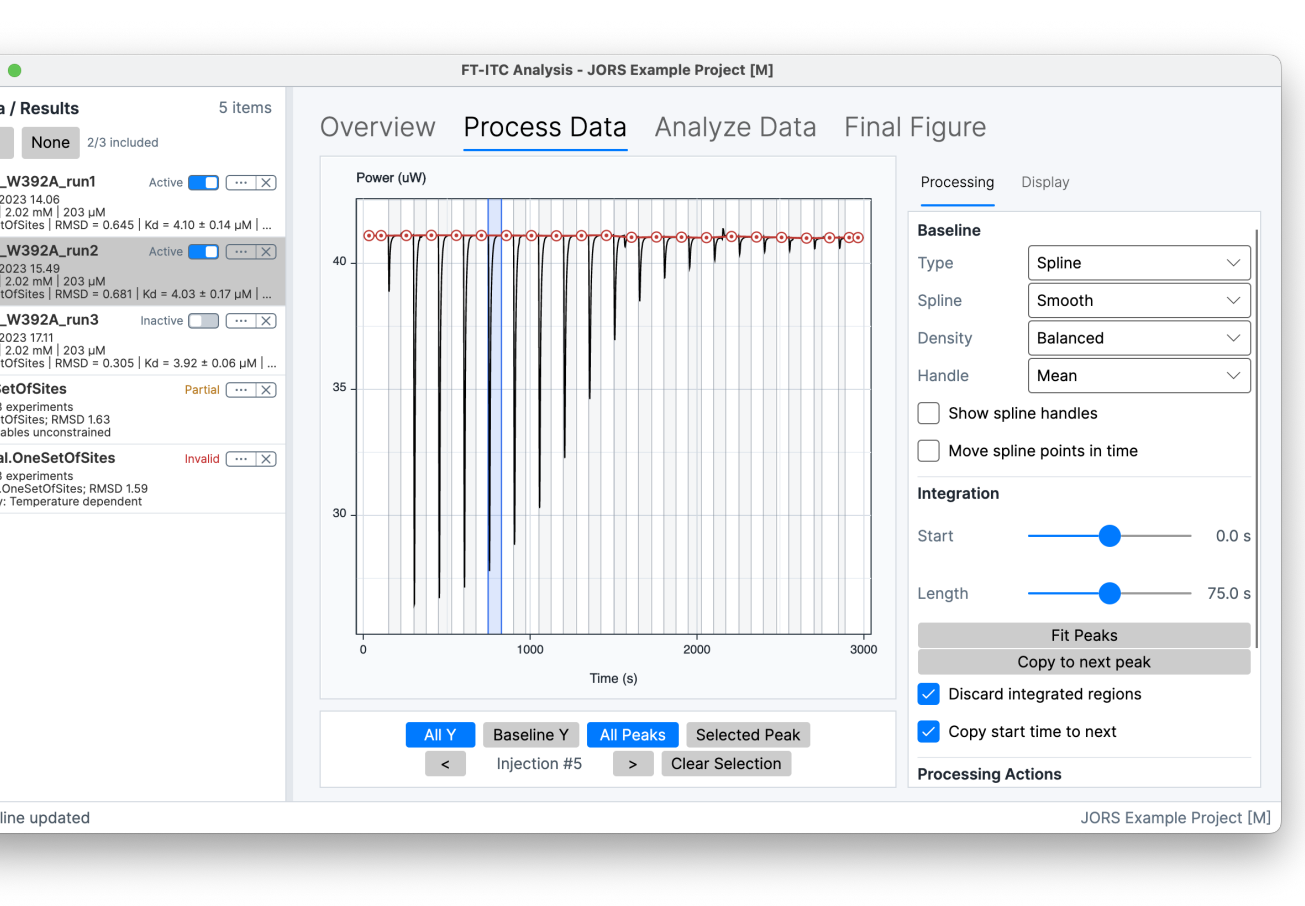
The **Process Data** workspace estimates and subtracts a baseline from the differential-power trace, integrates the corrected response for each injection, and estimates an uncertainty for the resulting heat. Polynomial and Segmented baselines provide repeatable, model-based treatment with less manual shaping when their assumptions suit the trace. Spline-point editing provides precise local control for difficult cases, while graphical boundary editing, **Fit Peaks**, and copying between injections accelerate integration-region adjustment.

These controls make processing efficient without deciding the scientific interpretation for you. Confirm that the chosen baseline represents the signal between injections and that each integration region captures the observed response.

Raw `.itc`, `.nitc`, `.ta`, and `.apj` imports use this workflow, as do Origin `.opj` imports that contain a usable time/power trace. Integrated-heat imports and Origin projects without a usable trace skip **Process Data**.

> **CALCULATION:**
>
> $$q_{i,\mathrm{raw}} = \int [P(t) - b(t)]\,\mathrm{d}t$$
>
> $$\Delta H_i = \frac{q_i}{n_i}$$
>
> $$n_i = c_{\mathrm{syr}} V_i$$
>
> The integral is evaluated between the injection's integration boundaries. Each recorded power value represents the trailing-period average ending at its timestamp, so fractional intervals at either selected boundary are included using the appropriate right-endpoint sample. $P(t)$ is differential power, $b(t)$ is the estimated baseline, and $q_{i,\mathrm{raw}}$ is the raw integrated heat. The heat q_i equals that raw value unless optional buffer subtraction changes it. c_{syr} is the syringe concentration, V_i is the injection volume, n_i is the injected amount, and ΔH_i is the molar heat.

Processing workspace



FT-ITC Analysis - JORS Example Project [M]

Data / Results — 5 items — 2/3 included

PRLR_W392A_run1 — Active — OneSetOfSites | RMSD = 0.645 | Kd = 4.10 ± 0.14 µM | ...
PRLR_W392A_run2 — Active — OneSetOfSites | RMSD = 0.681 | Kd = 4.03 ± 0.17 µM | ...
PRLR_W392A_run3 — Inactive — OneSetOfSites | RMSD = 0.305 | Kd = 3.92 ± 0.06 µM | ...
OneSetOfSites — Partial — Fit of 3 experiments — OneSetOfSites; RMSD 1.63 — All variables unconstrained
Global.OneSetOfSites — Invalid — Fit of 3 experiments — Global.OneSetOfSites; RMSD 1.59 — Affinity: Temperature dependent

Baseline
Type: Spline
Spline: Smooth
Density: Balanced
Handle: Mean
☐ Show spline handles
☐ Move spline points in time

Integration
Start: 0.0 s
Length: 75.0 s
☑ Discard integrated regions
☑ Copy start time to next

Injection #5

Baseline updated

Process Data combines the thermogram with baseline, integration, display, and navigation controls.

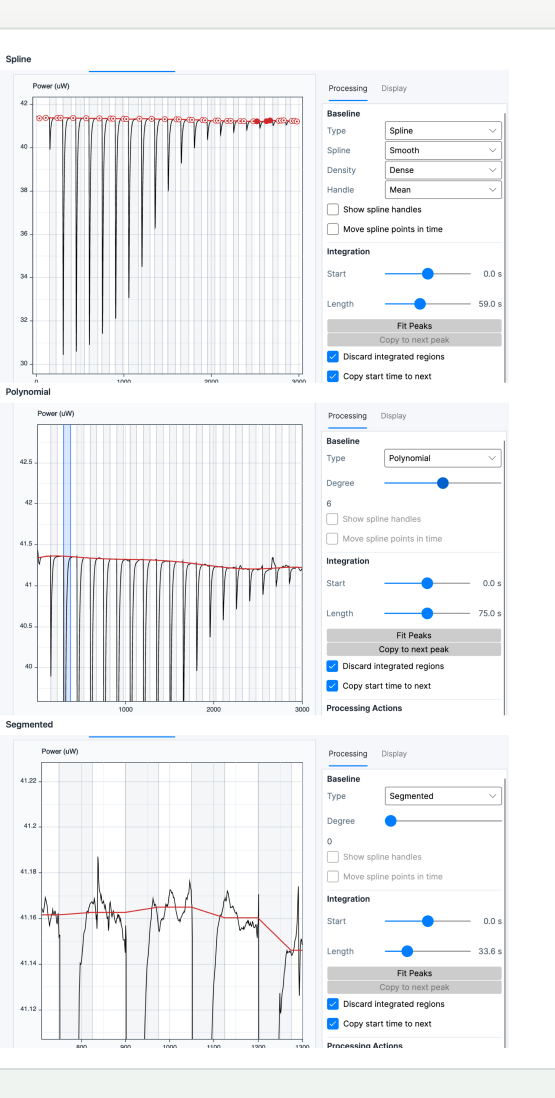
The **Processing** controls configure the baseline and integration regions. The **Display** controls show or hide the baseline, integration regions, corrected data, and cursor information.

With **All injections** selected, **Start** and **Length** apply to every injection. A selected graph region targets one injection, and double-clicking it focuses the graph on that peak. The previous and next controls move through the injections; **Clear Selection** returns to all injections.

The view controls separate horizontal and vertical scaling:

- **All Y** shows the complete power range, while **Baseline Y** emphasizes the baseline region.
- **All Peaks** shows the full injection series, while **Selected Peak** focuses on the current injection.
- Dragging an empty part of the graph zooms into the selected region.

Baseline models and editing



Spline, Polynomial, and Segmented provide different baseline representations for thermogram processing.

Spline

Spline places points in usable baseline regions and interpolates between them. It provides the most direct graphical control.

- **Linear** connects the spline points with straight sections. **Smooth** produces a continuously varying baseline and supports editable slope handles.
- **Sparse**, **Balanced**, and **Dense** change the number of automatically generated points.
- **Mean**, **Median**, and **Min volatility** determine the representative power used for an automatically generated point.
- **Show spline handles** displays the slope handles for a Smooth spline. **Move spline points in time** allows a point to be dragged horizontally as well as vertically.

Dragging a point corrects its position. A secondary-click on the graph adds a point **At Data** or **At Baseline**. A secondary-click on an existing point exposes **Lock** or **Unlock**, **Mark Linear** or **Unmark Linear**, and **Remove**. Marking neighboring points as linear makes the interval between them straight. Locked points are retained when automatic spline points are regenerated.

Balanced provides the middle spline-point density. Greater density increases local flexibility but can also follow noise or absorb part of an injection response.

Polynomial

Polynomial fits one polynomial across the complete thermogram and is suited to smooth global drift. **Degree** controls flexibility.

Polynomial behavior is least constrained at the beginning and end of the run, where a high degree can produce strong edge behavior. Additional degree increases flexibility whether it represents baseline drift or follows more of the trace.

Segmented

Segmented fits local constant, linear, or quadratic baseline behavior between integration regions and blends the local estimates across the run. It is suited to locally changing drift for which one global polynomial is too rigid. **Degree** selects the local behavior.

Local flexibility can follow genuine drift, but it also makes the result more dependent on the integration boundaries and on the baseline immediately around each peak.

Exclude integration regions from the baseline

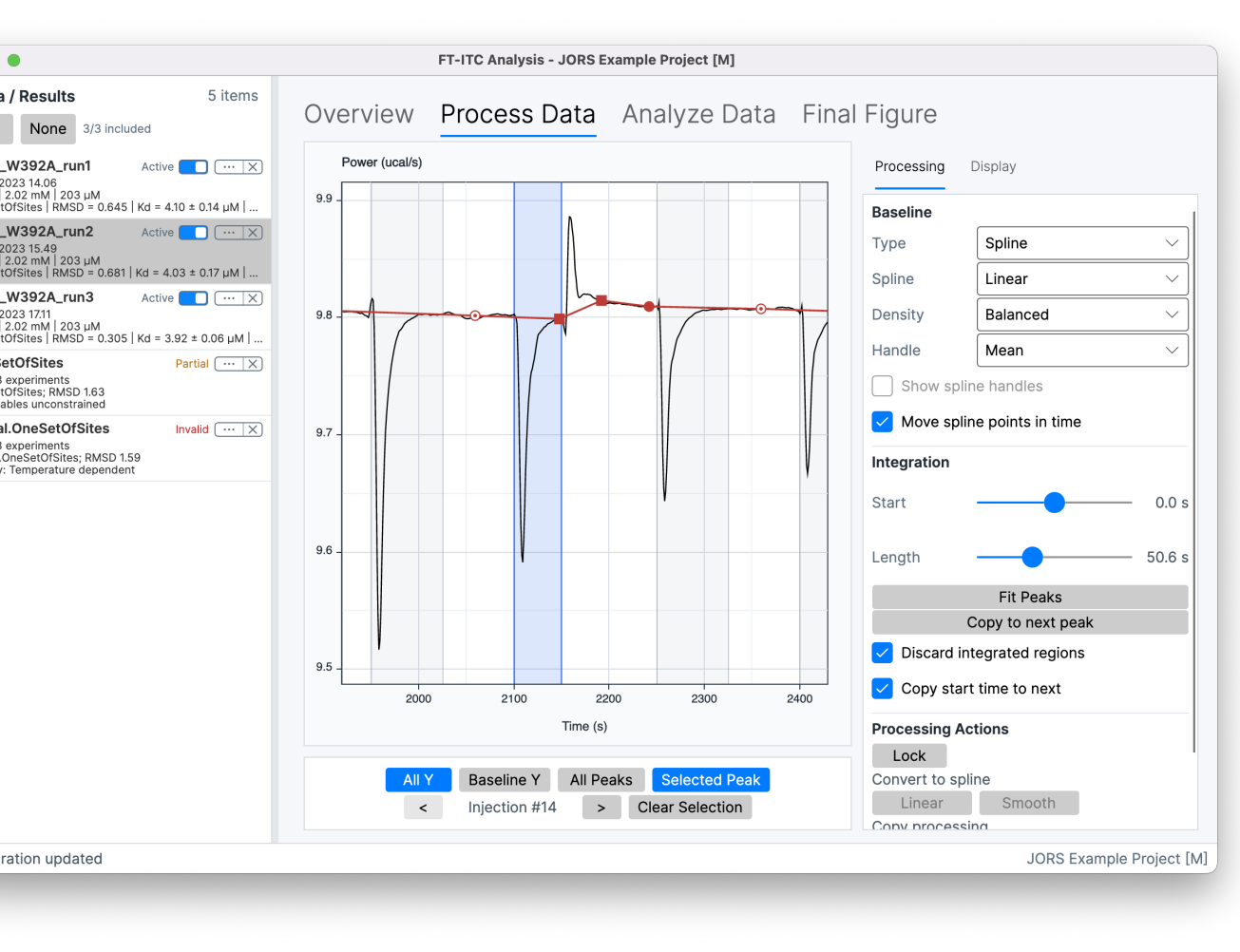
When **Discard integrated regions** is enabled, data inside the current integration regions are excluded when the baseline is recalculated. Moving a boundary can therefore change both the integrated area and the estimated baseline.

> **CAUTION:**
> Boundaries should follow the observed response and a consistent processing rule, not the heat expected from a model.

Convert to a spline

A Polynomial or Segmented baseline can be converted to a **Smooth** or **Linear** Spline when the automatic baseline is a useful starting point for graphical editing. Conversion changes the baseline representation and creates editable spline points.

Integration regions



FT-ITC Analysis - JORS Example Project [M]

Data / Results — 5 items — 3/3 included

Baseline
Type: Spline
Spline: Linear
Density: Balanced
Handle: Mean
☐ Show spline handles
☑ Move spline points in time

Integration
Start: 0.0 s
Length: 50.6 s
☑ Discard integrated regions
☑ Copy start time to next

Injection #14

Integration updated

Selecting an injection exposes its individual boundaries and navigation controls.

Each injection region has a start and an end boundary. **Start** sets the offset of the start boundary relative to the injection. The value displayed as **Length** positions the end boundary that many seconds after the injection begins; it is not a separate processing mode.

Either boundary can be dragged in the graph or adjusted with the controls. The application constrains the start and end to a valid interval within the injection scope and preserves a minimum separation between them.

Estimate end points with Fit Peaks

Fit Peaks estimates the end point of each injection from the decay of the baseline-corrected response. It changes the end boundaries and then integrates the resulting regions; it does not fit the integrated heats or select a persistent integration mode.

When peak fitting converges, the estimated boundaries replace the previous end points. If fitting fails or does not converge, the previous regions remain unchanged. Peak kinetics can vary across the titration, and the first injection can behave differently from the remaining series.

Copy a region to the next injection

Selecting an injection and choosing **Copy to next peak**, or pressing **Space**, copies its end boundary and advances to the next injection. **Copy start time to next** includes the start boundary in that operation.

Injection uncertainty

Each injection error bar represents an estimated ±1 standard deviation for that injection's molar heat. The estimate describes how local noise in the baseline-corrected thermogram propagates through the selected integration region. It is calculated independently for every injection, so the bars can vary across a titration.

The calculation uses baseline-corrected samples around the injection. It combines an estimate of local power noise with the temporal correlation between neighboring samples, the integration-region length, and uncertainty in the baseline level. A longer or noisier region will therefore often have a larger estimated uncertainty. When Buffer Subtraction is applied, the independent target and reference heat uncertainties are combined.

The correlation estimate uses only consecutive baseline sample pairs separated by at most twice the sampling interval. Both its product sum and its normalization use those same pairs, excluding a pair across a longer integration gap. Propagation assumes an AR(1) noise model, in which correlation decays geometrically with sample separation; this is a chosen approximation to the local noise.

Previously processed projects can retain heat SDs calculated with a normalization that included an unpaired sample across the gap. Reintegrate affected thermograms to update those SDs, then rerun affected weighted fits and parameter-uncertainty calculations before updating published tables or figures. These processing SDs supply fit weights; the weighted objective and displayed unweighted RMSD remain separate quantities.

> **CAUTION:**
> At least two surrounding baseline samples and two samples inside the integration region are required. When the estimate cannot be calculated, the application stores zero. Integrated-heat imports also lack a thermogram from which to estimate this uncertainty. A zero or absent error bar can therefore mean that no processing-derived estimate is available; it does not establish that the injection has no uncertainty.

The bars do not include uncertainty in cell or syringe concentration, fitted parameters, or model-derived confidence bands. They also do not quantify baseline-model choice, integration-boundary choice, calibration error, other instrumental effects, or other systematic uncertainty. Consequently, they do not validate the selected processing or constitute confidence intervals for the true heat.

Weight by injection error can use these estimates in the fitting objective. Concentration uncertainty is handled separately in supported resampling calculations. Both behaviors are described under Fitting calculation.

Propagate and lock processing

Active under **Copy processing** copies the selected experiment's processing to the other Active experiments. This can replace existing processing when a destination is unlocked. **New** targets experiments that do not yet have processing.

Processing can be propagated while the source is unlocked. Locked destinations are not overwritten. After propagation, each processor that should be protected can be selected and set to **Lock**. Locking disables baseline, spline-point, integration-region, and peak-fitting edits; **Unlock** makes changes available again.

CHAPTER 06

Single-experiment fitting

Fit one experiment, configure model and uncertainty options, control injection inclusion, and interpret fit diagnostics.

Last verified: 2026-09-23

Analyze Data in **Single experiment** mode fits the integrated heats of the selected Experiment Data. The fit uses the included injections together with the experiment concentrations, injection volumes, cell volume, temperature, and any model-specific information.

The inspector has four tabs:

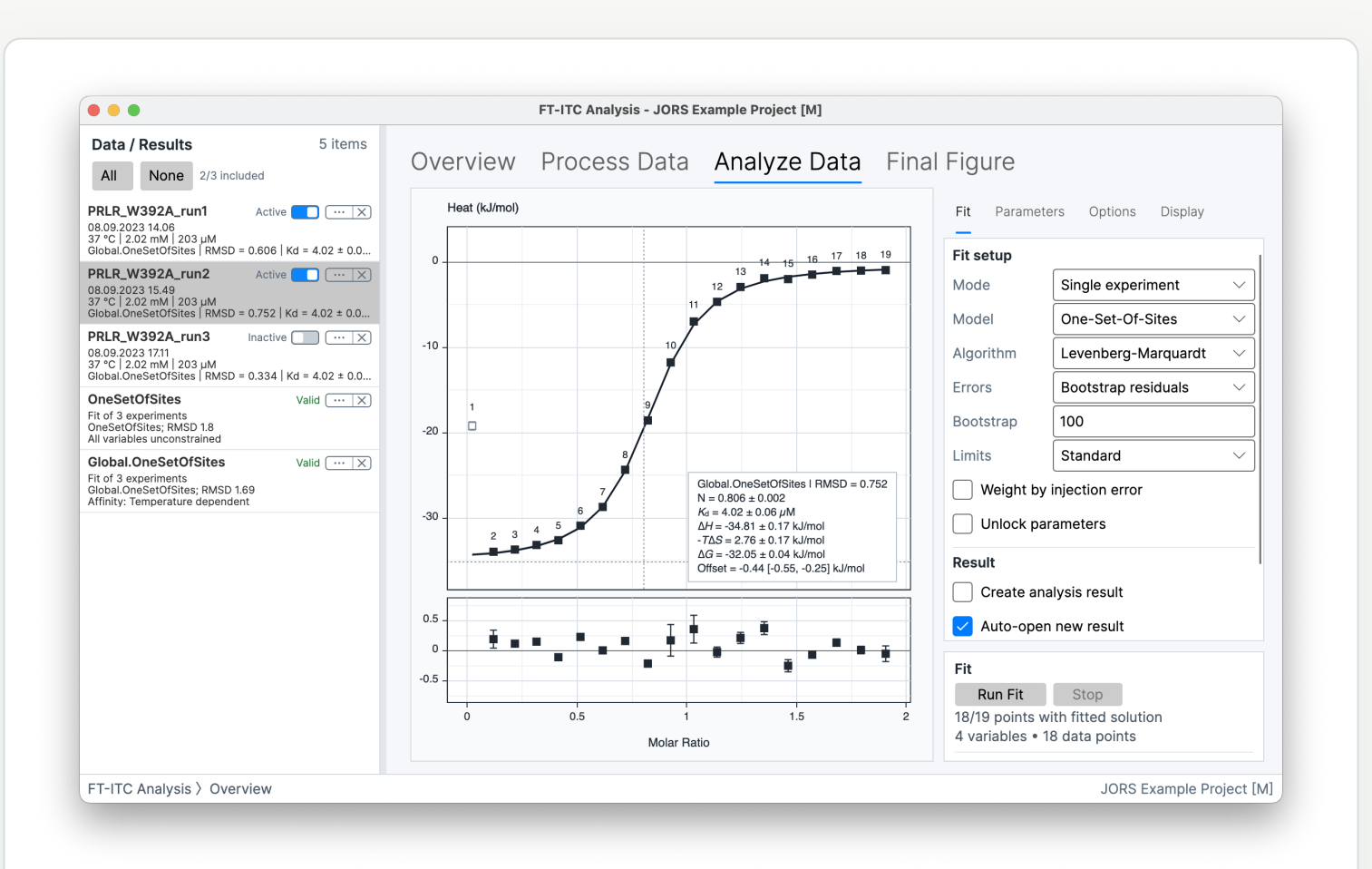
- **Fit** selects the model, optimizer, error-estimation method, limits, weighting, and result output.
- **Parameters** shows the model parameters and their starting or fixed values.
- **Options** contains settings specific to the selected model.
- **Display** controls the fitted curve and diagnostic information shown in the graph.

Profile-likelihood uncertainty

Profile likelihood estimates confidence intervals at a fixed 95% confidence level. After a successful primary fit, the method varies one fitted parameter at a time, holding it at each trial value while refitting the other free parameters. Both unweighted and weighted fits use an F-calibrated threshold based on how much the applicable residual sum of squares worsens relative to its best value. For weighted fits, the processing-derived peak-area standard deviations (SDs) remain unchanged and supply relative injection weights; the overall residual scale used to calibrate the interval is estimated separately and is never written back to an injection. The reported parameter value remains the primary best fit.

Graphs of integrated heats can display an envelope from available leave-one-out refits, although there are typically few such refits. Profile likelihood does not create a refit ensemble or a confidence band.

A complete interval is reported only when the threshold is crossed on both sides of the best-fit value. If the search reaches a parameter limit first, that side is recorded as censored and has no confidence endpoint. For a complete interval, the displayed `value ± SD` uses a symmetric display equivalent calculated from the asymmetric interval; it is not a sample standard deviation. Save the result in an `.ftxtc` project to preserve the profile diagnostics.



Fit setup
Mode: Single experiment
Model: One-Set-Of-Sites
Algorithm: Levenberg-Marquardt
Errors: Bootstrap residuals
Bootstrap: 100
Limits: Standard
☐ Weight by injection error
☐ Unlock parameters
Result
☐ Create analysis result
☑ Auto-open new result
Fit
Run Fit Stop
18/19 points with fitted solution
4 variables • 18 data points

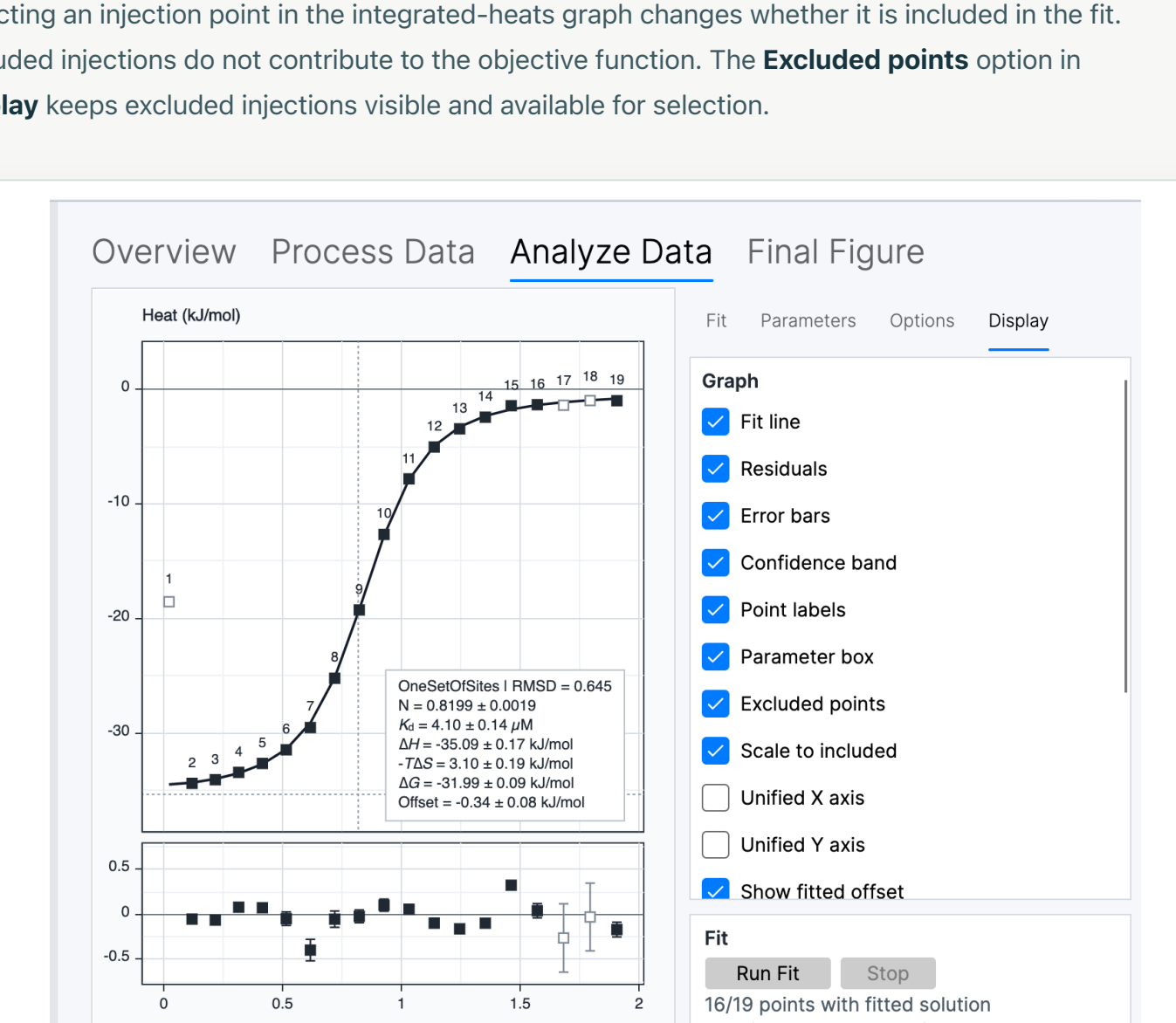
Global.OneSetOfSites | RMSD = 0.752
N = 0.806 ± 0.002
K_d = 4.02 ± 0.06 μM
ΔH = -34.81 ± 0.17 kJ/mol
-TΔS = 2.76 ± 0.17 kJ/mol
ΔG = -32.05 ± 0.04 kJ/mol
Offset = -0.44 [-0.55, -0.25] kJ/mol

Analyze Data combines the fitted-heats graph and residuals with the controls for configuring and running the fit.

Multiple-experiment fitting uses additional experiment selection and parameter constraints; see Multiple-experiment fitting.

Injection inclusion

Selecting an injection point in the integrated-heats graph changes whether it is included in the fit. Excluded injections do not contribute to the objective function. The **Excluded points** option in **Display** keeps excluded injections visible and available for selection.



Graph
☑ Fit line
☑ Residuals
☑ Error bars
☑ Confidence band
☑ Point labels
☑ Parameter box
☑ Excluded points
☑ Scale to included
☐ Unified X axis
☐ Unified Y axis
☑ Show fitted offset
Fit
Run Fit Stop
16/19 points with fitted solution
4 variables • 16 data points

OneSetOfSites | RMSD = 0.645
N = 0.8199 ± 0.0019
K_d = 4.10 ± 0.14 μM
ΔH = -35.09 ± 0.17 kJ/mol
-TΔS = 3.10 ± 0.19 kJ/mol
ΔG = -31.99 ± 0.09 kJ/mol
Offset = -0.34 ± 0.08 kJ/mol

Excluded injections remain visible when Excluded points is enabled and can be selected for inclusion again.

Changing injection inclusion does not rerun the fit. The fitted curve and parameters continue to represent the previous fit until **Run Fit** is used again. An inclusion change can also invalidate an Analysis Result that contains the experiment.

Injection bookkeeping: MicroCal, Ideal continuous mixing, and Discrete displacement

Injection bookkeeping determines the concentrations used by the fit and accounts for reaction heat carried out of the active cell by displaced solution. It does not change measured peak areas or baseline integration.

- **MicroCal** (default) uses the untruncated displaced-volume mass balance for concentrations and the existing endpoint displacement correction.
- **Ideal continuous mixing** models concentrations with ideal exponential mixing and accounts for displaced heat along the continuous mixing trajectory. It uses the exponential concentration law previously exposed as the Exponential preference, but its heat calculation is different. This is the convention previously labelled Dumas in the user interface.
- **Discrete displacement** uses discrete replacement bookkeeping: first displace a fraction of the previous cell mixture, then add and equilibrate the injection. This follows the discrete-injection formalism described by Freire, Schön and Velazquez-Campoy (2009), also implemented by pytc. FT-ITC retains its own equilibrium solvers.

For **MicroCal**, let u be cumulative injected volume divided by active cell volume, M_0 be the initial cell concentration, and C_s the syringe concentration. Starting with no ligand in the cell:

$$M = M_0(1 - u/2)/(1 + u/2)$$
$$X = C_s u/(1 + u/2)$$

Every delivered injection contributes to cumulative volume, including injections excluded from fitting. The ligand equation is the approximate expression used by FT-ITC for MicroCal processing. The displaced-volume assumptions remain approximate. Existing projects retain saved concentrations and fits until concentration reprocessing.

For one injection, let v be injection volume, V active cell volume, and Q the equilibrium binding heat content of the cell in joules. Ordinary binding models use

$$\text{MicroCal: } q = Q_{end} - Q_{start} + (v/V)(Q_{start} + Q_{end})/2$$
$$\text{Ideal continuous mixing: } q = Q_{end} - Q_{start} + (v/V)(Q_{start} + 4Q_{mid} + Q_{end})/6$$
$$\text{Discrete displacement: } q = Q_{end} - (1 - v/V)Q_{start}$$

For **Discrete displacement**, concentrations advance as $M_{end} = (1 - v/V)M_{start}$ and $X_{end} = (1 - v/V)X_{start} + (v/V)C_s$. Across a fixed-syringe segment, the retained fraction is the product of each injection's $(1 - v/V)$, not an exponential of cumulative volume. Every shot must be smaller than the cell volume; cumulative injected volume can exceed it. Zero-volume steps leave the state unchanged. No numerical integration or substeps are used. Displaced heat is treated consistently with this discrete replacement: the correction is $(v/V)Q_{start}$, using the pre-injection heat content, rather than MicroCal's average of the start and end heat contents.

The midpoint is evaluated halfway through the injection on the exponential concentration trajectory. Dissociation additionally accounts for dimer heat entering from the syringe; its legacy calculation is retained under MicroCal. Offsets are applied separately, as before.

Preferences > Processing > Injection bookkeeping selects the default for new data. **Experiment Details > Injection bookkeeping** explicitly switches an ordinary experiment and invalidates its fits without reintegrating measured heats. Changing the preference or editing a name/comment does not switch existing data. Older projects retain their saved concentrations and historical heat behavior; if their method is unknown, the selector displays **Saved processing — unchanged**. Choose a method explicitly before recalculating unknown saved concentrations. Rebuild tandem experiments through the tandem tool to change their method.

The ideal continuous mixing convention is the ideal-mixing limit inspired by Dumas (2022), with an FT-ITC finite-injection numerical integration. It does not implement that paper's imperfect-mixing/adjustable-volume model or a kinetic single-injection analysis. No option is assumed to be empirically superior. Simpson integration uses one fixed panel per injection, so unusually large injections or very sharp transitions can need additional scrutiny; there is no adaptive refinement. It requires three equilibrium states, whereas MicroCal and Discrete displacement require two.

Discrete displacement is useful when comparing analyses that use the same injection convention. It is not an imperfect-mixing correction. FT-ITC retains its own equilibrium solvers and offset convention, so results may differ from other programs even when the same displacement bookkeeping is selected. Fractional-stoichiometry two-site binding and monomer–dimer dissociation use FT-ITC-specific extensions of the convention. Neither fitted parameters nor background-heat conventions are automatically converted between programs.

Models

One-Set-Of-Sites

One-Set-Of-Sites represents one class of equivalent, independent binding sites. It fits stoichiometry, dissociation constant, binding enthalpy, and an injection-heat offset. The solution also reports thermodynamic quantities derived from the fitted affinity and enthalpy.

The **Options** tab provides **Use Syringe Correction** and **Stoichiometry**. Without syringe correction, the fitted N-value represents the apparent site stoichiometry. With syringe correction enabled, **Stoichiometry** fixes the number of cell-side sites and the fitted N parameter becomes the active syringe-concentration factor `alpha`.

The model cannot by itself distinguish a concentration error from other effects that change an apparent stoichiometry.

Two-Sets-Of-Sites

Two-Sets-Of-Sites represents two independent classes of sites. It fits separate stoichiometries, dissociation constants, and enthalpies for the two classes, together with a shared injection-heat offset.

Shared N-Values makes the two site classes use the same fitted stoichiometry. **Use Syringe Correction** instead fixes the first and second **Stoichiometry** values and fits one active syringe-concentration factor, `alpha`.

The two site labels are interchangeable: exchanging all parameters assigned to site 1 and site 2 describes the same physical model. A lower RMSD or optimizer objective alone does not establish that two distinguishable binding processes are supported by the experiment.

Sequential Binding Sites

Sequential Binding Sites represents two, three, or four ordered binding steps on a macromolecule in the cell. **Sequential binding steps** in the **Options** tab selects the number of steps. The model fits one macroscopic stepwise association constant and one molar step enthalpy for each transition, together with the ordinary molar injection-heat offset. It does not fit an N-value or syringe activity.

For step count n, let $\beta_0 = 1$, $\beta_i = \prod_{j=1\ldots i} K_j$, and let x be the free ligand concentration. The state weights and fractions are

CALCULATION:

$$w_i = \beta_i x^i$$
$$F_i = \frac{w_i}{\sum_{j=0}^{n} w_j}$$
$$\bar{\nu} = \sum_{i=0}^{n} iF_i$$
$$X_t = x + M_t \bar{\nu}$$

Here M_t and X_t are total macromolecule and ligand concentrations in the cell. The model solves the ligand balance as part of the fit and calculates the cell heat content from the population of every sequential state:

CALCULATION:

$$Q = VM_t \sum_{i=1}^{n} F_i \left(\sum_{j=1}^{i} \Delta H_j \right)$$

The reported K_i values are phenomenological, macroscopic step constants for the ordered transitions $M \rightarrow MX \rightarrow MX_2$ and so on. They are not microscopic intrinsic site constants. Step numbers therefore have a physical order, so fitted steps are never sorted or treated as interchangeable.

The macromolecule must be in the cell and ligand in the syringe. Reverse titrations with macromolecule in the syringe are outside this model. Multi-step fits can be weakly identifiable, especially when the concentration window does not populate every transition. Inspect parameter bounds, residuals, bootstrap uncertainty, and parameter correlations; an improved RMSD does not by itself establish the selected number of sequential steps.

Competitive Binding

Competitive Binding represents titration of a target ligand into a macromolecule that is initially in equilibrium with a prebound ligand in the cell. It fits the target ligand's stoichiometry, dissociation constant, binding enthalpy, and injection-heat offset.

The **Options** tab requires the pre-equilibrated competitor's **Total competitor** concentration, **Ligand Affinity**, and **Ligand Enthalpy**. **Total competitor** is the total analytical competitor concentration in the cell after pre-equilibration: free competitor plus competitor bound to the macromolecule. Do not enter only the initially bound complex. **From attributes** makes **Total competitor** use the corresponding value stored in the Experiment Data attributes instead of the value entered in the model options. **Ligand Affinity** and **Ligand Enthalpy** each have a separate **From attributes** option. Add a **Competitor properties** experiment attribute and select a one-set-of-sites Analysis Result to supply its global summary Kd and ΔH, including their uncertainty. Each experiment in a global fit can select its own result. The row shows the source status; hover over it or the selector for the full result name and a short Kd and ΔH summary with units and SD. A missing source can still use captured values when available. Opening the editor only previews values; the saved capture refreshes when a fit starts. The model also provides **Use Syringe Correction** and **Stoichiometry** with the same concentration-factor interpretation as One-Set-Of-Sites.

Model options remain available to edit even when no experiment is ready for fitting. When **From attributes** is selected, starting a fit copies each experiment's attribute value into its model option; the entered value is used only if **From attributes** is turned off. Different experiments in a global fit can therefore have different effective option values. Every included experiment must have the attributes required by the enabled selections. If any are missing, one error dialog lists the affected experiments and attributes so they can be added or the selections turned off. A fit that cannot start leaves any previously attached solution in place. **Update Result** reads the attributes again; existing results continue to use the values saved with their fits.

The **Total competitor** experiment attribute has separate concentration and SD fields in both desktop apps. Buffer, salt, and ionic strength concentration attributes have a value field without an SD field.

If the source result has temperature dependence enabled, its summary is evaluated at the experiment temperature. Otherwise its usual summary temperature is used, with a notice when the experiment temperature differs. The attribute keeps the values used by the fit: if the source result is removed, existing results remain valid and a new fit can use the captured values. A source removed before any values were captured must be restored or replaced before fitting. Updating the source result marks dependent analysis results stale; fitting again copies the updated summary. Residual bootstrap samples the copied affinity and enthalpy uncertainties separately, as it does for manually entered model options.

The **Ligand Affinity** and **Ligand Enthalpy** labels describe the pre-equilibrated competitor's properties. The fitted target affinity and enthalpy depend on those supplied properties. The competitor properties are model inputs, not quantities independently determined by this fit.

The reported apparent target K_d includes a competition factor calculated from the initial free competitor concentration. This calculation accounts for competitor already bound to cell sites rather than substituting the total competitor concentration. The apparent target K_d can therefore differ from its intrinsic value.

Dissociation

Dissociation represents dilution-driven monomer-dimer self-association. The syringe contains the macromolecule and the cell initially contains buffer. Dilution and mixing change the dimer population, and the model fits the association equilibrium through its reported dissociation constant, the association enthalpy per mole of dimer formed, and an injection-heat offset.

CALCULATION:

$$2M \rightleftharpoons D$$
$$K_a = \frac{[D]}{[M]^2}$$

Here, $[M]$ and $[D]$ are the monomer and dimer concentrations, and K_a is the association constant.

This is not a general model for dissociation of an arbitrary preformed complex or for other oligomerization schemes. It has no stoichiometry or syringe-correction options.

Thermodynamic relationships

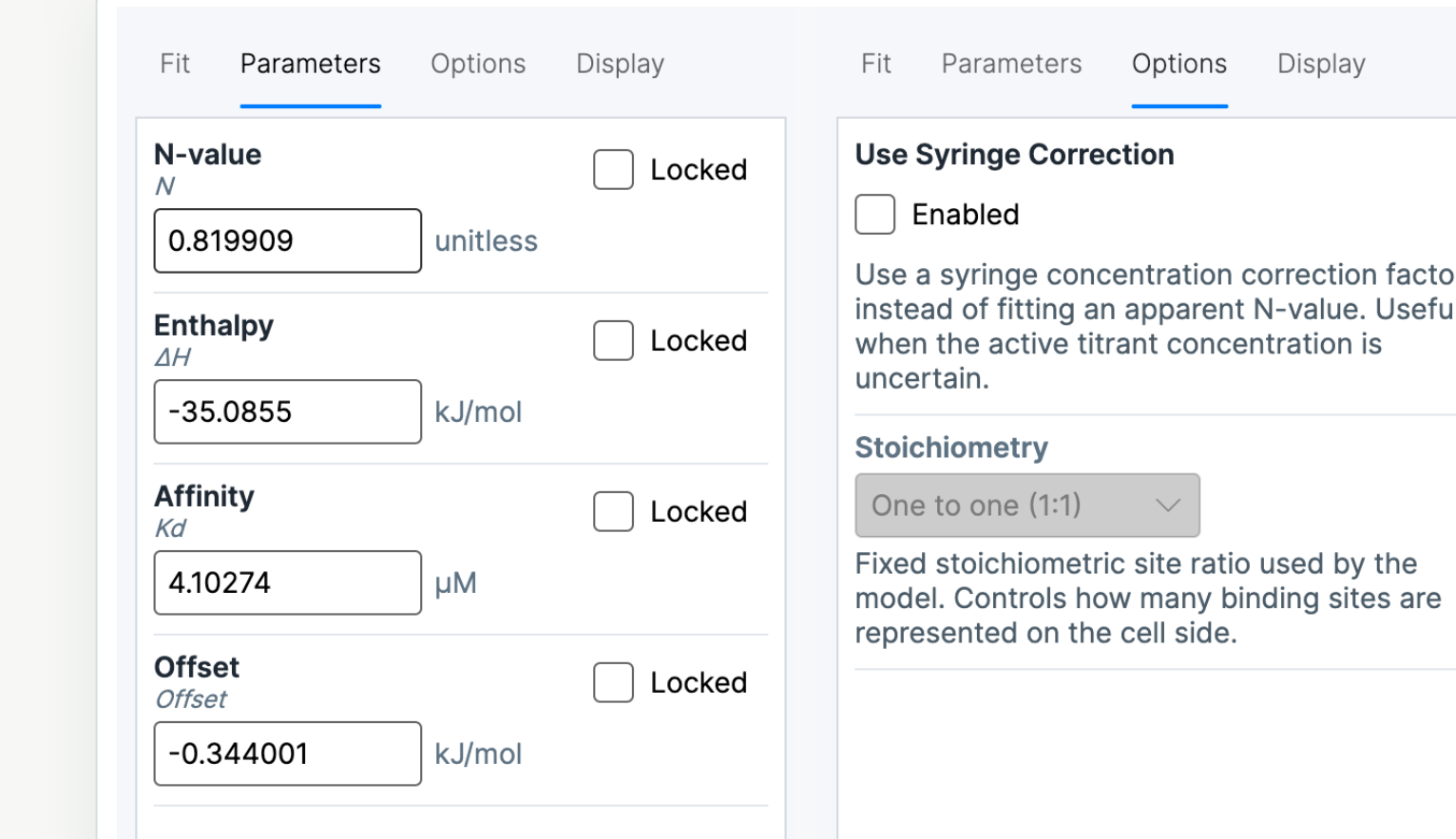
The application derives thermodynamic quantities from the fitted affinity and enthalpy. Temperature T is expressed in kelvin.

CALCULATION:

$$K_d = 1/K_a$$
$$\Delta G = RT\ln(K_d) = -RT\ln(K_a)$$
$$-T\Delta S = \Delta G - \Delta H$$

Here, R is the gas constant, and the final relationship matches the **−TΔS** quantity reported by the application.

Parameters and model options



N-value
N
0.819909 unitless
☐ Locked

Enthalpy
ΔH
-35.0855 kJ/mol
☐ Locked

Affinity
Kd
4.10274 μM
☐ Locked

Offset
Offset
-0.344001 kJ/mol
☐ Locked

Use Syringe Correction
☐ Enabled
Use a syringe concentration correction factor instead of fitting an apparent N-value. Useful when the active titrant concentration is uncertain.

Stoichiometry
One to one (1:1)
Fixed stoichiometric site ratio used by the model. Controls how many binding sites are represented on the cell side.

Parameters exposes parameter values and locks; Options contains settings specific to the selected model.

The application generates initial parameter values from the experiment and can reuse an attached fitted solution where applicable. Entering a value replaces the generated starting value for that parameter. Values are displayed in the current application units. Clearing a value returns the parameter to automatic initialization.

Locked holds a parameter at its displayed value during the primary fit. An unlocked parameter is adjusted by the optimizer. Locked values remain part of the model and affect every other fitted parameter even though they are not estimated by that fit.

Analysis choices are retained separately for the available fitting modes and models. **Restore defaults** clears the stored analysis inputs and reloads the fitting controls from the current preferences. It also resets the inspector's **Unlock parameters** setting. The action does not change preferences, including saved limits, result-output options, and display settings.

The **Limits** control selects a common parameter-bound policy:

- **Standard** uses the normal parameter bounds.
- **Expanded** permits a wider parameter range.
- **No limits** removes the configured parameter bounds.

A fitted value at a bound is not an interior estimate. It indicates that the reported value depends on the selected bound as well as on the data and model.

Before a fit starts, the application checks every free starting parameter against the currently selected limits. This includes values entered manually, values reused from an attached solution, and automatic values that become stale after the **Limits** policy changes. Bounds are inclusive, so a value exactly at a bound is allowed. Locked and globally determined parameters are not checked because they are not optimizer coordinates. An out-of-range value is marked in the **Parameters** inspector and **Run Fit** remains available so it can be corrected. The fit is blocked with a list of affected parameters; edit the value, clear it to restore the automatic default, or widen **Limits**. In a global fit, a local parameter that has no exposed global editor is identified by its experiment name in the fit status area; edit that experiment in single-experiment mode if needed.

Fitting calculation

Algorithm

Levenberg-Marquardt uses local derivative information and can be efficient when the starting values describe a suitable region of the fitting surface.

Nelder-Mead is a derivative-free simplex optimizer. It provides an alternative when the derivative-based method has difficulty converging from the chosen starting values. Agreement between optimizers does not by itself establish that the selected model is scientifically adequate.

Weight by injection error

Weight by injection error uses the integration uncertainty estimated during thermogram processing when calculating the fitting objective. Injections with larger estimated uncertainty consequently have less influence than injections with smaller estimated uncertainty.

CALCULATION:

$$r_i = q_{i,\text{obs}} - q_{i,\text{model}}$$
$$\text{RMSD} = \sqrt{\left[\frac{\Sigma\, r_i^2}{N}\right]}$$
$$\text{weighted objective} = \Sigma\,(r_i/\sigma_i)^2$$

Only included injections enter these sums. The value σ_i is the processing-derived uncertainty for injection i; weighting changes the fitting objective but does not remove systematic uncertainty.

The displayed RMSD is always calculated from the unweighted residuals, including after a weighted fit. The result tooltip also reports the optimizer objective separately: the raw residual sum of squares for unweighted fitting or the standardized residual sum of squares for weighted fitting. These are distinct diagnostics, so the RMSD value is not the objective value.

For a multiple-experiment result, the displayed global RMSD is pooled across every included injection in every member experiment. Each member row retains its own local RMSD, so the global value remains comparable when members contain different numbers of included injections.

If an included injection does not have a finite positive peak-area SD, the application uses the mean of the finite positive SD values from the other included injections. If none is available, it uses a small numerical fallback so the calculation remains defined. A substituted value prevents division by zero; it does not turn a missing processing estimate into a measured uncertainty.

The weighting describes the application's processing-derived uncertainty model. It does not account for every systematic source of experimental or processing uncertainty.

Parameter uncertainty

The **Errors** control determines whether the primary best fit is followed by repeated refitting:

- **None** retains the primary fit without resampling-based parameter uncertainty.
- **Bootstrap residuals** samples from the included injections' centered primary-fit residuals and adds each draw to a best-fit prediction. For an unweighted fit, it uses the raw residuals: injection peak-area SDs do not affect the synthetic peak areas or refits. This assumes the residual errors have roughly equal spread across injections. For an error-weighted fit, it divides each residual by its effective peak-area SD before centering and sampling, then multiplies each draw by the target injection's effective SD. Synthetic injections retain their stored peak-area SDs, so weighted refits use the same per-injection weighting inputs and fallback rule.
- **Leave-one-out** performs one deterministic refit for each deletion: one refit per included injection in a single-experiment analysis, or one refit per omitted experiment in a globally fitted multiple-experiment analysis. Concentrations, uncertain model options, and parameter locks are held at their primary-fit values so the resulting spread isolates deletion sensitivity.
- **Profile likelihood** fixes each fitted coordinate in turn and refits nuisance coordinates against the conditional likelihood. It uses the local objective for independent members and the complete objective for shared global coordinates; complete asymmetric crossings are retained as endpoints, while a bound reached before crossing is reported as censoring.

Bootstrap sets the requested number of residual-bootstrap iterations. It is enabled and used only for residual bootstrap; leave-one-out and profile likelihood have deterministic schedules and do not use this count. Only included injections supply residuals, and only retained usable refits enter the parameter distributions. Because residual-bootstrap sampling is with replacement, one residual can occur more than once in a synthetic dataset while another may not occur at all. The fit status distinguishes successful and failed refits.

Correlation diagnostics distinguish attempted refits, optimizer-usable refits, and the listwise-complete refits whose displayed coordinates are all finite. A high failed-refit fraction can make the retained ensemble selective; increasing the requested count improves Monte Carlo resolution but does not correct a systematically failing refit process.

Each replicate uses a fresh independent random stream; seeds are not stored, so rerunning a bootstrap does not reproduce the same random sequence.

When **Update Result** is used on a stored residual-bootstrap Analysis Result, its dialog shows the retained usable-refit count and offers the stored behavior plus larger supported presets up to 10,000 requested iterations. The update performs a fresh complete fit and bootstrap; it does not append samples to the saved distribution. Cancelling the calculation or completing it without any usable bootstrap refits preserves the previous Analysis Result.

The primary best-fit parameter remains the reported value. For a parameter with best-fit value $\hat{\theta}$ and values θ_b from B retained refits, the application summarizes the bootstrap distribution as follows:

CALCULATION:

$$\text{SD} = \sqrt{\frac{\sum_b \left(\theta_b - \hat{\theta}\right)^2}{B}}$$
$$95\%\ \text{CI} = \left[P_{2.5}(\{\theta_b\}), P_{97.5}(\{\theta_b\})\right]$$

SD is the root-mean-square deviation of the retained refits from the primary best fit. The confidence limits are the 2.5th and 97.5th percentiles of the retained refit distribution itself, so they need not be equally spaced around the best fit. Neither calculation replaces the best-fit value with the bootstrap mean or median.

The uncertainty display can show SD, the 95% confidence interval, both, or select between them automatically. This presentation rule is described under Uncertainty and evaluation temperature.

Concentration uncertainty

With **Concentration uncertainty** enabled in the **Fit** tab, the concentration SDs entered in **Details...** are propagated through residual-bootstrap calculations. The control is initialized from the corresponding preference and is active only for residual bootstrap; leave-one-out and profile likelihood keep primary concentrations fixed. Each nonzero fractional SD is the arithmetic standard deviation relative to the entered concentration. Synthetic clones draw a positive, mean-preserving lognormal multiplier: if the fractional SD is c, then $\sigma^2_{log} = \ln(1 + c^2)$, $\mu_{log} = -\sigma^2_{log}/2$, and the multiplier is $\exp(\mu_{log} + \sigma_{log}Z)$ for a standard-normal Z. Thus the multiplier has mean 1 and SD c, so cloned concentrations remain positive while preserving the entered arithmetic mean and SD. Explicit cell or syringe SDs take precedence over the automatic value configured in Preferences. These uncertainties affect the synthetic experiment concentrations used for bootstrap refits, not the concentrations used for the primary best fit.

Displayed parameter uncertainty

The bootstrap summary is first calculated for each fitted parameter coordinate. The application then converts that summary into the quantity shown to the user. For example, affinity is fitted as $\log_{10}(K_a)$ but is normally displayed as K_d. The displayed central value comes from the primary best fit, SD is propagated through the transformation, and the percentile limits are transformed and reordered as required.

Quantities calculated from more than one reported parameter, such as $-T\Delta S$, use the application's uncertainty-propagation rules for that calculation. Their displayed limits are therefore not necessarily the percentiles that would be obtained by recalculating the complete derived quantity independently for every bootstrap refit. The **Automatic** SD-or-CI decision is applied after transformation or propagation, separately for each displayed quantity.

Unlock parameters during error estimation

Locked parameters remain fixed during the primary fit. With **Unlock parameters** enabled, copies of those parameters are unlocked only for residual-bootstrap refits and can vary in the resampled solutions. Leave-one-out and profile likelihood always preserve the primary-fit locks.

This setting does not change or rerun the primary best fit. It changes only the parameter state used by the repeated error-estimation fits, and it has no effect when no fitted parameter is locked.

Fit execution and diagnostics

Run Fit starts the primary optimization and the selected error-estimation calculation. **Stop** requests cancellation of the active calculation. When fitting ends, the status reports the termination state, RMSD, iteration count, and elapsed time. A resampling calculation also reports its outcome and the number of successful and failed refits.

The **Display** tab exposes complementary diagnostics:

- **Fit line** shows the curve calculated from the current fitted solution.
- **Residuals** show the difference between each included observation and the fitted curve.
- **Error bars** show the processing-derived integration uncertainty.
- **Confidence band** shows uncertainty around the fitted curve when the solution contains suitable resampling results.
- **Excluded points** shows injections that do not contribute to the current fit.

These displays describe different aspects of the fitted solution. A small RMSD does not rule out systematic residual structure, poorly identified parameters, or dependence on model assumptions.

Store the fitted solution

With **Create analysis result** disabled, a successful single-experiment solution remains attached to its Experiment Data and is available in that experiment's analysis and figure workflows.

With **Create analysis result** enabled, a usable completed fit also creates a separate Analysis Result containing the experiment, model, fit settings, and solution. **Auto-open new result** determines whether the new result workspace opens immediately. A stopped or unusable fit does not create or replace an Analysis Result.

Fit availability and non-convergence

A single-experiment analysis requires processed or imported heats and at least three included injections with usable numerical values. The binding models also require nonzero cell and syringe concentrations. **Dissociation** uses the syringe concentration and does not require a macromolecule concentration in the initially buffered cell.

An unavailable model, failed termination, bound-limited solution, or failed resampling population can reflect missing or degenerate heat and concentration information, the supplied starting values, the selected limit policy, the optimizer's interaction with the fitting surface, model complexity, or weak parameter identifiability. Resampling can fail even when the primary fit succeeds because each refit presents a different or reduced dataset to the same model.

CHAPTER 07

Multiple-experiment fitting

Multi-dataset fitting across processed Active experiments, including constraints, shared options, diagnostics, and combined results.

Last verified: 2026-09-23

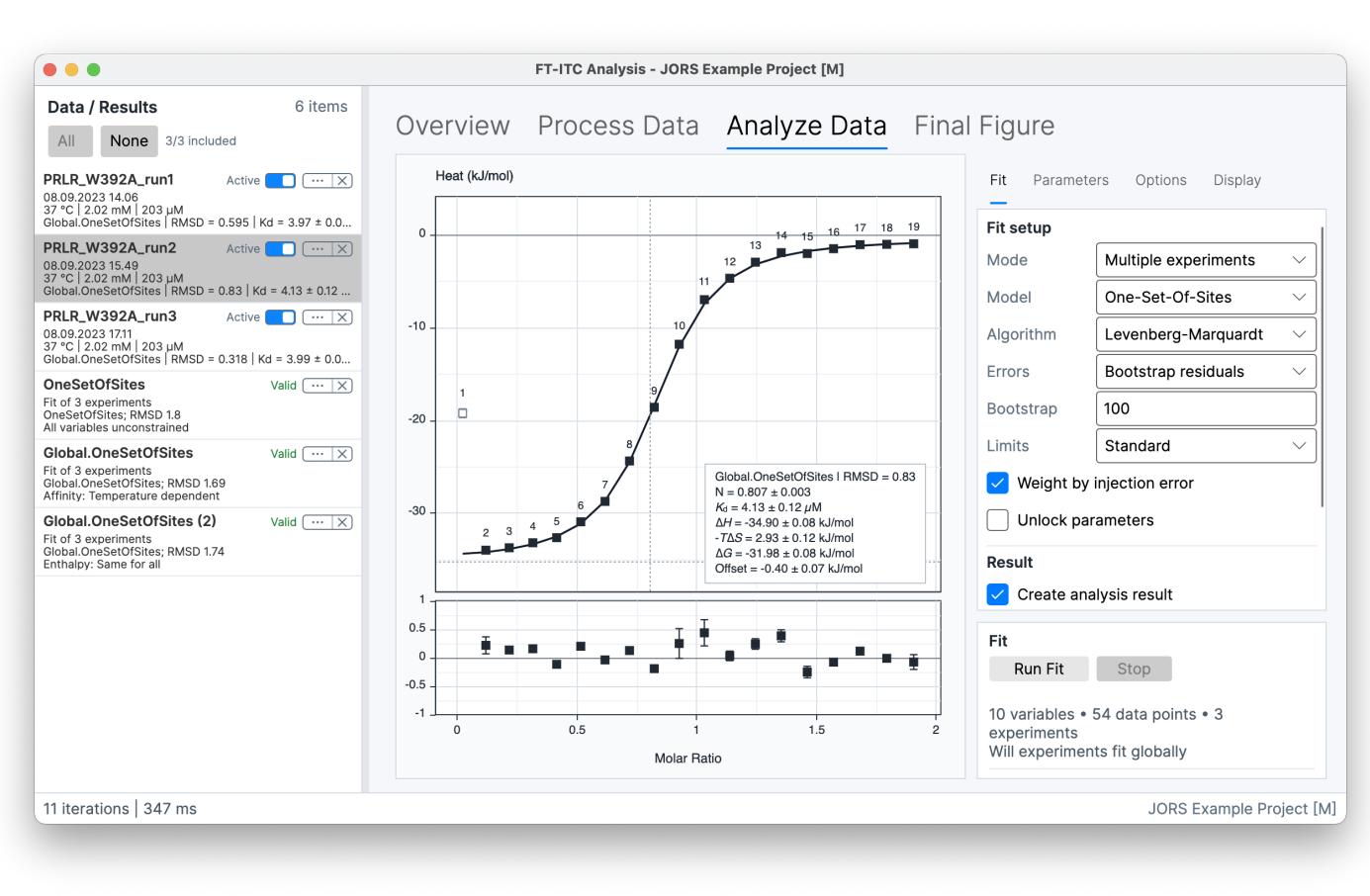
Multiple experiments is the multi-dataset mode in **Analyze Data**. It represents at least two processed **Active** experiments with one shared model context and one combined **Analysis Result**. When every constraint is **None**, the members are fitted independently. When one or more supported constraints are active, the solver performs one connected global optimization. Each experiment remains a member of the result, with member-specific parameters where the constraint state is **None**.

The model and model options apply across the active set. The resulting **Analysis Result** retains the member fits, any shared or temperature-dependent parameters, constraints, solver settings, diagnostics, and uncertainty output. Shared fitting controls are described in Single-experiment fitting.

Fit

The **Fit** tab identifies the **Mode** as **Multiple experiments** and exposes the shared **Model** selection. The solver controls include **Algorithm**, **Errors**, **Bootstrap**, **Limits**, **Weight by injection error**, **Concentration uncertainty**, and **Unlock parameters**. The available **Algorithm** values are **Nelder-Mead** and **Levenberg-Marquardt**; the error-estimation values are **None**, **Bootstrap residuals**, **Leave-one-out**, and **Profile likelihood**. For a globally fitted model, leave-one-out performs one deterministic refit per omitted experiment; profile likelihood uses the complete global objective and total observation/coordinate counts, while independent members use each member's local counts. Bootstrap count, concentration uncertainty, and parameter unlocking are disabled because they do not participate in deletion or profile schedules.

The **Result** controls describe whether a completed fit is stored as an **Analysis Result** and whether that result opens automatically. **Run Fit** and **Stop** are the fit controls. The status area records termination state, RMSD, iteration count, elapsed time, and error-estimation outcome when applicable. A multiple-experiment analysis is ready only when every member has usable processed data and the set contains at least two members.



PRLR_W392A_run1 Active
PRLR_W392A_run2 Active
PRLR_W392A_run3 Active
OneSetOfSites Valid — Fit of 3 experiments; OneSetOfSites; RMSD 1.8; All variables unconstrained
Global.OneSetOfSites Valid — Fit of 3 experiments; Global.OneSetOfSites; RMSD 1.69; Affinity: Temperature dependent
Global.OneSetOfSites (2) Valid — Fit of 3 experiments; Global.OneSetOfSites; RMSD 1.74; Enthalpy: Same for all

Global.OneSetOfSites | RMSD = 0.83
N = 0.807 ± 0.003
K_d = 4.13 ± 0.12 μM
ΔH = −34.90 ± 0.08 kJ/mol
−TΔS = 2.93 ± 0.12 kJ/mol
ΔG = −31.98 ± 0.08 kJ/mol
Offset = −0.40 ± 0.07 kJ/mol

Fit setup
Mode: Multiple experiments
Model: One-Set-Of-Sites
Algorithm: Levenberg-Marquardt
Errors: Bootstrap residuals
Bootstrap: 100
Limits: Standard
☑ Weight by injection error
☐ Unlock parameters

Result
☑ Create analysis result

Fit
Run Fit Stop
10 variables • 54 data points • 3 experiments
Will experiments fit globally

11 iterations | 347 ms

A shared enthalpy constraint connects the three Active experiments in one global fit.

Parameters

The **Parameters** tab contains the exposed fit parameters and, in multiple-experiment mode, the **Global constraints** section. Parameter rows show the current value, units, uncertainty information when available, and the parameter lock state. Values and locks control individual parameters. Global constraints define how those parameters are related across experiments.

The constraint states have these meanings:

STATE	MEANING
None	The parameter remains member-specific. Each experiment has its own fitted value.
Same for all	One common value is fitted for every member in the set.
Temperature dependent	A supported parameter is represented across the temperature series by the relationship exposed by the model.
Thermodynamically linked	Affinity shares ΔG at the fixed fit reference and derives ΔG(T) from the selected enthalpy relationship, including ΔCp when present.

The **Locked** state fixes an exposed parameter at its displayed value during fitting. **Locked** is not a constraint state and does not make a parameter common to the member experiments. A locked global value and a member-specific locked value therefore have different scopes.

For the core binding parameters, the available relationship states are model- and data-dependent:

PARAMETER	AVAILABLE STATES
Affinity	**Independent**, **Shared Kd**, **Shared ΔG**, or **Thermodynamically linked**
Enthalpy	**None** or **Same for all**; **Temperature dependent** is also available when the selected set exposes temperature dependence
N-value	**None** or **Same for all**
Offset	**None** or **Same for all**

The interface omits unsupported states for the current model. For affinity, **Independent** fits each member, **Shared Kd** fits one common dissociation constant, **Shared ΔG** shares one Gibbs-energy coordinate while Kd varies with temperature, and **Thermodynamically linked** derives the full temperature relationship from the selected enthalpy model. Profile linked intervals are approximate because fitted-coordinate correlations are omitted. Bootstrap and leave-one-out evaluate complete relationships, while Spolar–Record samples linked inputs independently in its Monte Carlo calculation.

For **Sequential Binding Sites**, the step count is one shared model option for the complete experiment set. The interface shows one **Affinity** constraint selector and one **Enthalpy** constraint selector, regardless of whether two, three, or four steps are active. A selected family style is applied to every active step while retaining a separate coordinate for each step: **Same for all** shares K_1 across members, separately shares K_2, and so on; it does not force the different steps to equal one another. **Temperature dependent** likewise uses one ΔG_i per affinity step and one reference $\Delta H_i/\Delta C_{p,i}$ pair per enthalpy step. Reducing the step count discards inactive step values and constraints; increasing it later creates new defaults for those steps.

Offset is an energy-per-mole-of-injectant correction. For each injection, its absolute heat contribution scales with the injected amount—the injection volume multiplied by the syringe concentration. With **Offset: None**, each experiment has its own fitted molar offset. With **Offset: Same for all**, the **Parameters** tab exposes one common molar offset for the complete set. That shared value has its own **Locked** control; locking it fixes the common value for every member.

CALCULATION:

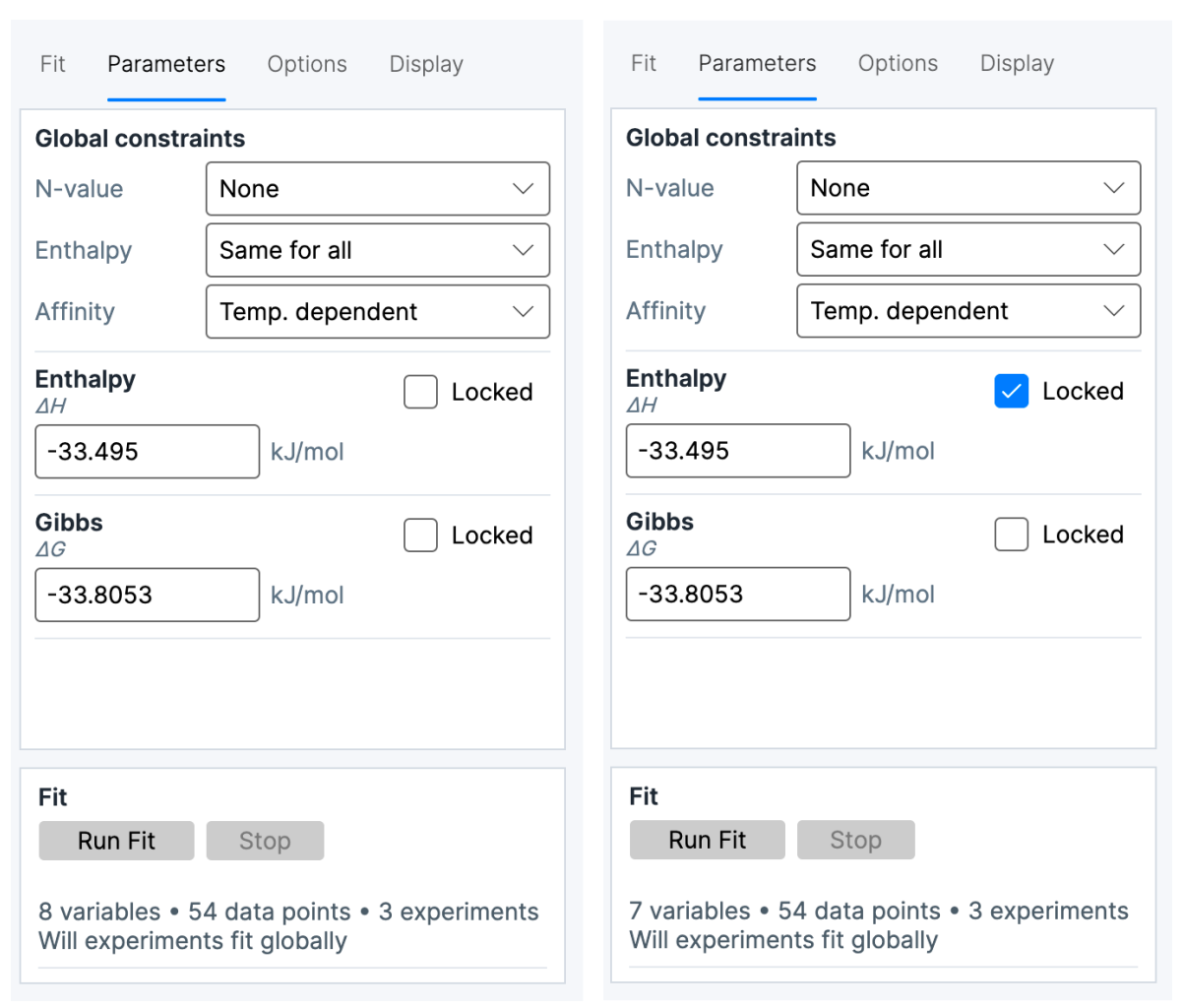
$$\Delta H(T) = \Delta H_{\text{ref}} + \Delta C_p(T - T_{\text{ref}})$$

$$\Delta G(T) = \frac{T}{T_{\text{ref}}}\Delta G_{\text{ref}} + \left(1 - \frac{T}{T_{\text{ref}}}\right)\Delta H_{\text{ref}} + \Delta C_p\left[\left(T - T_{\text{ref}}\right) - T\ln\left(\frac{T}{T_{\text{ref}}}\right)\right]$$

$$K_a(T) = \exp\left[-\frac{\Delta G}{RT}\right]$$

ΔH_{ref} is the enthalpy at the fixed fit reference, and ΔC_p is the equal-per-experiment regression slope when enthalpy is independent. The fit reference is fixed when the analysis is built; it is separate from the reporting temperature selected in preferences. A member's heat prediction continues to use its own fitted ΔH, while linked affinity uses the enthalpy trend. T and T_{ref} are absolute temperatures.

For independent enthalpies, the linked relationship gives every experiment equal weight, including repeated temperatures and locked enthalpies. Attaching profile results leaves the original best-fit values unchanged. Linked Gibbs and entropy intervals combine each fitted coordinate once, including its contribution through the enthalpy trend. Reported linked profile intervals are approximate propagation of the fitted coordinates and omit fitted-parameter covariance; bootstrap and leave-one-out results recompute the relationship from their participating members.



Global constraints
N-value: None
Enthalpy: Same for all
Affinity: Temp. dependent
Enthalpy ΔH ☐ Locked
-33.495 kJ/mol
Gibbs ΔG ☐ Locked
-33.8053 kJ/mol
Fit
Run Fit Stop
8 variables • 54 data points • 3 experiments
Will experiments fit globally

Global constraints
N-value: None
Enthalpy: Same for all
Affinity: Temp. dependent
Enthalpy ΔH ☑ Locked
-33.495 kJ/mol
Gibbs ΔG ☐ Locked
-33.8053 kJ/mol
Fit
Run Fit Stop
7 variables • 54 data points • 3 experiments
Will experiments fit globally

Global constraints define relationships between member parameters; Locked separately fixes an exposed value.

Options

The **Options** tab contains the options exposed by the selected model. In multiple-experiment mode, an option is a property of the shared model context and applies to every member. This includes model-specific concentration, stoichiometry, syringe-correction, or prebound-species settings when those options are exposed. The option values are preserved with the combined Analysis Result.

An option that changes which parameters are exposed can also change the available parameter rows and constraint states. For example, model options that share or replace stoichiometry alter which N-values are independently represented. The available rows always describe the active model and option combination.

Display

The **Display** tab controls the graph presentation for the active member set. Its controls cover the fitted line, residuals, error bars, confidence band, point labels, parameter box, excluded points, automatic scaling, unified axes, fitted offset, and fit-line interpolation. The parameter box can separately show model, fitted, and derived parameters.

Display settings affect the graph presentation, not the model, the member data, or the fitted values. The graph can therefore show the current member fits or connected global fit alongside member observations while the Parameters tab remains the source of the stored parameter and constraint state.

Combined Analysis Result

The combined **Analysis Result** contains the exact member experiments, the shared model and options, member parameters, any shared or temperature-dependent parameters, active constraints, solver and weighting settings, convergence diagnostics, and uncertainty results. Its result table can represent both global values and member-specific values. A selected member retains its own fitted curve and residual information even when a parameter is common across the set.

The status and diagnostics distinguish a successful solver termination from the quality and uncertainty of the fit. RMSD, iteration count, weighting, error-estimation method, and the number of successful uncertainty refits describe how the combined result was obtained. Wide or asymmetric uncertainty, a member with a markedly different residual pattern, or a member solution with an invalid status remains visible as member-level information rather than being hidden by a combined value. Result interpretation and validity are covered in Results and advanced analyses.

CHAPTER 08

Results and advanced analyses

Analysis Result views, validity, uncertainty presentation, evaluation temperature, and conditional advanced analyses.

Last verified: 2026-09-23

An **Analysis Result** is a stored fit for one or more experiments. It contains the model and options, constraints, solver state, weighting and uncertainty settings, member solutions, and a validity snapshot. The result workspace presents those stored values together with graph and analysis views. The fitting controls are described in Single-experiment fitting, and multi-dataset constraints are described in Multiple-experiment fitting.

Result views

The result view selector contains **Fit**, **Correlation**, and **Summary** for every Analysis Result. Thermodynamic temperature plots and parameter evaluation can show every active step in a sequential binding model. **Temperature** advanced analysis (Spolar Record method), **Salt**, and **Protonation** appear only when the selected model defines those analyses and the member metadata satisfy their additional requirements.

Summary presents the combined parameter graph and result table. The table can show fitted values, derived values, and the selected uncertainty representation for each stored solution.

Columns initially size to their headers and displayed values within compact limits; drag a header boundary to adjust an individual width. Wider tables retain their column widths and scroll horizontally, while spare space is assigned to the Experiment column. Molar-energy columns share one automatically resolved unit (or the fixed unit selected for result export), while ΔCp columns resolve independently.

Selecting a row makes that member the current result solution; the selection is retained by the result workspace and drives **Fit** and the local portion of **Correlation**. When switching directly between analysis results, the selected experiment is retained when the destination contains a member with the same experiment ID. Use **Preferences > General > Energy units** to choose joules or calories.

Advanced analyses can sample the stored values and uncertainty summaries from profile-likelihood results. This does not provide a bootstrap ensemble or a joint model of parameter covariance. Saved leave-one-out refits can provide an envelope on the integrated-heats graph.

Fit presents the saved fitted curve, residuals, error bars, confidence band, and excluded points for the selected member. The graph is read-only: it represents the stored solution and does not expose fit controls or alter the underlying experiment.

Correlation presents a matrix calculated from residual-bootstrap refits. Its availability, scope, and interpretation are described under Parameter correlation.

Inspector tabs

The result inspector has four tabs with shared labels across the supported desktop versions: **Summary**, **Analysis**, **Experiments**, and **Model**.

The **Summary** tab contains the result identity, model, member count, RMSD, information criteria, and solver diagnostics. The validity section reports **Analysis is valid**, **Partially invalid**, **Invalid**, or **Unknown status**, with reasons when the stored validity snapshot differs from current member inputs. Repeated desktop validity reasons affecting more than two experiments are summarized by category with an affected/total count; smaller or unrecognized reasons remain detailed. In the browser viewer, a valid result can also show **Saved result has analysis warnings** when a saved best fit or uncertainty refit reached a parameter boundary, or uncertainty refits reached an optimizer limit; these warnings remain visible alongside validity reasons. Solver information includes algorithm, iterations, whether injection-error weighting was used, error-estimation method, and bootstrap count.

In the browser viewer, RMSD is shown as a saved unweighted display diagnostic in μJ, separate from the weighted fitting objective. When the saved convergence record contains it, **Molar RMSD (kJ/mol)** is shown separately: the result summary uses the saved global metric, while an individual fit uses that fit's saved metric. Missing or non-finite saved values remain unavailable; member values are not averaged to reconstruct the result metric.

Information criteria

The **Information criteria** section reports the analysis-level AICc when it is available, otherwise AIC, together with the included observation count n and likelihood parameter count K. Both weighted and unweighted criteria estimate one residual-variance parameter, so $K = p + 1$, where p is the number of free fitted parameters. Weighted criteria use the injection integration errors as relative uncertainties and estimate one common variance multiplier from the standardized residuals. AICc is unavailable when $n \le K + 1$, in which case AIC is shown instead. If the likelihood cannot be evaluated, the displayed criterion shows its diagnostic reason.

When members were fitted independently, the result table also includes an **AICc / AIC** column. It reports each member's own criterion and is intended for comparing alternative models fitted to the same experiment, observations, response definition, and weighting mode. Values in different experiment rows are not comparable because they use different observations. AICc is shown when available, with AIC as the fallback when the small-sample correction is undefined; **Unavailable** means that the member likelihood could not be evaluated. Shared-parameter global fits expose only the analysis-level criterion and omit this member column. Neither AIC nor AICc establishes model adequacy.

The criteria use the saved result's included injections and response definition. At analysis level, residual statistics are pooled before estimating the variance, including when members were fitted independently. Unweighted criteria estimate one common variance across all members; weighted criteria estimate one common multiplier of the injection-error variances. Each independent member's own criterion estimates its variance separately. Consequently, neither pooled AIC nor pooled AICc is a sum of the member values.

With residuals r_i, raw residual sum of squares $RSS = \Sigma r_i^2$, integration-error SDs σ_i, and standardized residual sum of squares $Q = \Sigma(r_i/\sigma_i)^2$, the maximized likelihood terms are:

CALCULATION:

$$\text{Unweighted, estimated common variance:}\quad -2\log L = n\left[\log\left(\frac{2\pi\text{RSS}}{n}\right) + 1\right]$$

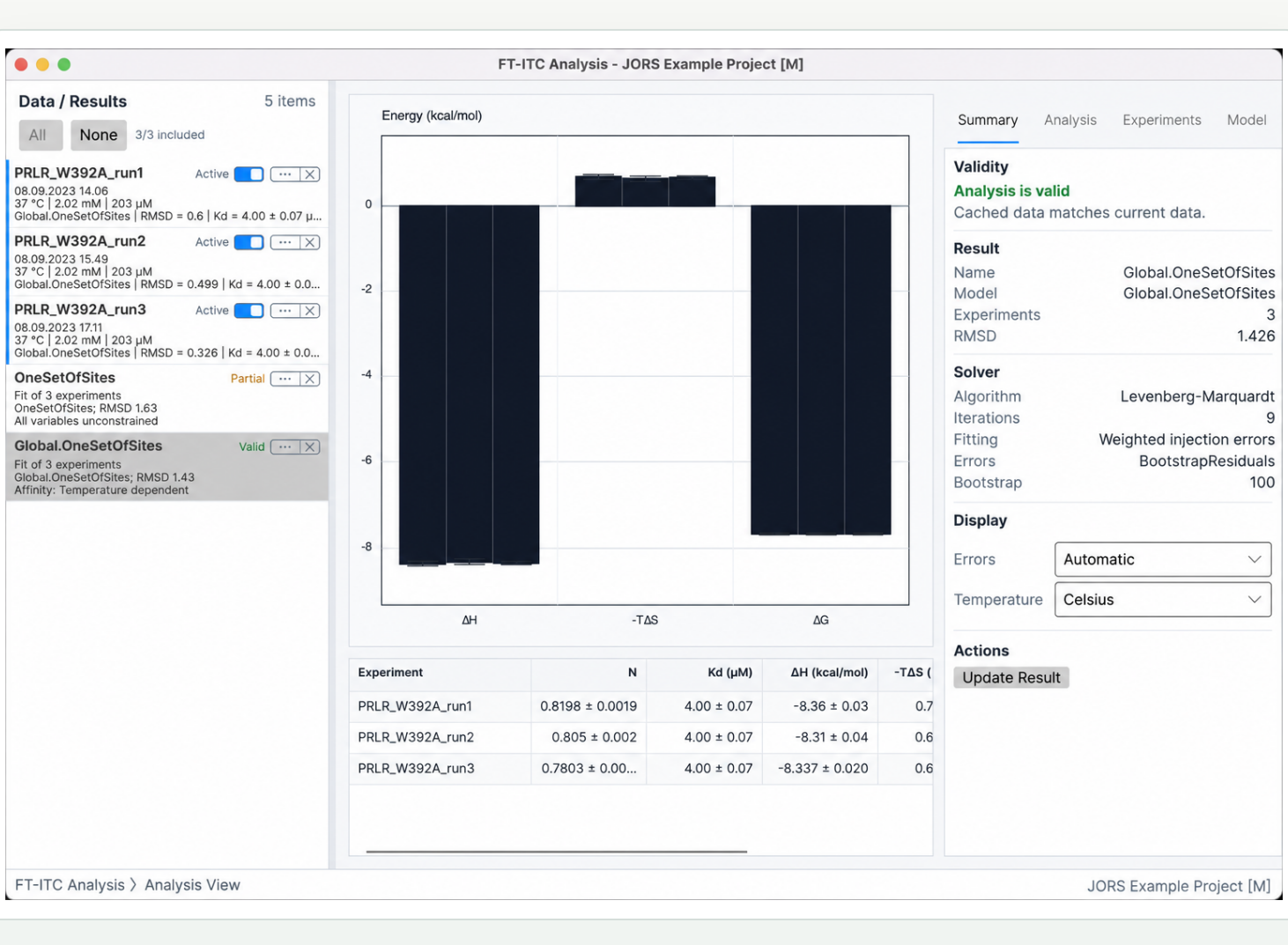
$$\text{Weighted, estimated variance multiplier:}\quad -2\log L = n\left[\log\left(\frac{2\pi Q}{n}\right) + 1\right] + \Sigma\log\sigma_i^2$$

The weighted variance multiplier is Q/n, calculated analytically without adding an optimizer variable or changing the stored integration errors or fitted parameters. The calculation uses the same injection SDs and the same per-experiment fallback for missing SDs as weighted fitting. Zero residual variance makes the estimated-variance likelihood unavailable. Weighted profile-likelihood intervals use these unchanged processing SDs as relative weights and apply the same overall residual-scale interpretation with an F-calibrated threshold. Neither calculation writes a model-derived value back to an injection SD.

The fitted parameter count p includes only parameters free in the saved global model. Shared coordinates count once; member-specific coordinates count once per member. For a member criterion, p includes only that member model's free fitted parameters. In both weighting modes, the reported values use $K = p + 1$, AIC $= -2\log L + 2K$, and AICc = AIC $+ 2K(K + 1)/(n - K - 1)$. This standard small-sample correction is an approximation for nonlinear ITC models.

Smaller values are preferred only when comparing models that use the same observations, response definition, and weighting mode. AIC and AICc do not establish model adequacy or replace residual and scientific checks. Prefer AICc when it is available.

Information criteria are recalculated when a saved project is opened. Weighted values from the former fixed-sigma calculation can therefore change; compare results calculated under the same likelihood convention.



PRLR_W392A_run1 Active
PRLR_W392A_run2 Active
PRLR_W392A_run3 Active
OneSetOfSites Partial — Fit of 3 experiments; OneSetOfSites; RMSD 1.63; All variables unconstrained
Global.OneSetOfSites Valid — Fit of 3 experiments; Global.OneSetOfSites; RMSD 1.43; Affinity: Temperature dependent

Experiment	N	Kd (μM)	ΔH (kcal/mol)	-TΔS (
PRLR_W392A_run1	0.8198 ± 0.0019	4.00 ± 0.07	-8.36 ± 0.03	0.7
PRLR_W392A_run2	0.805 ± 0.002	4.00 ± 0.07	-8.31 ± 0.04	0.6
PRLR_W392A_run3	0.7803 ± 0.00...	4.00 ± 0.07	-8.337 ± 0.020	0.6

Validity
Analysis is valid
Cached data matches current data.

Result
Name: Global.OneSetOfSites
Model: Global.OneSetOfSites
Experiments: 3
RMSD: 1.426

Solver
Algorithm: Levenberg-Marquardt
Iterations: 9
Fitting: Weighted injection errors
Errors: BootstrapResiduals
Bootstrap: 100

Display
Errors: Automatic
Temperature: Celsius

Actions
Update Result

Summary combines result validity, stored solver information, display controls, and the member parameter overview.

The **Analysis** tab contains the result view selector, parameter evaluation, and the analysis-specific controls and outputs. It is also the location of the uncertainty display, correlation information, and evaluation-temperature presentation associated with the selected view. Energy units are chosen consistently from the displayed central values and are also used for graph axes, errors, fitted bands, tooltips, and parameter lists.

The **Experiments** tab lists the result members and their stored status and condition information, including member temperature. The row selected in the result table determines which experiment appears in **Fit**; this tab provides that experiment's details.

The **Model** tab shows the stored model options, locked parameters and their fixed values, and the active constraints. A constraint with state **None** is not listed as an active global constraint; **Same for all** and **Temperature dependent** entries identify the relationships retained by the Analysis Result. Affinity constraints are labelled **Independent**, **Shared Kd**, **Shared ΔG**, or **Thermodynamically linked** to describe the fitted relationship precisely; enthalpy and other parameters retain their own constraint labels.

Uncertainty and evaluation temperature

The **Errors** display control provides **Automatic**, **Standard deviation**, **95% confidence interval**, and **SD + 95% CI**. **Standard deviation** presents the primary best-fit value with a symmetric ± SD; **95% confidence interval** presents that same best-fit value with its lower and upper confidence limits; and **SD + 95% CI** presents both. The central value is always the primary best fit, not the mean or median of the resampled values. Residual-bootstrap intervals use percentile limits; profile-likelihood intervals use the likelihood-threshold endpoints, with an equivalent symmetric scale for the SD display.

Automatic makes this choice separately for each reported quantity. Let L and U be its stored 95% confidence limits and $\hat{\theta}$ its primary best-fit value:

CALCULATION:

$$w_{\text{lower}} = \hat{\theta} - L;\quad w_{\text{upper}} = U - \hat{\theta}$$

$$\text{asymmetry} = \frac{|w_{\text{upper}} - w_{\text{lower}}|}{w_{\text{upper}} + w_{\text{lower}}}$$

Automatic shows the 95% confidence interval when both widths are positive and the asymmetry is at least 0.18; otherwise it shows SD. This is a display rule based on interval imbalance around the best fit, not a formal statistical test of distribution skewness.

The decision is made after a fitted coordinate has been transformed into the displayed quantity. A nonlinear transformation—such as fitting $\log_{10}(K_a)$ and displaying K_d—can therefore make the displayed interval asymmetric and cause **Automatic** to select CI for that quantity.

Changing **Errors** changes only how stored uncertainty is presented in tables, parameter evaluation, and graphs. It does not rerun the fit, change the best-fit parameters, or turn one error-estimation method into another. Bootstrap construction, parameter transformation, and the SD and percentile calculations are described under Parameter uncertainty.

The **Parameter Evaluation** section contains an evaluation **Temperature** field and the displayed thermodynamic quantities at that temperature. Temperature display can be **Celsius** or **Kelvin**. Changing the evaluation temperature updates the displayed quantities calculated from the stored model; it does not change injection heats or refit the result. Temperature-dependent values are meaningful together with their model, units, uncertainty representation, and evaluation temperature.

For locally fitted member parameters, summary values include individual parameter uncertainty. **Combined SD** is the square root of the between-experiment sample variance plus the mean individual variance; temperature trends use residual variance in place of sample variance. It describes combined spread, not the standard error of the mean, and observed spread may already contain fitting noise. **Approximate propagated interval** targets the average or evaluated trend: individual lower and upper 95% interval widths are propagated separately and combined with the observed-spread contribution. The best-fit central values remain unchanged, and asymmetric intervals can remain asymmetric. Exactly two observations at distinct temperatures propagate individual errors without estimating residual variance; additional observations, including replicates at those same temperatures, contribute residual spread. ΔCp propagates the corresponding slope uncertainty. These intervals do not have established 95% coverage, especially with few experiments, and covariance between experiments is omitted even for local parameters coupled through a global fit. Shared and temperature-constrained parameters retain their model-estimated uncertainty and CI95 meaning.

Parameter correlation

Correlation shows Pearson correlations between fitted parameter coordinates across residual-bootstrap refits. It requires **Bootstrap residuals**, at least 30 complete refits, and at least two parameters that vary across those refits. Parameters with no variation are omitted. Parameters fixed in the primary fit are also omitted unless **Unlock parameters** allowed them to vary during bootstrap error estimation; such parameters are marked with an asterisk and a warning that bootstrap parameter unlocking was enabled.

CALCULATION:

$$r_{jk} = \frac{\Sigma_b\left[\left(\theta_{bj} - \bar{\theta}_j\right)\left(\theta_{bk} - \bar{\theta}_k\right)\right]}{\sqrt{\left[\Sigma_b\left(\theta_{bj} - \bar{\theta}_j\right)^2\Sigma_b\left(\theta_{bk} - \bar{\theta}_k\right)^2\right]}}$$

r_{jk} is the Pearson correlation between fitted coordinates j and k. The index b runs over complete residual-bootstrap refits, and $\bar{\theta}$ is the corresponding mean coordinate.

For each off-diagonal cell, the application also reports an approximate 95% Monte Carlo precision interval using Fisher's transformation:

CALCULATION:

$$z = \operatorname{atanh}\left(r\right);\quad z_{\pm} = z \pm \frac{1.959964}{\sqrt{B - 3}};\quad r_{\pm} = \tanh\left(z_{\pm}\right)$$

B is the number of complete refits: usable refits with finite values for every displayed coordinate. The diagonal is the structural self-correlation $r = 1$ and has no Fisher precision interval.

Matrix values range from –1 to +1; pointing to a cell shows r and the Monte Carlo precision interval; an interval spanning zero marks the sign as unresolved at the current simulation precision. The result panel separately reports attempted, usable, failed, and complete refits. It warns when fewer than 100 complete refits make precision coarse, at least 20% of attempted refits failed, a usable refit lacked a finite displayed coordinate, a sign is unresolved, or the matrix is structurally rank limited. The correlation matrix remains unavailable below 30 complete refits. For a single-experiment result, the scope is **Single experiment**. A multiple-experiment result can show **Shared** coordinates alone or **Shared + selected local** coordinates when a member is selected. Shared and local labels identify the scope of each parameter.

Sequential correlations use the actual fitted coordinates. Affinity coordinates are labeled **log10 Ka1** through **log10 Ka4**, active enthalpy coordinates are included, and no N-value coordinate is added. A global result shows the shared per-step coordinates and, when requested, unconstrained coordinates for the selected member without duplicating constrained member values.

> INTERPRETATION:
>
> Correlation shows how fitted coordinates varied together under the residual bootstrap. The Fisher interval describes the finite-bootstrap Monte Carlo precision of the correlation coefficient r itself; it is not a confidence interval for either fitted parameter and does not replace the parameter uncertainty display. It does not establish identifiability, causality, model adequacy, or model validity. An interval spanning zero means only that the sign is unresolved at this Monte Carlo precision. Frequent refit failures can make the retained ensemble selective, and the Fisher interval does not account for those failures. Affinity is evaluated in the fitted coordinate system—$\log_{10}(K_a)$—rather than as the displayed K_d. A rank warning states the structural limit $rank \le B - 1$ for the displayed parameter count; it is distinct from numerical rank and scientific model validity.

Advanced analysis views

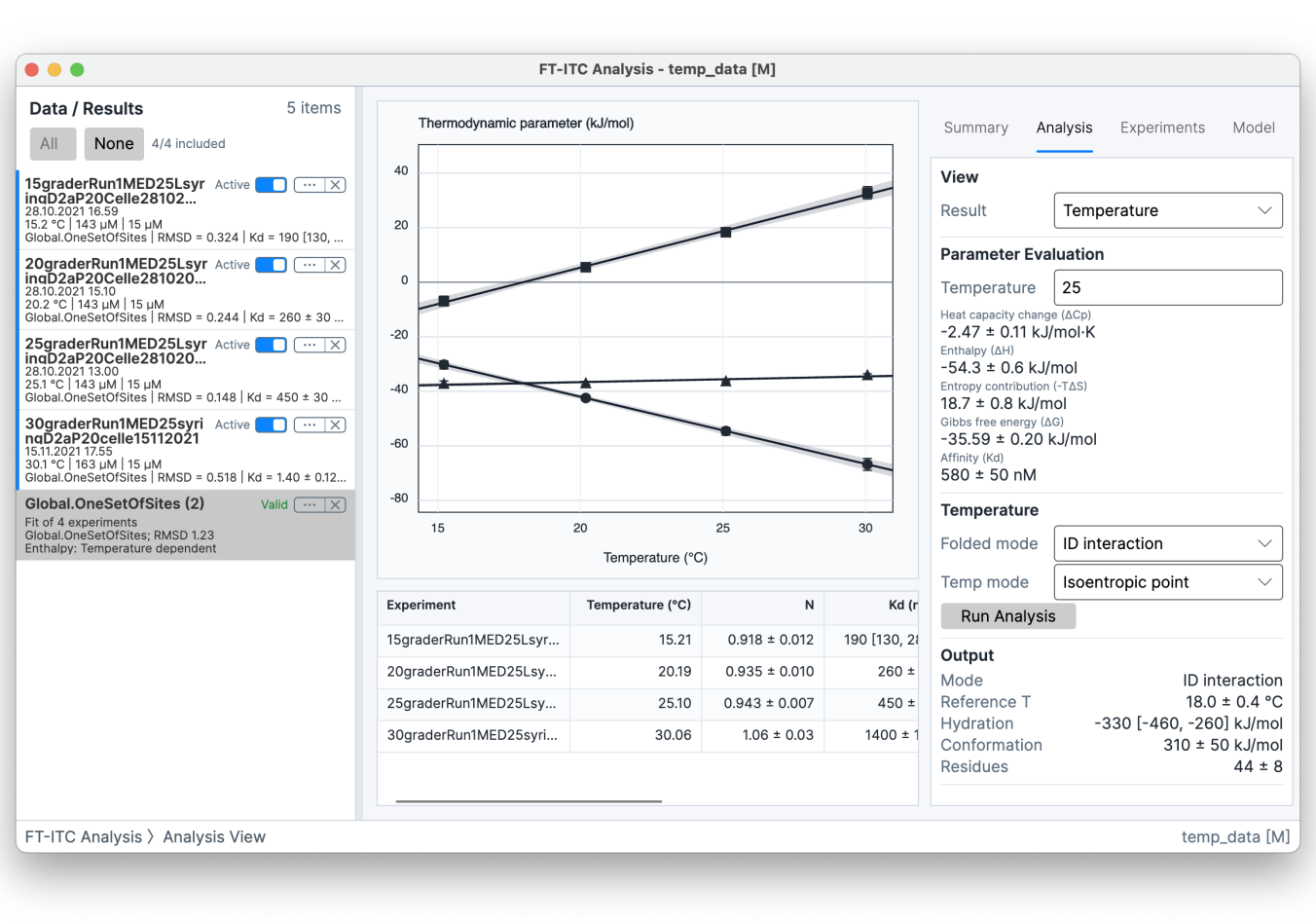
All advanced analyses require a **One-Set-Of-Sites** Analysis Result. Each analysis also requires the relevant variation in experimental conditions and the corresponding metadata. The advanced analyses operate on the stored member solutions and expose their own calculated outputs; they do not change the base fit parameters. A sequential result can still show its ordinary per-step ΔH, ΔG, –TΔS, Kd, and temperature-dependence presentation; it reports the Spolar Record method, protonation, and electrostatics as unsupported by that model rather than hiding the ordinary thermodynamic views.

Temperature

The **Temperature** view is available when the difference between the highest and lowest member temperatures exceeds the configured minimum span. Its temperature-analysis controls expose

Folded mode values **Globular** and **ID interaction**, together with **Temp mode** values **Isoentropic point**, **Mean temperature**, and **Reference temperature**. For a thermodynamically linked result, the isoentropic temperature is derived from the complete Gibbs, enthalpy, and heat-capacity relationship. Profile intervals are propagated approximately and omit fitted-coordinate correlations; bootstrap and leave-one-out uncertainty evaluates each saved relationship jointly, while the Spolar–Record Monte Carlo samples its linked inputs independently.

The stored temperature-analysis output includes reference temperature, hydration contribution, conformational contribution, and residue estimate. These values describe the selected folded and temperature-evaluation modes under the fitted temperature dependence. They remain conditional estimates of the stored model and member series rather than direct structural measurements.



15graderRun1MED25Lsyr ingD2aP20Celle28102… Active
28.10.2021 16.59
15.2 °C | 143 µM | 15 µM
Global.OneSetOfSites | RMSD = 0.324 | Kd = 190 [130, …

20graderRun1MED25Lsyr ingD2aP20Celle281020… Active
28.10.2021 15.10
20.2 °C | 143 µM | 15 µM
Global.OneSetOfSites | RMSD = 0.244 | Kd = 260 ± 30 …

25graderRun1MED25Lsyr ingD2aP20Celle281020… Active
28.10.2021 13.00
25.1 °C | 143 µM | 15 µM
Global.OneSetOfSites | RMSD = 0.148 | Kd = 450 ± 30 …

30graderRun1MED25syri ngD2aP20celle15112021 Active
15.11.2021 17.55
30.1 °C | 163 µM | 15 µM
Global.OneSetOfSites | RMSD = 0.518 | Kd = 1.40 ± 0.12…

Global.OneSetOfSites (2) Valid
Fit of 4 experiments
Global.OneSetOfSites; RMSD 1.23
Enthalpy: Temperature dependent

Experiment	Temperature (°C)	N
15graderRun1MED25Lsyr…	15.21	0.918 ± 0.012
20graderRun1MED25Lsy…	20.19	0.935 ± 0.010
25graderRun1MED25Lsy…	25.10	0.943 ± 0.007
30graderRun1MED25syri…	30.06	1.06 ± 0.03

View
Result: Temperature
Parameter Evaluation
Temperature: 25
Heat capacity change (ΔCp): -2.47 ± 0.11 kJ/mol·K
Enthalpy (ΔH): -54.3 ± 0.6 kJ/mol
Entropy contribution (-TΔS): 18.7 ± 0.8 kJ/mol
Gibbs free energy (ΔG): -35.59 ± 0.20 kJ/mol
Affinity (Kd): 580 ± 50 nM
Temperature
Folded mode: ID interaction
Temp mode: Isoentropic point
Run Analysis
Output
Mode: ID interaction
Reference T: 18.0 ± 0.4 °C
Hydration: -330 [-460, -260] kJ/mol
Conformation: 310 ± 50 kJ/mol
Residues: 44 ± 8

Temperature presents the stored temperature dependence together with its evaluation and folded-state analysis controls.

Salt

The **Salt** view requires a **Salt** attribute for every member and sufficient ionic-strength span across the member set. Its **Graph mode** values are **Affinity vs Salt**, **Debye-Huckel**, and **Counter Ion Release**. Counter-ion release is a salt analysis mode and is presented in the Salt view rather than as a separate advanced analysis.

The Salt output includes the extrapolated **Kd0** and **Counter ion** result when the analysis has a calculated fit. The graph mode determines whether the displayed dependence is expressed against salt, ionic strength using Debye-Huckel behavior, or ion activity for counter-ion release. The result is limited by the recorded salt identities, ionic-strength values, and the quality and span of the member affinity values.

CALCULATION:

$$\ln K_d(I) = \ln K_{d,0} + s\sqrt{I}$$

$$\ln K_d = b + n_{\text{ion}} \ln a_{\text{ion}}$$

I is ionic strength, *s* is the fitted sensitivity, a_{ion} is ion activity, and *b* is the fitted intercept. $K_{d,0}$ is the extrapolated value in the Debye–Hückel view. n_{ion} is the reported slope for Counter Ion Release; its sign is reported by the analysis and is not assigned an interpretation here.

Protonation

The **Protonation** view requires a **Buffer** attribute for every member and at least two distinct buffer identities. Its calculated output contains **Protons** and **Binding H**, with uncertainty when the stored analysis contains the corresponding uncertainty results.

The view relates the stored member binding enthalpies to the buffer protonation information associated with their conditions. The result is conditional on the recorded buffer identities, their temperature-dependent protonation enthalpies, and the model assumptions; it does not identify a particular residue or microscopic protonation event.

CALCULATION:

$$\Delta H_{\text{obs}} = \Delta H_{\text{bind}} + m\,\Delta H_{\text{buffer}}$$

The fitted slope is *m*. The application reports **Protons** as −*m*, while **Binding H** is the fitted intercept at zero buffer protonation enthalpy. This sign convention follows the application's protonation-enthalpy convention; it does not by itself assign a microscopic uptake or release mechanism.

Advanced-analysis values are supplemental views of a stored Analysis Result. Their availability and outputs are determined by the One-Set-Of-Sites model, member variation, metadata, selected graph or evaluation mode, and any completed uncertainty calculation. Result validity remains a separate indication of whether the stored fit inputs match the current project state. Figure and table output is covered in Figures and export.

The buffer registry contains approximations and some unknown quantities, so confirm that the chosen buffer values apply to the experiment. The TAPSO pKa temperature correction is a local approximation for 20–30 °C. Its protonated zwitterion is neutral, and imidazole's ionization heat-capacity slope is −9 J/(mol K). Recalculate affected derived analyses when comparing with results produced before these corrections.

CHAPTER 09

Figures and export

Configure final and supporting figures, print active graphs, and export data, results, figures, and complete analysis reports.

Last verified: 2026-09-23

FT-ITC Analysis keeps publication figures, numerical data, and fitted result tables as separate output types. **Final Figure** is an Experiment Data workflow. An Analysis Result does not open directly as a Final Figure; **Export Associated Final Figures…** on a result exports figures for the experiment solutions associated with that result.

Final Figure

The **Final Figure** workspace renders the selected Experiment Data as a publication figure. Its preview and PDF output reflect the selected experiment, processing state, fitted solution, and figure options. To export figures using the fits stored in an Analysis Result, use **Export Associated Final Figures…** on that result.

The **Export PDF** controls offer three scopes:

- **Selected** opens a save dialog for the displayed experiment figure and suggests the experiment name as the PDF filename.
- **Active** exports figures for Active Experiment Data. **All** exports figures for all Experiment Data in the project.

For **Active** and **All**, choose a parent folder and the app creates or reuses a subfolder named after the saved project file (without its extension). For an unsaved session, it uses the local export date (`yyyyMMdd`) as the folder name. Existing PDFs are kept; when a filename is already in use, the new figure gets a numeric suffix such as `Experiment (2).pdf`.

The result-list command **Export Associated Final Figures…** loads the result's member solutions into their experiments and writes one final-figure PDF per associated experiment. It is available for a result with exportable solutions. See Results and advanced analyses for result validity and stored member-solution behavior.

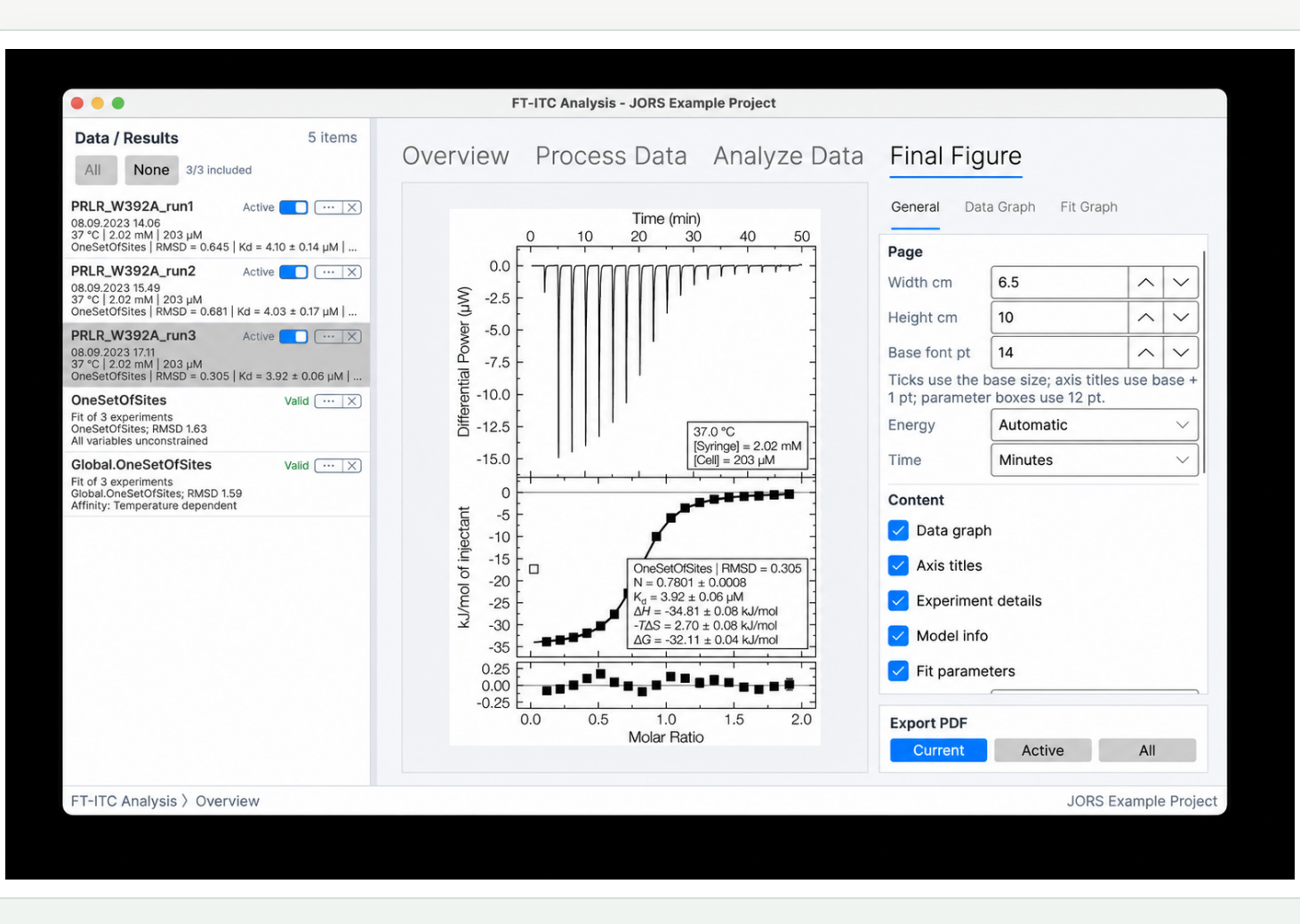
WHERE FINAL FIGURE SETTINGS ARE USED

- **Final Figure PDFs**, including Active, All, and **Export Associated Final Figures…**, use the current Final Figure appearance controls.
- **Supporting Figure…** starts with a snapshot of those controls when its window opens. This includes units, axis limits, visible traces and fits, and excluded-point automatic Y scaling. Its own plot size, grid, typography, panel labels, grouping, and information-box controls take precedence; later Final Figure edits do not update an already open Supporting Figure.
- **Analysis Report** uses a separate compact figure preset and its own energy and uncertainty selections. Final Figure axis limits and excluded-point automatic Y scaling do not carry over to report figures.

General

The **General** tab defines the page and common content. Page controls specify width and height in centimeters and the base font size. The **Energy units** selector provides **Automatic**, **J**, **kJ**, **cal**, and **kcal**. **Automatic** chooses a common molar-energy unit based on the figure's plotted and reported central values. A fixed choice uses the selected energy unit throughout the figure, except for the thermogram units described below. The selected family also controls thermogram units: differential power is **µW** or **µcal/s**, while integrated heat is **µJ** or **µcal**. Time controls set the displayed time unit. Content controls include the data graph, axis titles, experiment details, model information, fit parameters, and the information-box placement. The uncertainty selector provides **Automatic**, **SD**, **CI**, **SD + CI**, and **None**. **Automatic** uses the same per-quantity asymmetry rule described under Uncertainty and evaluation temperature.

The parameter controls determine which information appears in the information box: thermodynamic, derived, or offset parameters; temperature; concentrations; injection delay; instrument; and user-defined attributes. The information box is descriptive figure content and does not alter the underlying fit.



PRLR_W392A_run1 Active
08.09.2023 14.06
37 °C | 2.02 mM | 203 µM
OneSetOfSites | RMSD = 0.645 | Kd = 4.10 ± 0.14 µM | …

PRLR_W392A_run2 Active
08.09.2023 15.49
37 °C | 2.02 mM | 203 µM
OneSetOfSites | RMSD = 0.681 | Kd = 4.03 ± 0.17 µM | …

PRLR_W392A_run3 Active
08.09.2023 17.11
37 °C | 2.02 mM | 203 µM
OneSetOfSites | RMSD = 0.305 | Kd = 3.92 ± 0.06 µM | …

OneSetOfSites Valid
Fit of 3 experiments
OneSetOfSites; RMSD 1.63
All variables unconstrained

Global.OneSetOfSites Valid
Fit of 3 experiments
Global.OneSetOfSites; RMSD 1.59
Affinity: Temperature dependent

Page
Width cm: 6.5
Height cm: 10
Base font pt: 14
Ticks use the base size; axis titles use base + 1 pt; parameter boxes use 12 pt.
Energy: Automatic
Time: Minutes

Content
☑ Data graph
☑ Axis titles
☑ Experiment details
☑ Model info
☑ Fit parameters

Export PDF
Current | Active | All

Final Figure combines a live publication preview with page, content, and PDF export controls.

Data Graph

The **Data Graph** tab controls the differential-power trace. Power and time axis titles, tick density, and explicit minimum and maximum values define the axes. **Corrected data** selects the baseline-corrected trace when one is available. **Shared power axis** applies one power-axis range across the active experiments represented by a figure set.

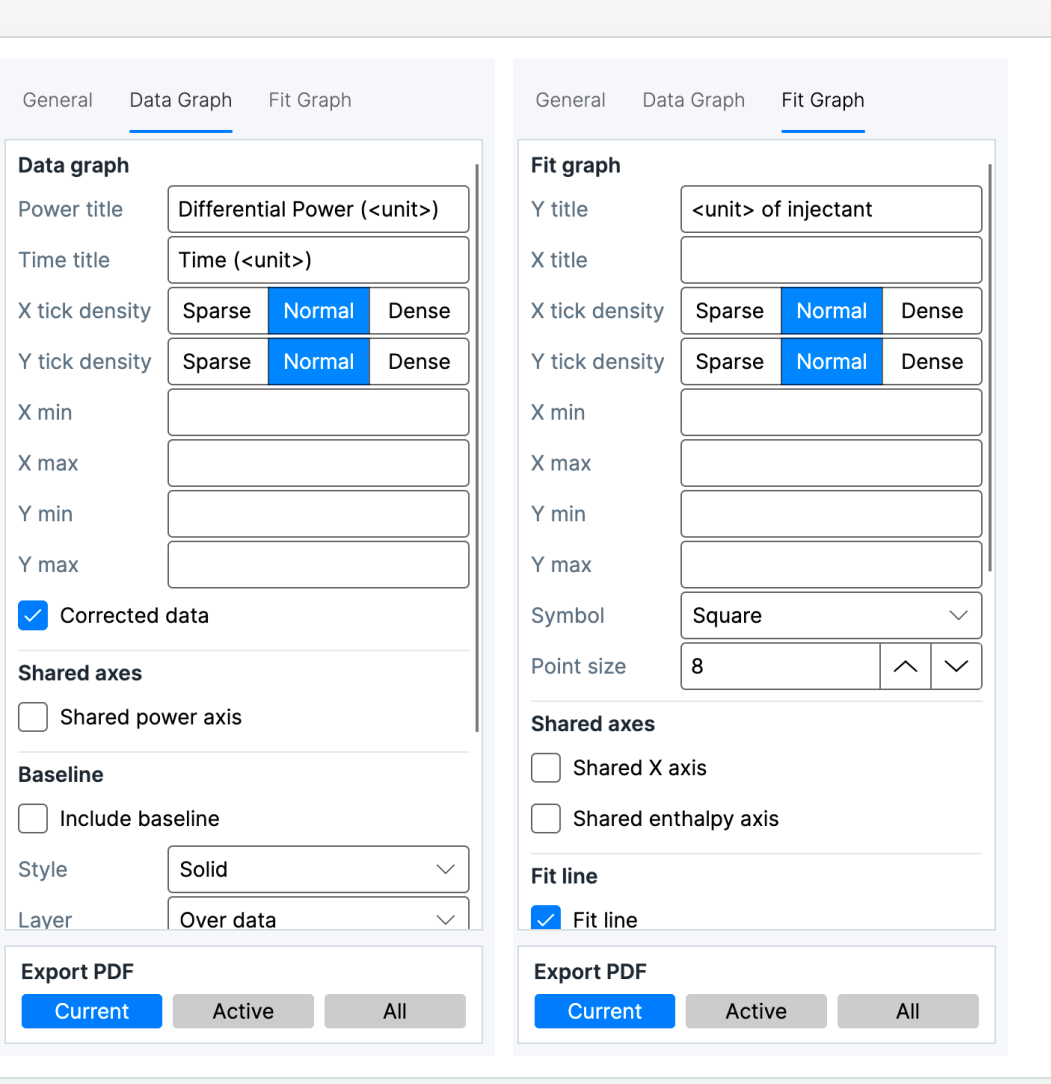
Baseline controls expose the baseline, its **Solid** or **Dashed** style, **Under data** or **Over data** layer, and line width. **Integration ranges** displays the integration intervals as **Bar**, **Fill**, or **Endpoint lines**. These overlays describe processing and do not recalculate integration.

Fit Graph

The **Fit Graph** tab controls the integrated-heats and model panels. Enthalpy and molar-ratio axis titles, tick density, and explicit ranges define the axes. Symbols are **Square** or **Circle**, with an independent point size. **Shared X axis** and **Shared enthalpy axis** unify the corresponding ranges across active experiments; the residual-axis range follows the shared enthalpy setting when configured for unified residual axes.

Include excluded points in automatic Y scaling lets excluded injections affect the automatic integrated-heat and residual axis ranges. It starts from the application preference. Explicit axis limits take precedence.

Fit line displays the fitted binding curve with a selected width and **Smooth**, **Spline**, or **Linear** smoothness. **Show residuals graph** adds the residual panel, and **Residual gap** separates it visually from the fit panel. The display controls include the zero enthalpy line, confidence band, error bars, excluded points, excluded error bars, and offset-corrected heats. Error bars use processed injection uncertainties; confidence bands require bootstrap uncertainty in the fitted solution; fit lines and residuals require a fitted solution.



Data graph
Power title: Differential Power (<unit>)
Time title: Time (<unit>)
X tick density: Sparse | Normal | Dense
Y tick density: Sparse | Normal | Dense
X min:
X max:
Y min:
Y max:
☑ Corrected data
Shared axes
☐ Shared power axis
Baseline
☐ Include baseline
Style: Solid
Layer: Over data
Export PDF
Current | Active | All

Fit graph
Y title: <unit> of injectant
X title:
X tick density: Sparse | Normal | Dense
Y tick density: Sparse | Normal | Dense
X min:
X max:
Y min:
Y max:
Symbol: Square
Point size: 8
Shared axes
☐ Shared X axis
☐ Shared enthalpy axis
Fit line
☑ Fit line
Export PDF
Current | Active | All

Data Graph and Fit Graph provide separate controls for the thermogram and fitted-heats panels.

Numerical data export

File > Export Data… opens the numerical export dialog. **Export Selected Data…** invokes the same export for the selected experiment. The data scope is one of **Selected experiment**, **Active experiments**, or **All experiments**.

The format list contains:

- **Thermogram Data** — a `.csv` file containing time and power samples, with the **Export baseline-corrected trace** option when baseline-corrected samples exist.
- **Integrated Peaks** — a `.csv` file containing injection-axis values, molar heats, their SD, fitted values, and residuals when available, with **Export offset-corrected peaks** when a fitted solution is available.
- **Combined Data** — a `.csv` file placing thermogram samples and integrated-peak columns side by side; raw, corrected, fitted, and residual columns are included when available.
- **MicroCal / SEDPHAT** — a `.dat` file with MicroCal-style `DH`, `INJV`, `Xt`, `Mt`, `XMt`, `NDH`, `DY`, and `Fit` columns. `DH` is in µcal, `INJV` is in µL, concentrations are in mM, and `NDH`, `DY`, and `Fit` are in cal/mol. The first injection uses `--` for normalized heat and residual, following the conventional excluded-first-injection layout.
- **pytc** — a `.dh` file containing the pytc-compatible injection and metadata fields.
- **ITCsim** — a `.csv` file containing ITCsim-compatible injection data and metadata, with offset-corrected peaks available when a fitted solution exists.

Molar-energy unit is available for Integrated Peaks, Combined Data, and ITCsim: choose **J/mol**, **kJ/mol**, **cal/mol**, or **kcal/mol**. Each export dialog starts at kJ/mol for the Joules energy preference or kcal/mol for the Calories preference; a choice applies to that export only. Integrated Peaks and Combined Data identify the chosen unit in the heat, SD, model, and residual column headers. ITCsim keeps its `PeakHeat` column name and records the unit as `#EXPINFO ENERGYUNIT J`, `KJ`, `CAL`, or `KCAL`; those tokens denote energy per mole. Peak area is an absolute heat in J and is distinct from the exported molar heat.

Other output units remain format-specific. Thermogram samples use seconds and watts; MicroCal/SEDPHAT and pytc use their documented concentration, volume, temperature, and heat conventions. Fitted columns and correction controls are disabled when the selected data do not contain the corresponding processed or fitted state. Export defaults are described in Settings and defaults.

These exports are not project backups. General `.csv` and `.tsv` exports cannot be reopened through **File > Open…**. A MicroCal / SEDPHAT `.dat` export or pytc `.dh` export can be reopened as integrated-heat input, but doing so restores neither a thermogram nor the FT-ITC project state and requires choosing the source heat unit. Select **microcalorie** for the `DH` column when reopening a newly generated MicroCal / SEDPHAT export. Historical FT-ITC `.dat` exports may instead require **joule**, matching the unit used when they were created. Save an `.ftxtc` project when the analysis must remain editable.

Analysis Result Exporter

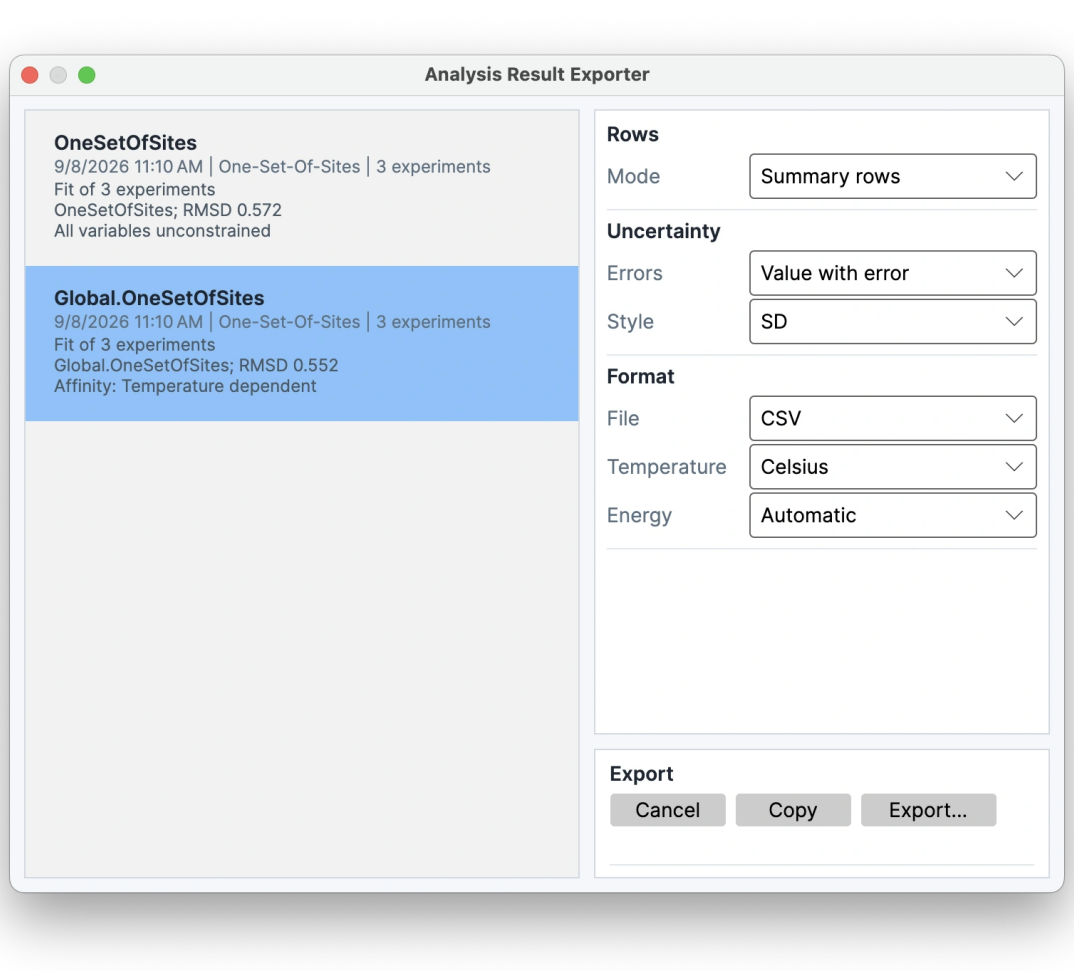
Analysis Result Exporter… builds a table from one or more selected Analysis Results. **Summary rows** produces one row per result; **All replicate rows** produces one row per member fit. Error layout is **Value with error** or **Separate columns**. Uncertainty style is **SD**, **CI**, or **SD + CI**. The file format is **CSV** or **TSV**. **Energy units** provides **Automatic**, **J**, **kJ**, **cal**, and **kcal**. Automatic chooses one stable molar-energy unit across all selected results and one independent unit for ΔCp columns; a fixed choice uses the selected energy unit consistently. Temperature presentation is **Celsius** or **Kelvin**.

The configured table is available through **Copy** and **Export…**. Copy places the same delimited text on the clipboard; Export writes it to a file. Profile-likelihood SD columns use the equivalent display scale implied by their stored endpoints and are labeled as such in exported metadata; the exact lower/upper endpoints remain available in separate-column mode. The exporter does not include the parameter-correlation matrix, which is calculated for display rather than stored as a result-table field.

Summary rows use the same Parameter Evaluation calculation at its default evaluation temperature: the configured reference temperature for a temperature series, otherwise the mean experiment temperature. The exported temperature column labels that evaluation temperature. Local summary rows report **Combined SD** and an **Approximate propagated interval**, using the same calculation described under Parameter Evaluation. Summary interval columns have neutral `_interval_lower` and `_interval_upper` suffixes; individual rows retain their CI95 columns. Kd is derived from the summarized Gibbs energy rather than averaged directly, and ΔCp columns are included for applicable temperature-dependent results. No results are pooled across separate analyses.

Result tables, clipboard output, final-figure metadata, and viewer output create thermodynamic columns from the fitted model shape. A sequential result therefore exports one Kd, ΔH, ΔG, and −TΔS value for every active step, including steps 3 and 4, rather than truncating the result to two interactions. The fixed step count is included as model information. Values remain in their displayed families; exporting does not convert macroscopic sequential constants into microscopic site constants.

Each affinity column chooses its concentration unit independently from its own magnitude. For example, Kd1 may be shown in nM while Kd4 is shown in µM. The column header and its values always use the same unit; this is display scaling only and does not change fitted or persisted values.



OneSetOfSites
9/8/2026 11:10 AM | One-Set-Of-Sites | 3 experiments
Fit of 3 experiments
OneSetOfSites; RMSD 0.572
All variables unconstrained

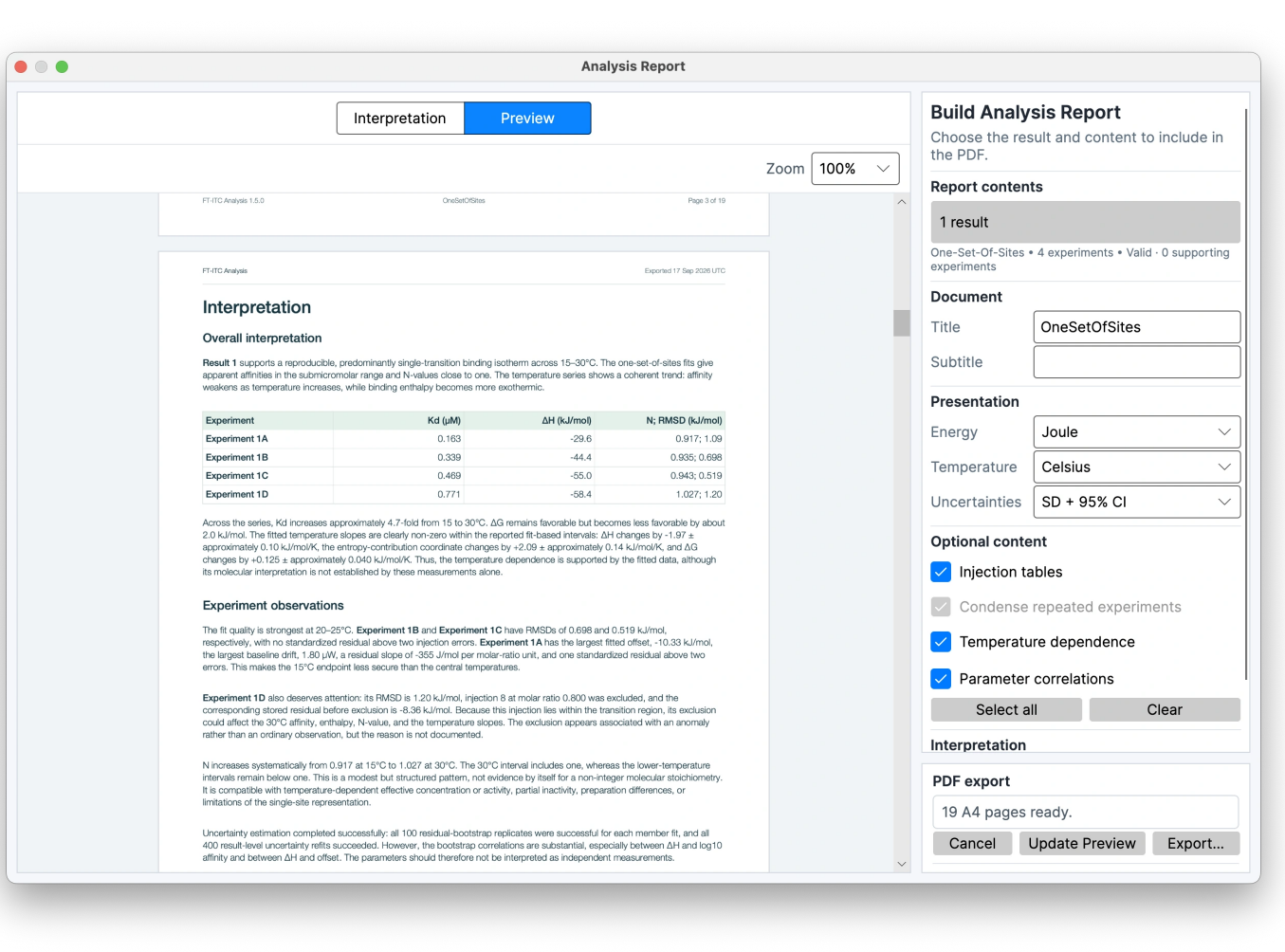
Global.OneSetOfSites
9/8/2026 11:10 AM | One-Set-Of-Sites | 3 experiments
Fit of 3 experiments
Global.OneSetOfSites; RMSD 0.552
Affinity: Temperature dependent

Rows
Mode: Summary rows
Uncertainty
Errors: Value with error
Style: SD
Format
File: CSV
Temperature: Celsius
Energy: Automatic
Export
Cancel | Copy | Export…

Analysis Result Exporter configures a delimited table for clipboard or file output.

Analysis Report

Tools > Analysis Report… creates a complete, printable report from one or more saved Analysis Results. The same tool is available as **Export Analysis Report…** in result-specific menus, where that result is initially selected. Use **Select report contents…** to choose results and optional supporting experiments together. The picker follows application order; applying the same ordered selection again restores its saved study context and approved interpretation. At least one result is required.



Interpretation
Overall interpretation

Experiment	Kd (µM)	ΔH (kJ/mol)	N; RMSD (kJ/mol)
Experiment 1A	0.163	-29.6	0.917; 1.09
Experiment 1B	0.339	-44.4	0.935; 0.698
Experiment 1C	0.469	-55.0	0.943; 0.519
Experiment 1D	0.771	-58.4	1.027; 1.20

Experiment observations

Build Analysis Report
Choose the result and content to include in the PDF.
Report contents
1 result
One-Set-Of-Sites • 4 experiments • Valid • 0 supporting experiments
Document
Title: OneSetOfSites
Subtitle:
Presentation
Energy: Joule
Temperature: Celsius
Uncertainties: SD + 95% CI
Optional content
☑ Injection tables
☐ Condense repeated experiments
☑ Temperature dependence
☑ Parameter correlations
Select all | Clear
Interpretation
PDF export
19 A4 pages ready.
Cancel | Update Preview | Export…

Analysis Report combines a paginated preview with controls for report contents, presentation, interpretation, and PDF export.

Report contents and layout

The report is a multipage A4 portrait PDF with margins suitable for printing. A report containing multiple results begins with a combined result index and concise table of contents on the first page, followed by the report-level interpretation and one self-contained chapter for each result. Results are referenced as **1**, **2**, and so on, and their experiments as **1A**, **1B**, **2A**, **2B**, and so on. These labels are derived from the saved result and experiment order, so figures, captions, and interpretation evidence use the same stable references. No experiments or scientific estimates are merged across results. Each chapter overview arranges the experiments in compact, borderless figure panels labeled with their references and names. The report automatically chooses three to five columns, centers incomplete rows, and suppresses repeated axis titles.

The analysis summary contains a compact grouped thermodynamic bar chart for every saved member and active binding step. Its whiskers show only the saved 95% confidence interval from the solution distribution or profile-likelihood calculation; the symmetric SD approximation remains available in report tables but is not superimposed on this plot. Each experiment section places a wider, baseline-focused view of the uncorrected thermogram—with its saved baseline and compact horizontal markers for the integration windows—beside a top-aligned Final Figure at its standard size containing the complete thermogram, followed closely by fitted and derived parameters. **Fit details** also reports dimensionless c-values where the fitted model defines them. The one- and two-site Wiseman values use `c = N[cell]₀/Kd`; syringe-correction fits use the fixed site count rather than the fitted syringe active fraction. Sequential fits report `[cell]₀/Kd` for each active step, while competitive fits report the apparent c-value based on `Kd_app`. When **Injection tables** is selected, a table records every saved injection, its inclusion state, volume, concentrations, analysis-axis value, integrated heat and uncertainty, fitted heat, and residual. This processing presentation is an automatic report preset and does not add another figure control. A concise notice replaces the processing graph when raw thermogram samples are unavailable. Selected shared correlations follow the analysis summary; selected member correlations follow that experiment's parameter table, using a compact native matrix rather than a dedicated page.

Directly selected experiments that are not represented by a selected result appear once in a report-level **Supporting experiments** chapter after the result chapters and are labeled **S1**, **S2**, and so on. The report uses only saved content: a raw thermogram with any saved baseline and integration boundaries, finite integrated heats explicitly labeled as having no fit, experiment metadata, notes about processing or correction exceptions, dates with a verified source, attributes, comments, and the report-wide optional injection table. Missing stages receive concise availability notices. Experiments already represented by a checked result remain visible but unavailable as supporting selections. Recorded buffer-subtraction references may be selected automatically according to the application setting, but remain ordinary optional selections; the report does not judge whether a reference is an appropriate blank. Report creation never fits, reintegrates, or performs subtraction.

Configure the report

Choose a title, an optional subtitle, energy and temperature units, and an **Uncertainties** style. The subtitle appears as secondary, regular-weight text below the report title. Reports default to **SD + 95% CI**. The selected style controls uncertainty in tables and applicable analysis plots. The thermodynamic summary bar chart always shows available saved 95% confidence intervals, as described above; it does not add SD whiskers. This selection is remembered only for the current application session. Final Figure injection errors and fitted confidence bands retain their established scientific meanings.

The subtitle editor accepts longer descriptions. It wraps text and scrolls vertically so you can review it before previewing the report.

Optional content includes report-wide **Injection tables** and **Condense repeated experiments** switches and the completed analysis types available in any selected result. Each option applies to every eligible result chapter; chapters without that saved content simply omit it. The injection-table switch is on by default and applies consistently to every experiment. When the same experiment occurs in a later result, condensed mode retains both figures, comments, source file, date with a verified source, instrument, temperature, concentrations, and all result-specific fitted content, but replaces repeated secondary acquisition and processing details with a reference to its first occurrence. The **Parameter correlations** option includes the shared matrix and all available member matrices alongside their corresponding summaries. Available analyses are selected by default; **Select all** and **Clear** change that analysis list together. Temperature dependence is included once per eligible result with its saved fit and parameter summary, while saved electrostatics, protonation, and Spolar outputs are presented without recomputation. Report creation never reruns fitting or advanced analysis.

Write and preview an interpretation

The builder opens in **Interpretation** mode with an editor for your remarks. Text wraps automatically. Interpretation Markdown supports `##` section headings, `###` subsection headings, bullets, `**bold**`, and `*italic*` emphasis. Use the **Interpretation / Preview** control above the workspace to move between writing and the rendered report. Selecting **Preview** automatically builds the PDF when it has not yet been built or is out of date; a current preview is reused. **Update Preview** forces a rebuild. Editing while a preview exists marks it as out of date without repeatedly rendering while you type. **Export PDF…** rebuilds stale content and saves the completed PDF. Results with warnings, outdated inputs, or partially invalid status remain exportable and carry a prominent notice. A structurally unusable result shows a validation error and cannot be previewed or exported.

Generate an automated interpretation

Online generation privacy: Generate sends selected names, comments, conditions, fits and injection evidence, plus your question or context, to the FT-ITC online interpretation service, which uses OpenAI to generate a response. Thermograms start unchecked each time the generator opens and require an explicit opt-in. The service retains usage metadata without automatic expiry; full scientific evidence and generated prose are not included in that metadata. Provider and infrastructure retention are separate; no deletion deadline is guaranteed. Canceling does not retract sent data. Local processing, fitting, saving and export work without generation. Data flow and retention.

The inspector's **Interpretation** section reports whether the text is manual, automatically generated, user edited, or out of date. **Edit** returns focus to the large editor. **Generate interpretation…** opens a dialog with optional **Main question** and **Additional context** fields. Use the question to focus the assessment and the context to explain the experiment. The cross-platform dialog labels its preset selector **Interpretation depth**. Requesting generation sends the selected report evidence and these inputs to the online interpretation service. Compressed thermograms are unchecked each time the generator opens; if **Include compressed thermograms** is available for your access, you can opt in for that request.

Generation assesses all selected results and supporting experiments together, using the same references as the PDF. Results remain independent, including alternative fits of the same experiment. Supporting experiments contribute available observations and metadata without being assigned a fitted result or assumed control role. Generation uses existing processing, integration, residual, and uncertainty evidence where available; it does not rerun integration, baseline fitting, or model fitting. When compressed thermograms are included, they contain minimum and maximum power values for each 15-second interval, plus an aligned fitted baseline when available. Exact peak timing, the order of values within each interval, and the detailed waveform are unavailable. The dialog shows the compact package size against your current request limit and blocks generation when the estimated request is too large. An opted-in thermogram is retained in the request; a size or model-context error is shown instead of silently omitting it.

When generation succeeds, the desktop app sends an **Interpretation ready** system notification. Whether it appears depends on operating-system notification support and permissions. The draft remains available in the generation dialog for review.

For fitted result members, the interpretation evidence includes the same model-specific c-values shown in **Fit details**. They remain attached to that result member rather than to shared experiment-source evidence. Supporting experiments have no c-value because the report does not assign them a fit.

When several result members have identical source data and processing, the request can share that evidence between them. Each result retains its own fitted values, uncertainty, validity, and constraints. Sharing source evidence does not imply independent replication or equivalent fits.

If the selection contains outdated or failed analyses, a warning gives you an opportunity to review them before continuing. The report distinguishes saved fit inputs and estimates from current observations; some comparisons are unavailable when the saved fit cannot be matched to the current data. An interpretation can identify potential concerns, but it does not certify the experiment or establish a mechanism. The service may suggest references from details such as a paper title, journal, or year, but it does not search the web.

Review and edit the returned draft, then choose **Use in report** to replace the approved interpretation. The draft focuses on observed issues and targeted follow-up checks, using references such as **Result 2** and **Experiment 1B**. It avoids routine warnings based solely on modest deviations in fitted N-values or missing blank titrations.

The interpretation service is asked to keep straightforward assessments to about 500–600 words or fewer, with roughly 1,000 words available for complex collections. These are writing targets, not acceptance limits. Longer text flows onto additional pages without automatic shortening or regeneration. Text length, headings, and Markdown formatting do not prevent you from accepting a draft.

Markdown tables require a header and separator row. They support bold or italic cell text and left, center, or right alignment. Cells wrap, and table headers repeat across pages. Use a few columns to keep tables readable. Unsupported formatting may appear as literal text in the report.

Canceling, closing the dialog, or encountering a service error leaves previously approved text unchanged. Study context, the thermogram setting, approved text, and generation details are saved with the report. Changes to selected results, supporting evidence, context, or settings can mark an approved automated interpretation as out of date. Older approved interpretations remain readable even when their freshness cannot be verified. The report distinguishes manually written interpretations from automatically generated text and subsequent edits.

The **Summary** generation task produces a compact factual report without knowledge-base retrieval or capability-code quota usage. Local report editing, saved interpretations, and PDF export remain available if automated interpretation generation is unavailable.

Save a local copy of interpretation evidence

Use **Save package** in the generation dialog to save a local ZIP of the evidence, question, context, and settings selected for that request. Saving the ZIP does not contact the interpretation service or run new fits. The archive contains scientific data and context; review it before sharing.

Troubleshoot automated interpretation

If generation fails, note the visible error and use **Copy Support Report** when contacting support. Local report editing and PDF export remain available if online generation is unavailable.

Choose the appropriate output tool

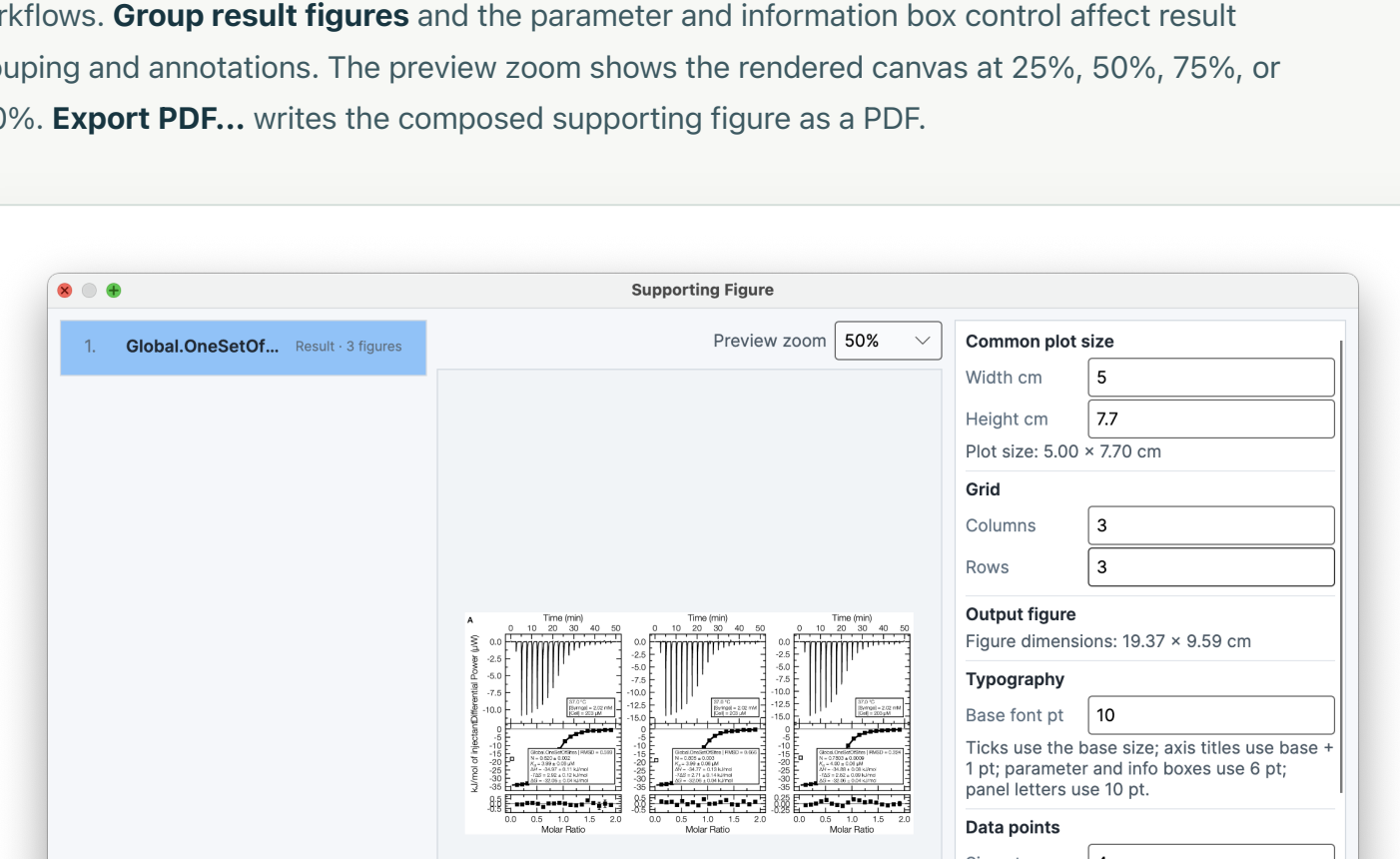
The Analysis Report is distinct from the other output tools:

- **Analysis Result Exporter** writes compact CSV or TSV tables for numerical reuse.
- **Final Figure** writes a publication figure for an experiment, while **Export Associated Final Figures...** writes one figure per result member.
- **Supporting Figure** composes selected figures into one multi-panel canvas.
- **Analysis Report** assembles one or more saved results as self-contained chapters with figures, detailed tables, diagnostics, optional advanced plots, and source and analysis history.

Supporting Figure

Supporting Figure... composes figures from Experiment Data and Analysis Results into a multi-panel canvas. The **Figure order** list defines source order; **Add...**, **Remove**, **Up**, and **Down** change the composition. The source picker can filter available experiment and result figures by name or type.

Common plot size specifies width and height in centimeters. Grid controls specify columns and rows. Typography controls define base font size, point size, line weight, and tick style. **Panel letters** adds stable A, B, ... identifiers. **Panel titles** optionally adds the experiment name after that identifier (for example, **A. Experiment name**) and is off by default for existing Supporting Figure workflows. **Group result figures** and the parameter and information box control affect result grouping and annotations. The preview zoom shows the rendered canvas at 25%, 50%, 75%, or 100%. **Export PDF...** writes the composed supporting figure as a PDF.



Preview zoom: 50%

Common plot size
Width cm: 5
Height cm: 7.7
Plot size: 5.00 × 7.70 cm

Grid
Columns: 3
Rows: 3

Output figure
Figure dimensions: 19.37 × 9.59 cm

Typography
Base font pt: 10
Ticks use the base size; axis titles use base + 1 pt; parameter and info boxes use 6 pt; panel letters use 10 pt.

Data points
Size pt: 4

Lines and ticks
Weight: 1 pt · Standard ticks

Export Figure
3 panels ·

Supporting Figure arranges the three member figures from an Analysis Result on a shared multi-panel canvas.

Printing

File > Print prints the active graph or figure through the operating system's print workflow. The active target can be the overview thermogram, processing graph, analysis graph, result graph—including an available **Correlation** matrix—or Final Figure, depending on the selected workspace. The operating-system print dialog supplies the available printer and PDF destinations; the graph content comes from the active application view.

CHAPTER 10

Tools

Design simulated titrations, subtract buffer controls, and merge standard or back-mixed tandem experiments.

Last verified: 2026-09-23

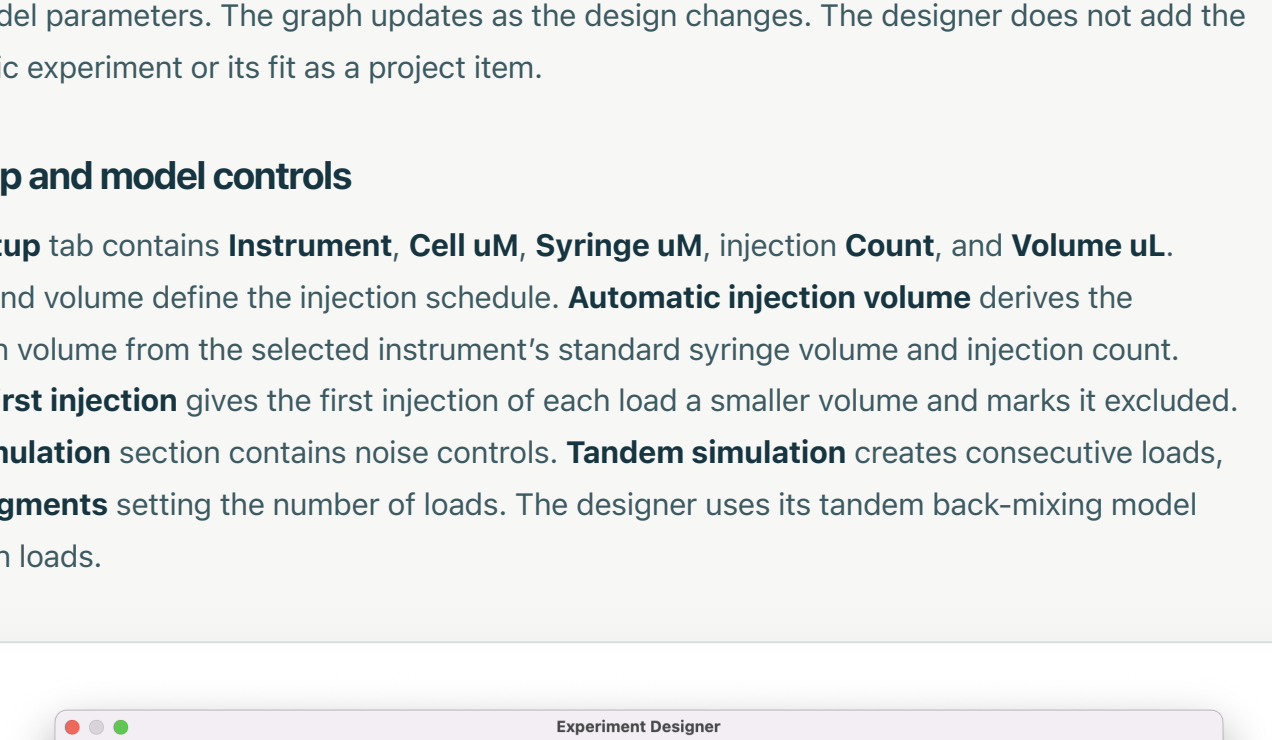
The **Tools** menu contains **Experiment Designer...**, **Buffer Subtraction...**, and **Experiment Merger...**. Each tool handles its output differently: Experiment Designer keeps simulation and fitting inside its window, Buffer Subtraction stores a correction on target experiments, and Experiment Merger creates a new processed Experiment Data item. Source and target selection uses the project state described in Workspace.

Experiment Designer

Experiment Designer... creates a synthetic titration from instrument, concentration, injection, and model parameters. The graph updates as the design changes. The designer does not add the synthetic experiment or its fit as a project item.

Setup and model controls

The **Setup** tab contains **Instrument**, **Cell uM**, **Syringe uM**, injection **Count**, and **Volume uL**. Count and volume define the injection schedule. **Automatic injection volume** derives the injection volume from the selected instrument's standard syringe volume and injection count. **Small first injection** gives the first injection of each load a smaller volume and marks it excluded. The **Simulation** section contains noise controls. **Tandem simulation** creates consecutive loads, with **Segments** setting the number of loads. The designer uses its tandem back-mixing model between loads.


Experiment Designer
Heat (kcal/mol)
Molar Ratio
OneSetOfSites | RMSD = 0.073
N = 0.994 [0.985, 1.000]
Kd = 103 ± 8 nM
ΔH = -7.15 [-7.28, -7.07] kcal/mol
-TΔS = -2.38 ± 0.08 kcal/mol
ΔG = -9.53 ± 0.05 kcal/mol
Offset = -0.03 ± 0.03 kcal/mol
Setup
Model
Apply / Fit

Instrument
Instrument: MicroCal ITC200
Syringe volume: 39.5 uL
Cell volume: 204.0 uL

Concentrations
Cell uM: 10
Syringe uM: 100

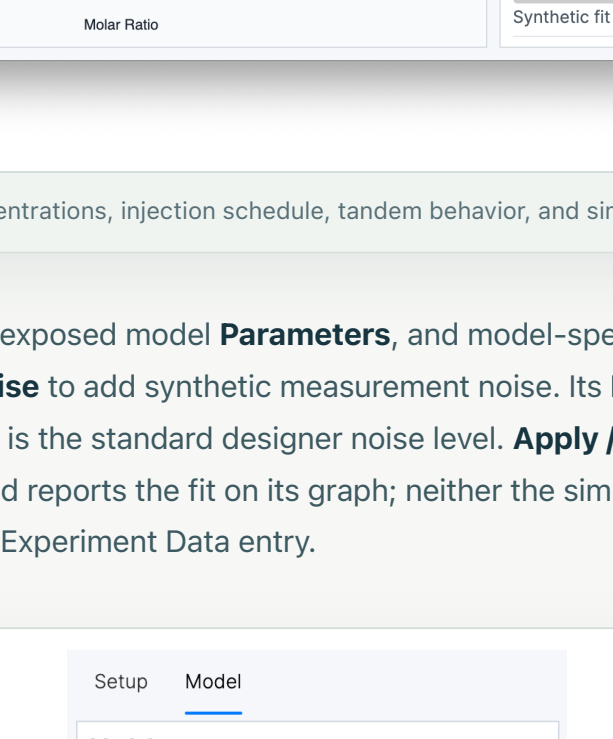
Injections
Count: 20
Volume uL: 2
☑ Automatic injection volume
☑ Small first injection
1 × 0.5 uL, 19 × 2.053 uL

Tandem
☐ Tandem simulation

Simulation
☑ Simulate noise
Synthetic fit complete.

Setup defines the instrument, concentrations, injection schedule, tandem behavior, and simulation state.

The **Model** tab contains **Type**, exposed model **Parameters**, and model-specific **Options**. In the Setup tab, enable **Simulate noise** to add synthetic measurement noise. Its **Noise level** slider ranges from **0.1×** to **5.0×**; **1.0×** is the standard designer noise level. **Apply / Fit** fits the synthetic data in the designer window and reports the fit on its graph; neither the simulation nor this fit becomes an Analysis Result or Experiment Data entry.



Model
Type: One-Set-Of-Sites

Parameters
N-value
N
0.994042 unitless

Enthalpy
ΔH
-7.15288 kcal/mol

Affinity
Kd
0.102797 µM

Simulation
☑ Simulate noise
Synthetic fit complete.

Model exposes the synthetic model type and its parameter values.

Buffer Subtraction

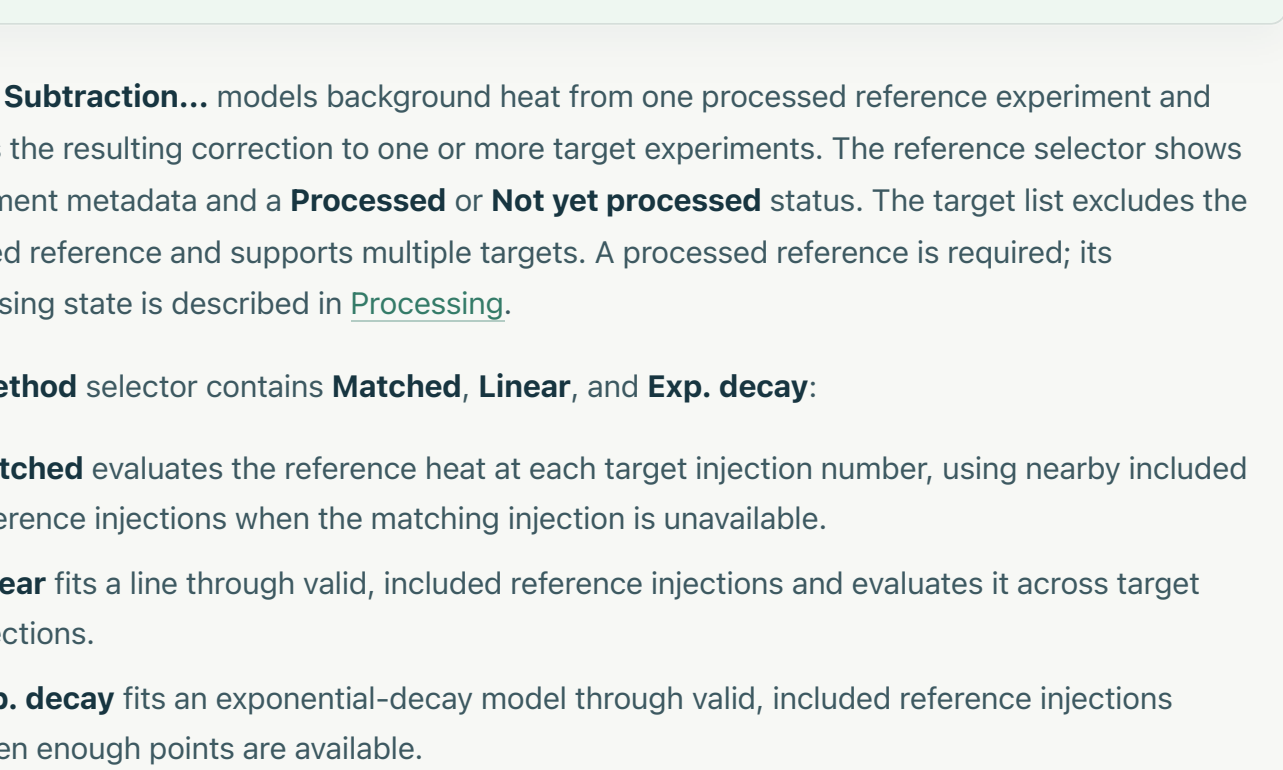
> **BEFORE YOU BEGIN:**
> Process the reference experiment and every target experiment. Buffer subtraction operates on their integrated injection heats.

Buffer Subtraction... models background heat from one processed reference experiment and applies the resulting correction to one or more target experiments. The reference selector shows experiment metadata and a **Processed** or **Not yet processed** status. The target list excludes the selected reference and supports multiple targets. A processed reference is required; its processing state is described in Processing.

The **Method** selector contains **Matched**, **Linear**, and **Exp. decay**:

- **Matched** evaluates the reference heat at each target injection number, using nearby included reference injections when the matching injection is unavailable.
- **Linear** fits a line through valid, included reference injections and evaluates it across target injections.
- **Exp. decay** fits an exponential-decay model through valid, included reference injections when enough points are available.

The preview graph shows reference and target heats and the selected subtraction model. Reference-point inclusion changes the points available to the fitted methods. **Focus Y axis on buffer data** changes only the preview range. A continuous model line appears for **Linear** and **Exp. decay**; **Matched** is represented by injection-level reference values.



Reference
Experiment: hepes-blank
Method: Linear
☑ Focus Y axis on buffer data
13.08.2026 18.56
25 °C | [Syringe] 1000 µM | [Cell] 100 µM
Processed

Targets
hepes-01
13.08.2026 18.56 | 25 °C | 1000 µM
hepes-02
13.08.2026 18.56 | 25.2 °C | 1000 µM
hepes-03
13.08.2026 18.56 | 25 °C | 1000 µM

Apply

Buffer Subtraction previews the selected reference model and target experiments before storing the correction.

> **CALCULATION:**
>
> $$q_{i,\mathrm{corr}} = q_{i,\mathrm{target}} - q_{i,\mathrm{ref}}$$
>
> The selected method determines the reference value evaluated for injection i. $q_{i,\mathrm{target}}$ is the target heat, $q_{i,\mathrm{ref}}$ is the corresponding reference value, and $q_{i,\mathrm{corr}}$ is the corrected heat used for fitting and export.

Apply stores the reference and method on each target. The corrected heats are then used by downstream fitting and export while the original integrated heats remain unchanged. The reference Experiment Data becomes inactive. Changes in its processing or injection inclusion update the target corrections. The subtraction is project data and can affect the validity of dependent results.

Experiment Merger

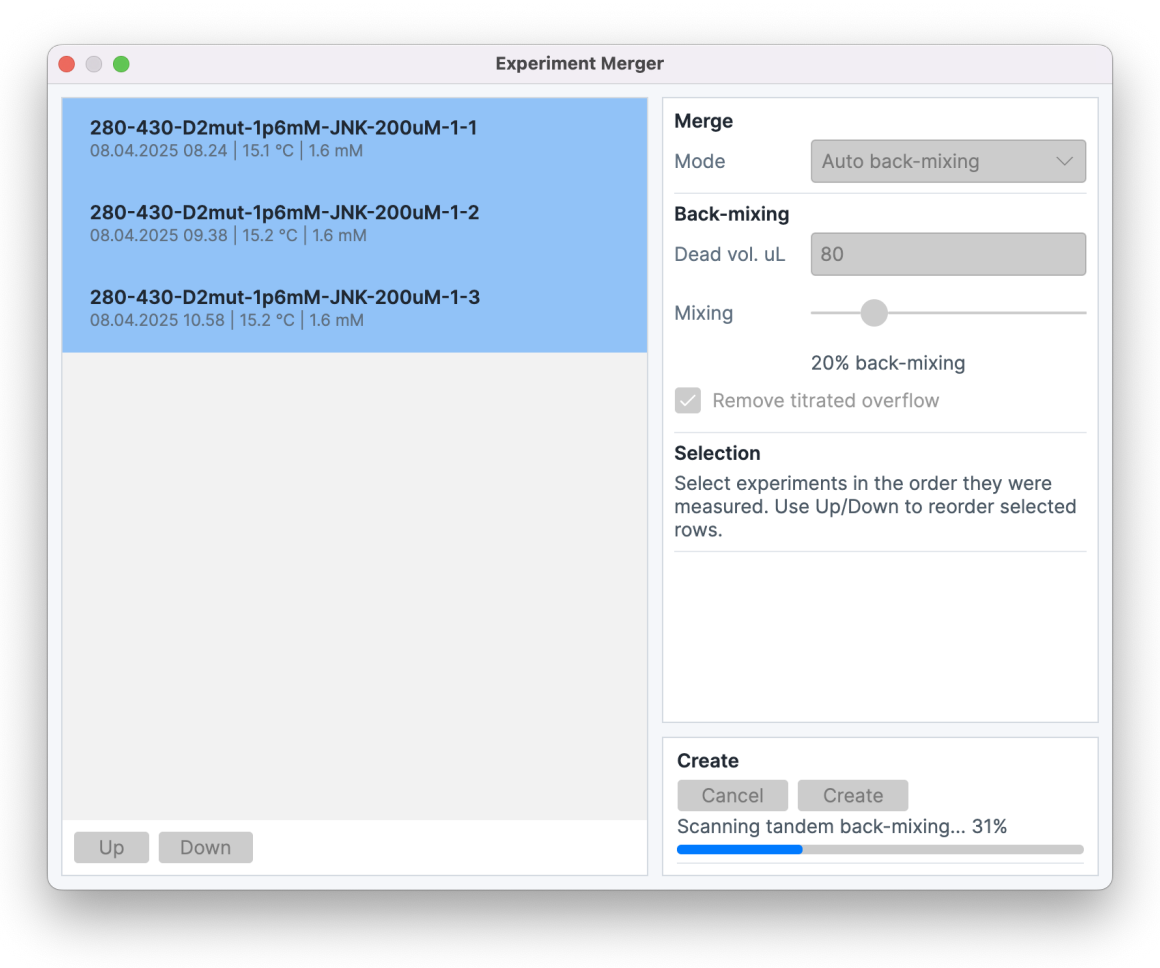
> **BEFORE YOU BEGIN:**
> Process each source experiment before merging. The merger uses each source's baseline-corrected thermogram when processing is available; otherwise, the application can fall back to the raw thermogram. The newly merged Experiment Data is processed automatically.

Experiment Merger... joins two or more eligible thermogram experiments from consecutive segments of a tandem titration. The source list contains thermograms that are not already tandem experiments. Selection order defines segment order; **Up** and **Down** reorder selected rows.

The merge **Mode** selector contains:

- **Simple tandem**, which concatenates the segments using the standard concentration progression without a user-selected back-mixing correction.
- **Fixed back-mixing**, which applies one configured mixing fraction at every segment transition. With three or four selected experiments, you can instead set an individual fraction for each reload.
- **Auto back-mixing**, which estimates the reload mixing fractions from the selected experiments and is available for up to five source experiments.

Back-mixing controls include **Dead vol. uL**, the **Mixing** fraction, and **Remove titrated overflow**. Dead volume represents the filling-stem or overflow volume above the active cell volume. The overflow control records whether titrated overflow was removed between segments. In Fixed mode, the shared slider supplies the fraction. With three or four experiments selected, enable the individual-fractions option to set **Reload 1**, **Reload 2**, and, for four experiments, **Reload 3** separately. Auto mode estimates the values from the selected data; these are model-based estimates rather than direct measurements of mixing.



280-430-D2mut-1p6mM-JNK-200uM-1-1
08.04.2025 08.24 | 15.1 °C | 1.6 mM
280-430-D2mut-1p6mM-JNK-200uM-1-2
08.04.2025 09.38 | 15.2 °C | 1.6 mM
280-430-D2mut-1p6mM-JNK-200uM-1-3
08.04.2025 10.58 | 15.2 °C | 1.6 mM

Merge
Mode: Auto back-mixing

Back-mixing
Dead vol. uL: 80
Mixing
20% back-mixing
☑ Remove titrated overflow

Selection
Select experiments in the order they were measured. Use Up/Down to reorder selected rows.

Create
Scanning tandem back-mixing... 31%

Auto back-mixing scans possible transition corrections for an ordered tandem experiment set.

Create produces a new processed Experiment Data item. Its thermogram samples are time-shifted and concatenated, its injection sequence retains segment boundaries, and its segment metadata stores the calculated starting active-cell and active-titrant concentrations. The merged item's comments record the selected tandem mode and back-mixing parameters. Source experiments remain separate and are not changed by creation; the new item is marked as a tandem experiment and is not eligible as a later merger source. It is a snapshot and does not update if its source experiments are subsequently edited. The resulting item can be fitted through Analyze Data.

> **INTERPRETATION:**
> A configured or automatically fitted back-mixing fraction is a model-based correction, not a direct measurement of liquid mixing that occurred between runs.

Tandem injection-displacement correction

The **Injection bookkeeping** preference selects **MicroCal**, **Ideal continuous mixing** or **Discrete displacement** for newly constructed tandem experiments, in both simple concatenation and back-mixing modes. Existing tandem experiments retain their saved method; rebuild through the tandem tool to change it. Let u be cumulative injected volume divided by active cell volume. The reference curves without back-mixing are

$$A_M(u) = (1 - u/2) \,/\, (1 + u/2)$$
$$B_M(u) = u \,/\, (1 + u/2)$$

for MicroCal, and

$$A_E(u) = \exp(-u)$$
$$B_E(u) = 1 - \exp(-u)$$

for the ideal continuous mixing concentration law. A is the retained fraction of the original cell material and B is the cell concentration of syringe material relative to its syringe concentration. This convention also accounts for displaced heat along the mixing trajectory, starting from each segment's recorded initial state; see Injection bookkeeping.

For an injection advancing the history from u_0 to u_1, define

$$r = A(u_1) \,/\, A(u_0)$$
$$M_1 = rM_0$$
$$L_1 = rL_0 + C_s[B(u_1) - rB(u_0)]$$

Here M and L are the macromolecule and ligand concentrations in the active cell and C_s is the syringe concentration. After a segment transition, the concentrations produced by the active/dead-volume mixing model provide the starting state, while u retains the cumulative injection history. Without back-mixing, repeated application of these equations reproduces the reference curves exactly.

For **Discrete displacement**, use the individual shot retention $r = 1 - v/V$, giving $M_1 = rM_0$ and $L_1 = rL_0 + (1 - r)C_s$. Within each segment, concentrations are reconstructed from its starting state and the running product of shot retentions. The active/dead-volume conservation and inter-segment mixing equations are unchanged. Heat uses the recorded segment starting state without evaluating previous injection heats.

The MicroCal reference curves are from Malvern Instruments, *MicroCal ITC Analysis Software Using Origin User Manual*, MAN0577-02-EN-00 (20 May 2015), section 12.3.1, equations 2 and 4, using the manual's approximate ligand expression. The equations assume the manual's average displaced concentrations; FT-ITC extends them to handle the starting concentrations recorded for each tandem segment. MicroCal bookkeeping has a limit when the cumulative injected volume approaches twice the active cell volume, so choose another method for longer sequences. Ideal continuous mixing has no corresponding cumulative-volume limit. Discrete displacement requires each individual injection to be smaller than the cell volume, without a cumulative-volume cap.

CHAPTER 11

Settings and defaults

Configure shared defaults for display, processing, fitting, and export.

Last verified: 2026-09-23

Preferences... contains **General**, **Processing**, **Fitting**, and **Export**. Settings labeled as defaults provide starting values for new or reset work. Display settings affect presentation; export settings affect generated tables and figures. Project-specific values stored in a project remain distinct from application preferences. Project and recovery behavior is covered in Installation, files, and projects.

Both desktop applications use the same built-in values for first launch and Restore Defaults, including a **20,000-iteration optimizer limit** and **10% automatic concentration SD**. Saved custom values remain in effect, including values between or beyond the slider presets. Choosing **Apply** also preserves settings that are not shown in that application's Preferences window.

Restore Defaults stages the built-in values in the window. **Apply** saves the staged values as application preferences. **Cancel** closes the window without saving staged edits.

General

SETTING	EFFECT
Energy units	Select **Joule** or **Calories**. Displayed values automatically use the base or kilo prefix from the finite central values being shown; empty groups default to kJ or kcal.
Concentration unit	Shared default unit for concentration entry, parameter display, and supported result-table concentration fields; format-specific data exports use their documented units.
Designer instrument	Shared default instrument for the Experiment Designer and its instrument-specific volumes.
Number precision	Controls numeric presentation: **Strict**, **Standard**, **Single decimal**, or **All decimals**. It does not set export decimal places.
Uncertainty display	Shared uncertainty presentation: **Automatic**, **Standard deviation**, **Confidence interval**, **SD + confidence interval**, or **None**. Automatic uses the confidence interval for a quantity whose stored 95% interval is materially asymmetric around its best-fit value; otherwise it uses SD. This changes presentation, not the underlying fit or calculated uncertainty.
Reference temperature (°C)	Temperature used where a result or derived quantity is evaluated at a reference temperature.
Minimum temperature span (°C)	Minimum temperature variation required for temperature-dependent result analyses.
Minimum salt span (mM)	Minimum ionic-strength variation required for salt-dependent result analyses.
Include buffer in ionic-strength calculation	Includes buffer contribution when ionic strength is calculated.
Check for updates and online resources on launch	Controls launch-time online checks; it does not disable local analysis.
Confirm remove/delete actions	Controls confirmation before remove or delete commands.
Automatically discard injections outside the thermogram range	Controls automatic removal of orphan injections when data are loaded.
Enable autosave	Enables periodic recovery copies.
Interval (minutes)	Sets the autosave interval. The built-in default is **5 minutes**.
Maximum files	Sets the number of autosave files retained. The built-in default is **10 files**.
Prompt to recover after an interrupted session	Controls whether the application presents an available autosave recovery after an interrupted session. Recovery prompting is enabled by default.
Open Autosave Folder	Opens the folder containing autosave recovery files.

Energy family, concentration unit, number precision, and uncertainty presentation are shared application settings where the corresponding control is available. The energy family changes displayed values but does not change the units used by supported interchange formats. Automatic display uses kJ or kcal when a value group is empty, zero, non-finite, or at or above the 100-unit threshold; fixed publication/result-export overrides are configured in their respective dialogs.

Processing

SETTING	EFFECT
Injection bookkeeping	Selects **MicroCal** (default), **Ideal continuous mixing** or **Discrete displacement** for new data. Ideal continuous mixing models exponential concentrations and accounts for displaced heat along the mixing trajectory; Discrete displacement uses discrete replacement without numerical integration. Change an existing ordinary experiment explicitly in Experiment Details; rebuild a tandem experiment to change its method. Saved fits are not automatically migrated.
Buffer subtraction	Sets the default buffer-subtraction model: **Matched**, **Linear**, or **Exp. decay**.
Discard integration regions for baseline	Controls whether existing integration regions are excluded from baseline construction.
Reprocess integrated heats on load	Sets the initial **Recalculate concentrations and ratios** choice for `.dat` and `.aff` imports. You can change it for an import without changing this preference. Recalculation uses the imported injection volumes and experiment concentrations; it does not create a thermogram or repeat baseline correction and peak integration. `.dh` concentrations are always calculated.
Point density	Sets the default spline point density: **Sparse**, **Balanced**, or **Dense**.
Handle mode	Sets the default spline handle calculation: **Mean** or **Median** in the preferences window.
Allow spline point time dragging by default	Controls whether spline points can be moved in time by default.
Copy integration start with selected region	Controls whether copying an integration region also copies its start time.

Processing preferences provide defaults for new processors. Processing values already stored with an experiment are not replaced merely by changing a preference.

Fitting

SETTING	EFFECT
Default solver	Sets the starting optimizer: **Nelder-Mead [SIMPLEX]** or **Levenberg-Marquardt**.
Error estimation	Sets the default uncertainty method: **None**, **Bootstrap residuals**, **Leave-one-out**, or **Profile likelihood**. The built-in method is **Bootstrap residuals**.
Bootstrap iterations	Sets the number of residual-bootstrap refits. Leave-one-out uses one refit per deletion and profile likelihood does not use this count. The built-in count is **100**.
Optimizer tolerance	Sets the solver tolerance preset: **Fast**, **Relaxed**, **Balanced**, **Strict**, or **Very Strict**. The built-in default is **Balanced**.
Max iterations	Sets the maximum number of optimizer iterations. The built-in default is **20,000**, both at first launch and after Restore Defaults is applied.
Parameter limits	Sets the default parameter-limit policy: **Standard**, **Extended**, or **No limit**.
Use injection-error weighted fitting	Controls weighting of injection observations by their estimated errors.
Include concentration uncertainty in bootstrap	Includes concentration uncertainty in residual-bootstrap resampling. Leave-one-out keeps concentrations fixed.
Automatic concentration SD (%)	Sets the automatic fractional concentration SD used when concentration-uncertainty handling is enabled.
Create single-experiment analysis result	Controls creation of an Analysis Result after a usable single-experiment fit. Disabled in the built-in defaults.
Create global analysis result	Controls creation of a combined Analysis Result after a usable multiple-experiment fit. Enabled in the built-in defaults.
Auto-open new analysis result	Controls whether a newly created result is opened automatically.

Bootstrap method and count are shared fitting defaults; concentration sampling and parameter unlocking apply only to residual bootstrap. If profiling is canceled or fails, the successful primary fit is retained. Whether an update replaces an existing result with partial profile output depends on the stored update policy. Fit-specific settings captured in an Analysis Result remain part of that result. See Single-experiment fitting for model and uncertainty interpretation.

Automated interpretation access

To request an account with additional interpretation options, use the FT-ITC interpretation service registration page. If eligible, submit your name and email, accept the Terms and acknowledge the Privacy Notice, then verify your email; the access code is sent separately by email. Keep the code confidential. If it is exposed or does not arrive, contact `support@ft-itc.org`.

Without a code, automated interpretation uses the public **Default** depth. Enter the account code and choose **Verify Access** to load the options available to your account. When additional presets are available, choose one from **Interpretation depth**. Preferences show available account details and remaining quota. The code stays masked and is checked when an interpretation is generated.

Export

SETTING	EFFECT
Selection	Sets the default export scope: **Selected experiment**, **Active experiments**, or **All experiments**.
Decimals	Sets the number of decimal places in exported numeric tables.
Export baseline-corrected data	Includes baseline-corrected data in data exports.
Export fit points with peaks	Includes fitted peak points with exported peak data.
Molar ratio	Includes molar-ratio values in exported tables.
Injection info	Includes injection volume, delay, peak error, integration length, and temperature columns.
Concentrations	Includes cell and syringe concentration columns.
Included state	Includes the injection-inclusion state.
Peak heats	Includes integrated peak heats.
Fit values	Includes fitted values.
Width cm and **Height cm**	Set the default final-figure dimensions. The built-in dimensions are **6.5 × 10 cm**.
Publication font	Selects the publication figure font where the platform provides this selector.

Show residual graph	Includes the residual graph in final figures.
Show residual graph gap	Includes the gap separating residual and fit panels.
Unify residual graph axis	Uses a common residual-axis scale across applicable panels.
Fit line	Sets fit-line smoothing: **Smooth**, **Spline**, or **Linear**.
Show parameter box by default	Includes the parameter information box in new final figures.
Show experiment details by default	Includes experiment metadata in the figure information box.
Show model info by default	Includes model information in the figure information box.
Auto axes ignore excluded/bad points	Excludes points marked excluded or bad when automatic axes are calculated.
Thermodynamic parameters	Includes thermodynamic parameters in final-figure information.
Offset parameter	Includes the fit offset in final-figure information.
Derived parameters	Includes derived parameters in final-figure information.
Temperature	Includes temperature in final-figure information.
Concentrations	Includes concentrations in final-figure information.
Injection delay	Includes injection delay in final-figure information.
Instrument	Includes instrument information in final-figure information.
Attributes	Includes experiment attributes in final-figure information and limits their display to **Used in analysis**, **All**, or **None**.

Export preferences affect newly generated exports and figure defaults; they do not rewrite an existing export or a stored figure configuration. See Figures and export for output formats and units.

The energy-unit preference also sets the initial molar-energy unit in the Integrated Peaks, Combined Data, and ITCsim export dialogs: kJ/mol for Joules or kcal/mol for Calories. You can choose another unit for each export without changing the preference.

> **PLATFORM NOTE:**
> The **Publication font** selector is available on Windows and Linux, with choices of **Native**, **Inter**, and **Liberation Sans**. Some macOS releases use the native publication renderer and do not show this selector.

CHAPTER 12

Glossary and equations

Terminology, keyboard conventions, calculation symbols, equations, and references.

Last verified: 2026-09-23

Workspace terms

TERM	MEANING IN FT-ITC ANALYSIS
Experiment Data	One imported or application-created ITC dataset, including its details and analysis state.
Data / Results	The workspace list containing experiments and stored Analysis Results.
Selected	The item currently shown or targeted by a selection-specific command.
Active experiments	Experiments included in a multi-experiment operation.
Solution	A fitted model state for one experiment.
Analysis Result	A stored fit result with one or more member solutions and the fit state used to create it.
Attribute	Experiment metadata used for organization or analysis context.
Valid result	An Analysis Result whose recorded fit inputs still match the current project state.
Correlation view	Pearson correlations between fitted parameter coordinates across residual-bootstrap refits that have finite values for every displayed parameter, with diagnostics for simulation precision.
Pearson correlation	A value from −1 to +1 describing linear co-variation between two fitted coordinates.
Monte Carlo precision interval	Approximate Fisher-transformation interval describing how precisely the correlation coefficient *r* is estimated from the finite retained-refit ensemble; it is not uncertainty for either fitted parameter or a model-validity assessment.
AIC / AICc	Analysis-level criteria for comparing saved fits with the same observations, response definition, weighting mode, and likelihood convention; smaller values are preferred. Both modes estimate one residual-variance parameter; weighted criteria use integration errors as relative uncertainties. AICc uses the standard small-sample approximation for nonlinear fits and is preferred when available. Independently fitted members also show a per-member AICc / AIC column for same-experiment comparisons. Neither criterion establishes model adequacy.

Uncertainty terms

TERM	MEANING IN FT-ITC ANALYSIS
Best-fit value	Parameter value from the primary optimization. Resampling estimates uncertainty around this value but does not replace it with a resampling mean or median.
Standard deviation (SD)	Root-mean-square deviation of retained resampling values from the primary best-fit value.
95% confidence interval (CI)	The 2.5th and 97.5th percentiles of the retained resampling distribution. It can be asymmetric around the best-fit value.
Automatic uncertainty display	Shows CI for a materially asymmetric stored interval and SD otherwise. It selects a presentation separately for each reported quantity.
Displayed parameter uncertainty	Fitted-coordinate uncertainty after conversion or propagation into the quantity shown to the user. The Automatic display decision is applied to this displayed quantity.
Profile likelihood	Conditional profiling at a fixed 95% confidence level. Unweighted and weighted intervals use an F-calibrated worsening relative to the best residual score. Weighted intervals retain processing-derived peak-area SDs as unchanged relative injection weights and estimate the overall residual scale separately. A parameter bound reached before crossing is censoring, not a confidence endpoint. The reported primary fit remains unchanged.
Profile equivalent SD	Symmetric `value ± SD` display scale computed from a complete asymmetric profile interval; it is a display equivalent, not a Gaussian sampling SD.

Keyboard conventions

The application command modifier is **Command** on macOS and **Ctrl** on Windows or Linux.

SHORTCUT	FUNCTION
Command/Ctrl + O	Open data or a project.
Command/Ctrl + S	Save the current project.
Command/Ctrl + Shift + S	Save the project under another name.
Command/Ctrl + E	Open data export.
Command/Ctrl + P	Print the active graph or figure.
Command/Ctrl + Z	Restore the most recently removed item.
Command/Ctrl + I	Invert the Active Experiment Data set.
Command/Ctrl + comma	Open Preferences.
Command/Ctrl + Q	Quit FT-ITC Analysis.
F1	Open **Help and Guide** on Windows and Linux. On macOS, open Help through the **?** menu command.
Enter	Open **Details** for the selected Data/Results item on Windows and Linux.
Left / Right	Change the selected injection in **Process Data**.
Space	Copy the selected integration length to the next injection in **Process Data**; the start is also copied when that processing option is enabled.

Thermodynamic symbols

SYMBOL	MEANING
N	Stoichiometry or site-number term under the selected model.
α	Syringe-side active-concentration correction when syringe correction is enabled.
K_a	Association constant.
K_d	Dissociation constant.
K_i	Macroscopic association constant for sequential transition *i*.
β_i	Cumulative sequential association product $\prod_{\mathrm{j}=1\ldots\mathrm{i}} K_\mathrm{j}$.
F_i	Fraction of macromolecule in sequential state MX_i.
$\bar{\nu}$	Mean ligand occupancy in the sequential model.
ΔH	Binding enthalpy change.
ΔG	Gibbs free-energy change.
$-T\Delta S$	Entropic contribution reported by the application at the evaluation temperature.
ΔC_p	Heat-capacity change in a supported temperature-dependent analysis.
q	Integrated injection heat.
σ	Standard uncertainty or spread, according to context.
Offset	Energy-per-mole-of-injectant correction included in the modeled injection heat.

Calculation symbols

SYMBOL	MEANING
$P(t)$	Differential power at time t in a thermogram.
$b(t)$	Estimated baseline at time t.
n_i	Amount injected in injection i.
r_i	Residual for injection i: observed heat minus model heat.
r_jk	Pearson correlation between fitted parameter coordinates j and k.
θ_bj	Value of fitted coordinate j in complete bootstrap refit b.
σ_i	Processing-derived uncertainty for injection i.
R	Gas constant.
T	Absolute temperature, in kelvin, for thermodynamic relationships.
I	Ionic strength used in salt analysis.
a_ion	Ion activity used in Counter Ion Release analysis.
m	Fitted slope in the Protonation relationship; the application reports **Protons** as $-m$.
n_ion	Counter-ion slope reported by Counter Ion Release analysis.
RSS	Raw residual sum of squares, Σr_i^2, over included injections.
Q	Standardized residual sum of squares, $\Sigma(r_\mathrm{i}/\sigma_\mathrm{i})^2$. Weighted AIC/AICc estimates a common variance multiplier as Q/n.
L	Gaussian likelihood used for AIC/AICc.
p	Number of free fitted parameters in the saved global model; shared coordinates count once and member coordinates count per member.
K	AIC/AICc likelihood parameter count: $p + 1$ in both weighting modes, including one estimated common residual variance or one estimated multiplier of the injection-error variances.

Equation index

RELATIONSHIP	LOCAL EXPLANATION
Integration and molar normalization	Processing
Monomer–dimer association	Dissociation
Sequential binding polynomial and ligand balance	Sequential Binding Sites
Thermodynamic conversion and **−TΔS**	Thermodynamic relationships
Fit residuals, RMSD, and weighting	Weight by injection error
AIC, AICc, Gaussian likelihood, and parameter counting	Information criteria
Temperature-dependent constraints	Multiple-experiment fitting
Parameter correlation (Pearson residual bootstrap)	Parameter correlation
Bootstrap SD and percentile confidence interval	Parameter uncertainty
Automatic uncertainty display	Uncertainty and evaluation temperature
Salt dependence	Salt
Protonation dependence	Protonation
Buffer correction	Buffer Subtraction
Tandem injection-displacement correction	Tandem injection-displacement correction

Units are those shown by the application or export.

References

- FT-ITC Analysis repository
- FT-ITC Analysis website
- FT-ITC Project Viewer
- Software DOI: 10.5281/zenodo.14832177
- Malvern Instruments, *MicroCal ITC Analysis Software Using Origin User Manual*, MAN0577-02-EN-00 (20 May 2015), section 12.3.1, equations 2 and 4.
- **Help > Citation** for the current paper citation, versioned software citation, and BibTeX
- Philippe Dumas, *Isothermal titration calorimetry in the single-injection mode with imperfect mixing*, European Biophysics Journal 51, 77–84 (2022). The application's **Ideal continuous mixing** option is inspired by the paper but uses ideal mixing with a finite-injection numerical approximation; it does not implement the paper's imperfect-mixing model.
- Ernesto Freire, Arne Schön and Adrián Velazquez-Campoy, *Isothermal titration calorimetry: general formalism using binding polynomials*, Methods in Enzymology 455, 127–155 (2009). The application's **Discrete displacement** option uses this convention for concentrations and displaced-volume heat compensation, retaining FT-ITC's equilibrium solvers and offset convention.

Scientific claims and model-specific methods require the primary literature appropriate to the experiment and analysis.